

UNIVERSITY OF ZIMBABWE

SHORT-TERM MODELING AND ANALYSIS OF ZIMBABWE GOLD (ZIG) EXCHANGE RATES: AN EMPIRICAL STUDY USING CHANGE POINT ANALYSIS, TIME SERIES TECHNIQUES AND VOLATILITY MODELING

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To my caring mom Mrs Mtape,
my loving brother Tendai,
and everyone who has made it possible
for me to be here.

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List of Abbreviations

Abbreviation	Meaning
ARIMA	Auto-Regressive Integrated Moving Average
EGARCH	Exponential Generalized Autoregressive Conditional Heteroskedasticity
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
LSTM	Long Short-Term Memory
ML	Machine Learning
RBZ	Reserve Bank of Zimbabwe
RMSE	Root Mean Square Error
SVM	Support Vector Machine
USD	United States Dollar
ZiG	Zimbabwe Gold (currency)
ZWD	Zimbabwe Dollar
ZWL	Zimbabwe RTGS Dollar

Abstract

This dissertation looks into the short-term dynamics of the Zimbabwe Gold (ZiG) exchange rate after it was introduced in April 2024. It focuses on structural breaks, forecasting accuracy, and volatility assessment. The research uses a combination of change point analysis, time series modeling, and statistical testing to assess the currency's stability and behavior during its early months using daily exchange rate data from the Reserve Bank of Zimbabwe.

Using change point detection techniques such as the Chow Test, CUSUM, Bayesian, and Bai-Perron approaches, notable changes in the ZiG/USD exchange rate were found. On September 27, 2024, a large structural break was noted, which corresponded with a change in government policy. Time series models such as ARIMA, SARIMA, LSTM, Random Forest, and XGBoost were employed to forecast exchange rate movements over a 30-day horizon. Machine learning models, particularly Random Forest and XGBoost, demonstrated superior performance over traditional approaches in capturing non-linear and complex exchange rate patterns. The ARCH-LM test was performed to evaluate volatility; however, the findings showed no substantial volatility, hence the GARCH and EGARCH models were not developed.

The study provides useful insights for investors, policymakers, and financial analysts attempting to navigate the uncertainties of Zimbabwe's evolving monetary system. The findings highlight the ZiG's sensitivity to economic and policy shocks and the significance of adaptive, data-driven modeling techniques in forecasting emerging currency behaviors. Further research is advised to examine the ZiG's long-term effects on Zimbabwe's economy and to improve the forecasting models' robustness as more data becomes available.

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Chapter 1

Introduction

1.1 Introduction

Exchange rates have a significant impact on trade balances, investment choices, and general financial stability, making them essential markers of a nation's economic health. Changes in these rates have a big impact on inflation and foreign investment. In order to manage the intricacies of global finance and preserve economic stability, economists and decision-makers must closely examine exchange rates.

An important turning point in Zimbabwe's economic history was the April 2024 introduction of the Zimbabwe Gold (ZiG) currency. In order to stabilize the nation's economy, which has traditionally experienced hyperinflation and currency volatility, the ZiG was introduced. The ZiG, which is supported by gold and foreign reserves, seeks to rebuild confidence in Zimbabwe's currency by providing a more reliable substitute for earlier monetary systems. Early assessments, however, indicate that the ZiG exchange rate has already seen significant fluctuations, beginning with a period of stability and ending with a strong increase in September 2024. This change points to underlying causes affecting the stability and volatility of the currency that should be investigated further.

The goal of this study is to fully examine the dynamics of the ZiG/USD exchange rate, with an emphasis on volatility analysis, change point detection, and short-term modeling. Using daily exchange rate data from the Reserve Bank of Zimbabwe (RBZ) since the introduction of the Zimbabwean dollar (ZiG), this study seeks to analyze the stability of the exchange rate, identify notable changes, and forecast short-term variations. In order to help policymakers make well-informed decisions about currency stability and economic resilience, the study looks at the levels and volatility of the ZiG exchange rate.

This study will fill in knowledge gaps on the ZiG's behavior, particularly as a recently established currency in a fragile economic climate. It will also emphasize how crucial it is to identify changes in exchange rate dynamics in order to make timely policy adjustments. By doing this, this study adds important empirical information about the ZiG and how it interacts with other economic variables. The project framework will be outlined in this chapter's conclusion, along with how next chapters will fulfill the study goals and support broader efforts to promote economic stability in Zimbabwe.

1.2 Research Background

Significant changes have occurred throughout Zimbabwe's currency history, including times of hyperinflation and continuous attempts to stabilize the economy. Introduced in 1970, the Rhodesian Dollar (RHD) was the currency of Southern Rhodesia until 1980, when the country gained independence and adopted the Zimbabwe Dollar (ZWD) (Global Financial Data, 2024). At first, the ZWD was more valuable than the USD, but over time, steep inflationary pressures reduced its value, leading to hyperinflation in 2007 and 2008, when prices doubled every day at their highest point (Chen, 2023). The demonetization of the ZWD in 2009 and the implementation of a multi-currency system, mostly based on the US dollar, were the results of this severe economic instability. While the multi-currency system offered stability, it constrained Zimbabwes monetary policy flexibility and exposed the country to USD exchange rate volatility, which often misaligned with local economic demands (Makoni, 2019).

Zimbabwe tried to reestablish a sovereign currency in 2019 by combining bond notes and electronic money into the RTGS Dollar (ZWL). Despite being intended to bring about monetary independence, the ZWL was viewed with suspicion because of Zimbabwe's past currency problems. Low confidence and inflation continued to be major obstacles to stability, making attempts to float the ZWL unsuccessful (Muronzi, 2019). The economy consequently kept using the USD for day-to-day transactions, which reduced the ZWL's efficacy and indicated the country's reliance on foreign exchange (Nyathi & Dube, 2021).

In April 2024, the Zimbabwe Gold (ZiG) currency was introduced as the most recent attempt to fix Zimbabwe's monetary system. The ZiG is partially backed by gold reserves, in contrast to previous currencies, with the goal of fostering public confidence and establishing a stable, market-reflective monetary system within the framework of a flexible exchange rate. Despite supply and demand-driven fluctuations, the ZiG strives for stability with its gold backing (The Herald, 2024). At first, the ZiG/USD exchange rate hovered around 13 to 14 ZiG per USD. However, the official rate changed from 13.9 to 24.4 ZiG per USD on September 27, 2024, when the Reserve Bank of Zimbabwe depreciated the currency by 43%.

New exchange rate pressure brought on by rising food import prices and declining mineral export sales was blamed for this devaluation. The central bank chose to implement a currency adjustment instead of using reserves to preserve the prior rate in order to relieve this strain (Jusa, 2024). In light of shifting economic pressures, this incident emphasizes how crucial it is to investigate the underlying exchange rate dynamics of the ZiG.

Exchange rate analysis has emerged as a vital instrument for Zimbabwe's economic stability, as changes impact inflation, trade balances, and overall economic resilience. According to Yussif et al. (2021), exchange rate volatility can have a substantial impact on a country's import and export performance suggesting that changes may discourage foreign commerce or affect trading partners' pricing strategy. It is crucial to comprehend the ZiG's volatility in order to evaluate its impact on trade dynamics and economic stability.

This study's mainstay, time series analysis, offers a potent method for simulating short-term trends and patterns in currency swings. Clusters of volatility that represent market responses to economic situations can be captured when combined with volatility modeling, especially GARCH approaches. In line with the

objectives of this study, Yussif et al. (2021) point out that GARCH and EGARCH models are especially good at capturing abrupt changes in exchange rate behavior. For policymakers overseeing a new currency such as the ZiG, these insights can provide a projection of future times of stability or volatility, which is essential (Adams & Weiss, 2018; Lee & Kim, 2022).

Additionally, change point analysis is now useful for identifying changes in financial data series, which are frequently brought on by policy modifications or economic shocks. In their study, Manner, Rodrguez, and Stckler (2024) demonstrate how change point analysis can detect notable changes in financial market volatility and dependence, underscoring its importance in comprehending the interdependencies of economic variables, especially in times of crisis. Change point identification in ZiG data may highlight important structural changes that are essential for currency stability initiatives, especially in light of Zimbabwe's unstable economic climate.

Although there has been some research on emerging market currencies, there is a noticeable lack of studies that are specifically to Zimbabwe, particularly with regard to its new currency, the ZiG. By employing daily exchange rate data to provide a thorough understanding of ZiG's short-term dynamics, this study seeks to close this gap. This study will uncover volatility patterns and possible changes in the stability of the ZiG by using time series analysis, volatility modeling, and change point analysis. This will provide important insights into Zimbabwe's policy framework. The ZiG's role in fostering economic recovery and strengthening monetary sovereignty in Zimbabwe is greatly supported by these findings.

1.3 Statement Of The Problem

An important attempt was made to stabilize Zimbabwe's economy - previously beset by chronic hyperinflation and currency instability - through the introduction of the Zimbabwe Gold (ZiG) currency in April 2024. While designed to restore confidence in the nation's monetary system, the ZiG has exhibited concerning fluctuations against the US dollar, raising serious questions about its stability as a monetary instrument. Of particular concern is the currency's sudden spike in value; Chikohomero (2024) documents a dramatic surge from 13-14 ZiG/USD between April and August 2024 to a peak of 24 ZiG/USD by September 2024. This rapid appreciation undermines the ZiG's reliability as a stable medium of exchange, creates uncertainty for long-term investment planning, and casts doubt on whether its gold reserves provide adequate protection against market volatility.

Policymakers' need to identify early behavioral changes in the currency's performance adds to this difficulty. Real-time exchange rate turning point detection may allow for preventative measures to lessen sudden economic shocks, according to Lustig (2021). These uncertainties present serious obstacles to Zimbabwe's overall economic recovery, which is heavily reliant on the establishment of a reliable and stable currency, as the Synergia Foundation (2019) paper emphasizes. Efforts to draw in foreign investment and restore confidence among domestic businesses may stall if the short-term volatility patterns of the ZiG are not well understood, which would further impede the country's progress toward sustainable growth.

Lack of empirical study of the ZiG's dynamics, especially in time series trends, volatility modeling, and change point detection, is a crucial gap that exacerbates these difficulties. While machine learning ap-

proaches provide deeper insights into non-linear patterns, change point analysis aids in identifying significant changes in the currency's behavior, and traditional time series methods can show macroeconomic trends. Evidence-based stabilization methods would be informed by a more robust framework for evaluating the ZiG's responsiveness to market forces, which would be provided by integrating these methodologies.

In light of this, this study aims to conduct a thorough analysis of the ZiG exchange rate through the use of change point detection, volatility modeling, and time series techniques. The research attempts to improve knowledge of the ZiG's function in Zimbabwe's monetary system by spotting trends in exchange rate behavior and the causes of its swings, ultimately assisting with policies that promote economic recovery and fortify monetary sovereignty.

1.4 Research Questions

The study by the researcher intends to address the questions that follow:

1. What are the historical trends in the ZiG exchange rates against the USD since its introduction in April 2024?
2. Can changes in the ZiG exchange rate's behavior be accurately detected using change point analysis, and what do these changes suggest about the stability of the currency?
3. Which change point analysis technique (like Chow Test, CUSUM, Bai-Perron, Bayesian) is most effective in identifying structural breaks in the ZiG/USD exchange rate?
4. Which time series modeling approaches are most capable of accurately predicting future changes in the ZiG/USD exchange rates?
5. Does the ZiG/USD exchange rate exhibit statistically significant volatility during the study period, and if so, what models best capture its dynamics?
6. What effects might the results have on Zimbabwe's economic strategy?

1.5 Research Objectives

This study's main objective is to use change point analysis, time series methods and volatility modeling to examine the short-term dynamics of the Zimbabwe Gold (ZiG) exchange rates.

The particular goals are:

1. To examine the ZiG exchange rates' historical patterns in relation to the USD since their launch in April 2024.
2. To identify key changes in the ZiG exchange rate's behavior using multiple change point analysis techniques and evaluate their effects on currency stability.
3. To compare the performance of the change point analysis techniques and determine the most effective method for detecting structural breaks in the ZiG/USD exchange rate

4. To develop and apply both traditional time series models (like ARIMA) and machine learning-based approaches (like Long Short-Term Memory (LSTM) networks) to predict future changes in the ZiG/USD exchange rate, thereby assessing and comparing their effectiveness in capturing short-term exchange rate dynamics.

5. To assess the presence and patterns of volatility in the ZiG/USD exchange rate using statistical tests (e.g., ARCH-LM, Engles Test) and, if detected, model its dynamics using GARCH-family approaches.

6. To offer suggestions to decision-makers on the findings' significance for Zimbabwe's economic policy.

1.6 Research Hypothesis

This study will be guided by the following research hypotheses:

- **H1:** The ZiG exchange rates versus the USD have shown notable historical tendencies since April 2024.
- **H2:**The Bai-Perron method will prove most effective for structural break detection in ZiG/USD rates, outperforming Chow, CUSUM, and Bayesian approaches.
- **H3:**Machine learning models (like XGBOOST) outperform traditional time series models in forecasting ZiG/USD exchange rate fluctuations
- **H4:**The ZiG/USD exchange rate exhibits statistically significant volatility patterns during the study period as measured by ARCH-family tests.
- **H5:** The study's conclusions will have a big impact on Zimbabwe's economic policies, especially when it comes to plans for economic recovery and currency stabilization.

1.7 Significance of the Study

This study's thorough examination of exchange rate dynamics adds significantly to our understanding of Zimbabwe's gold-backed currency (ZiG). An important gap in the literature on developing market currencies, especially for commodity-pegged monetary systems, is filled by this study, which is the first empirical analysis of the ZiG/USD exchange rate since its launch in April 2024.

The results offer investors and financial markets useful standards for evaluating the ZiG's performance throughout its first phase of circulation. In Zimbabwe's changing financial landscape, identifying historical trends and structural breaks provides essential points of reference for risk assessment and trading strategies. Market players are further equipped with data-driven tools for short-term exchange rate prediction through the comparative evaluation of forecasting models.

The study's methodological comparison of change point detection strategies (Chow, CUSUM, Bai-Perron, and Bayesian) establishes best practices for spotting structural breaks in currencies with little historical data, advancing currency analysis. An empirical basis for determining whether the ZiG's gold-backed architecture has the desired stabilizing impact is provided by the volatility analysis framework, which offers a rigorous

method of evaluating exchange rate stability through ARCH-family tests. Because it makes it possible to compare the ZiG objectively to earlier monetary regimes, this component is especially useful in light of Zimbabwe's history of currency volatility.

The report offers the Reserve Bank of Zimbabwe evidence-based measures for assessing the efficacy of monetary policy at the policy level. Decisions regarding market interventions are informed by the volatility assessment methodology, while the structural break analysis provides possible signs of policy impact. These contributions, which provide empirical evidence for currency management policies, are especially pertinent as Zimbabwe adopts its new monetary system.

Finally the study lays the groundwork for future investigations into the ZiG and related financial breakthroughs. A methodological standard for examining new currency instruments in emerging nations is established by the combination of conventional econometric techniques with machine learning methodologies. This study offers the framework for monitoring the ZiG's development and assessing the long-term sustainability of gold-backed currencies in economies that are prone to inflation as additional data becomes available.

1.8 Organization of the Study

This study starts by presenting the research question concerning the Zimbabwe Gold (ZiG) currency and describing its importance in relation to short-term exchange rate dynamics, volatility modeling, and economic stability. The ZiG is placed within the larger framework of currency research in Chapter 2, which also offers a thorough literature assessment, highlighting any gaps that the study seeks to fill. Chapter 3 provides a full description of the methodology used, including sampling plans, data gathering methods, and analytic procedures, which support the methods used. In Chapter 4, the data analysis and results are presented. Time series analysis and change point identification are used to evaluate the volatility patterns of the ZiG.Chapter 5 ends with a summary of the results, their implications for investors, stakeholders, and policymakers, as well as suggestions for further study to further our understanding of how the ZiG has affected Zimbabwe's economic prospects.

1.9 Chapter Conclusion

The groundwork for the study of the Zimbabwe Gold (ZiG) currency has been established in this chapter, which also emphasizes the currency's importance in relation to Zimbabwe's economy and exchange rate forecasts. The study is to contribute to a better understanding of the behavior of the currency in a changing economic environment by analyzing the short-term dynamics of the ZiG with an emphasis on volatility patterns, time series modeling, and change point analysis. Highlighting the significance of this study, the chapter describes how the findings would benefit different stakeholders, such as investors, companies, and politicians. A thorough literature analysis will be provided in the following chapter, which will place this research within the body of existing scholarship and highlight any gaps that this study aims to fill. This will highlight the importance and urgency of investigating the dynamics of the ZiG.

Chapter 2

Literature Review

2.1 Introduction

This chapter examines the body of research to lay the groundwork for examining the short-term exchange rate dynamics of the Zimbabwe Gold (ZiG) currency, with an emphasis on volatility, change points, and pertinent statistical methods such as time series analysis and machine learning techniques. In light of the paucity of data on ZiG's performance since its launch in April 2024, this review places this study within larger exchange rate modeling discussions by examining theoretical frameworks and empirical investigations on comparable currency systems. The chapter ends with a conceptual framework that establishes the basis for an empirical investigation of the ZiG/USD exchange rate and connects the findings of the literature to the goals of the study.

2.2 Defination of Terms and Concepts

2.2.0.1 Exchange Rate

According to Hamilton (2018), an exchange rate is the price of one currency stated in terms of another currency or group of currencies. It is essential to international finance because it affects investment flows, trade balances, and economic stability (Zori, 2019). The USD-ZiG exchange rate is particularly important since the USD is widely utilized in Zimbabwean foreign exchange, affecting pricing, economic transactions, and financial stability, making it a vital tool for comprehending Zimbabwe's economic dynamics.

2.2.0.2 Change Point Analysis

It is a statistical technique for identifying notable changes in a time series, like abrupt variations in the mean, variance, or trend (Taylor, 2022). According to Manner et al. (2023), this method is useful for spotting structural changes in data that could point to significant policy or economic developments. Change point analysis, for example, might be used to assess if the September 2024 depreciation was a statistically significant shift in the ZiG/USD exchange rate, providing information about the stability of the currency and the efficacy of Zimbabwe's monetary policy.Its general framework;

1. **Define the Time Series:** Let Y_t be the time series of exchange rates at time t for $t = 1, 2, \ldots, T$.

2. **Model Parameters:** Assume there are k segments in the time series, defined by change points at $\tau_1, \tau_2, \ldots, \tau_{k-1}$, which partition the time series into segments.

3. **Likelihood Function:** The likelihood function can be defined as:

$$L(\theta) = \prod_{j=1}^{k} L_j(\theta_j) \tag{2.1}$$

   where $L_j(\theta_j)$ is the likelihood of segment j given its parameters θ_j.

4. **Change Point Detection:** To find the optimal change points, a common approach is to maximize the likelihood function or minimize the residual sum of squares (RSS) of the segments:

$$\text{RSS} = \sum_{j=1}^{k} \sum_{t \in S_j} (Y_t - \hat{Y}_{jt})^2 \tag{2.2}$$

   where S_j is the set of indices for segment j, and $\hat{Y}_{jt}$ is the predicted value for segment j.

5. **Statistical Test:** A statistical test, such as the Bayesian Change Point Detection or the CUSUM test, can be employed to evaluate the significance of detected change points.

H3: Change point analysis is a useful tool for spotting notable changes in the ZiG exchange rates behavior, which can point to unstable times.

2.2.0.3 Time Series Analysis

Time series analysis is a technique that enables analysts to see how variables change over time by examining a sequence of data points gathered at regular intervals (Salesforce, Inc., 2022). In addition to revealing relationships between data points, this method offers insights into cyclical habits, trends, and seasonal patterns. For short-term currency rate modeling and forecasting, key models include ARIMA (Autoregressive Integrated Moving Average), Seasonal ARIMA (SARIMA), and Exponential Smoothing (ETS).By examining past data, these models create forecasts and give investors and policymakers important information about expected future changes in the ZiG/USD exchange rates.

2.2.0.4 ARIMA Model

ARIMA stands for Autoregressive Integrated Moving Average. According to Noble (2024), "it's a technique for time series analysis and for forecasting possible future values of a time series." ARIMA is used when a metric is recorded at regular intervals, ranging from fractions of a second to daily, weekly, or monthly periods (MastersInDataScience.org, 2023).

ARIMA models consist of the following components:

- **Autoregressive (AR)**: Denotes that an observation plus a predetermined number of lag observations are used in the model.
- **Integrated (I)**: Indicates how raw observations are differencing to make the time series stationary.

- **Moving Average (MA)**: This method uses a moving average model applied to lagged observations to model the relationship between an observation and residual errors.

The general formula for an ARIMA model, denoted as ARIMA(p, d, q), is:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \ldots + \phi_p Y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-q} + \epsilon_t \tag{2.3}$$

where:

- Y_t: The current value of the time series at time t.
- c: Constant term (if applicable).
- p: The order of the Autoregressive (AR) part, with parameters $\phi_1, \phi_2, \ldots, \phi_p$ that relate past values $Y_{t-1}, Y_{t-2}, \ldots, Y_{t-p}$ to the current value.
- d: The Integrated (I) component, indicating the number of differences needed to make the series stationary.
- q: The order of the Moving Average (MA) part, with parameters $\theta_1, \theta_2, \ldots, \theta_q$ that relate past errors $\epsilon_{t-1}, \epsilon_{t-2}, \ldots, \epsilon_{t-q}$ to the current value.
- ϵ_t: The error term (random shock) at time t.

Explanation of the Terms:

- **AR(p)**: The model includes p lagged values of Y_t, each weighted by ϕ coefficients.
- **I(d)**: The data is differenced d times to achieve stationarity.
- **MA(q)**: The model includes q lagged error terms, each weighted by θ coefficients.

2.2.0.5 SARIMA

SARIMA (Seasonal Auto-Regressive Integrated Moving Average) is an extension of the ARIMA (Autoregressive Integrated Moving Average) model that incorporates seasonality in addition to the non-seasonal components (Santra, 2023). It is denoted as ARIMA(p, d, q)(P, D, Q)[s] and the formula is:

$$Y_t = c + \phi_1 Y_{t-1} + \ldots + \phi_p Y_{t-p} + \theta_1 \epsilon_{t-1} + \ldots + \theta_q \epsilon_{t-q} + \Phi_1 Y_{t-s} + \ldots + \Phi_P Y_{t-Ps} + \Theta_1 \epsilon_{t-s} + \ldots + \Theta_Q \epsilon_{t-Qs} + \epsilon_t \tag{2.4}$$

where:

Non-seasonal components:

- p, d, q: Orders of the Autoregressive (AR), Integrated (I) (number of differences), and Moving Average (MA) terms, respectively.
- ϕ_i: Parameters of the non-seasonal AR terms.
- θ_j: Parameters of the non-seasonal MA terms.

Seasonal components:

- P, D, Q: Orders of the seasonal AR, differencing, and MA components.
- s: The seasonal period (e.g., 12 for monthly data with yearly seasonality).
- Φ_i: Parameters of the seasonal AR terms.
- Θ_j: Parameters of the seasonal MA terms.
- Y_t: The value of the time series at time t.
- ϵ_t: Error term (white noise) at time t.
- c: Constant term (if included).

2.2.0.6 Machine Learning Time Series Model

Ruiz (2019) claims that these algorithms examine and forecast time-dependent data by utilizing machine learning approaches to identify patterns, trends, seasonality, and abnormalities. Since they can effectively manage complicated dependencies and non-linear interactions, models such as Random Forest, XGBoost, and LSTM networks are especially useful for short-term forecasting (Kurze, 2023).

H4: Using both conventional time series modeling methods (like ARIMA) and machine learning-based strategies (like LSTM), future fluctuations in the ZiG/USD exchange rates can be accurately forecasted.

2.2.0.7 Random Forest

Random Forest is a machine learning method that generates an ensemble of several decision trees in order to arrive at a single, more accurate prediction or outcome (Donges, 2024).

2.2.0.8 LSTM

LSTM (long Short Term Memory)is "an advanced form of Recurrent Neural Network, is crucial in Deep Learning for processing time series and sequential data"(Lawrence,2023). Because of this feature, LSTMs are especially useful for applications like forecasting, where it's crucial to identify patterns across time.

2.2.0.9 XGBOOST

A "scalable, distributed gradient-boosted decision tree (GBDT) machine learning library" (NVIDIA, 2023). XGBoost is well-known for its parallel tree boosting capabilities and is frequently used for forecasting jobs in addition to regression, classification, and ranking challenges. XGBoost is a useful tool for forecasting exchange rates and other financial time series variables because it efficiently models and forecasts time series data by utilizing its capacity to manage complicated dependencies and non-linear patterns(Wijaya,2023).

2.2.0.10 Volatility

Volatility is defined as the degree of fluctuation in a currency's value over time, as indicated by the standard deviation of exchange rate returns (Ilzetzki et al.,2023). The daily volatility formula is expressed as:

$$\text{Daily Volatility} = \sqrt{\text{Variance}} \tag{2.5}$$

where variance is calculated as:

$$\text{Variance} = \frac{1}{n}\sum(P_{\text{av}} - P_i)^2 \tag{2.6}$$

In this formula, P_i represents the daily exchange rate, P_{av} denotes the mean exchange rate, and n is the number of daily exchange rates.

To assess volatility on an annual basis, daily volatility is often scaled by the square root of 252, the approximate number of trading days in a year:

$$\text{Annualized Volatility} = \sqrt{252} \times \text{Daily Volatility} \tag{2.7}$$

Substituting the daily volatility formula, we have:

$$\text{Annualized Volatility} = \sqrt{252} \times \sqrt{\frac{1}{n}\sum(P_{\text{av}} - P_i)^2} \tag{2.8}$$

Exchange rate volatility is a crucial measure in evaluating financial risk, as it may indicate economic uncertainty or a currencys sensitivity to external shocks (Taba, Sarca & Yksel, 2020).

2.2.0.11 Volatility Modeling

Volatility modeling refers to the statistical techniques used to estimate and forecast fluctuations in exchange rates over time, enabling analysts to assess potential risks associated with currency movements (Epaphra, 2020). Models such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity) and EGARCH (Exponential GARCH) are particularly valuable in short term exchange rate analysis as they capture volatility clustering, providing crucial insights for investors and policymakers in assessing market risks and currency stability (Epaphra, 2020).

H2: Throughout the study period, the ZiG exchange rates show patterns of volatility that may be statistically examined and described.

2.2.0.12 GARCH Model

The GARCH model, which stands for Generalized Autoregressive Conditional Heteroskedasticity, "is a statistical model used to predict future volatility based on past data"(Ruppert, 2011). It estimates current volatility based on past periods' information, specifically focusing on the previous period's error terms and variances. The GARCH(1,1) model can be expressed with the following formulas:

Mean Equation:

$$r_t = \mu + \epsilon_t \tag{2.9}$$

Variance Equation:

$$\sigma_t^2 = \alpha_0 + \alpha_1\epsilon_{t-1}^2 + \beta_1\sigma_{t-1}^2 \tag{2.10}$$

In these equations: r_t represents the return at time t. μ is the mean return. ϵ_t is the error term (i.e., the shock). σ_t^2 is the conditional variance at time t. α_0, α_1, and β_1 are coefficients that need to be estimated.

2.2.0.13 EGARCH Model

The EGARCH model extends the GARCH framework to account for asymmetries in volatility, meaning it can capture situations where positive and negative shocks have different impacts on volatility. The EGARCH(1,1) model can be expressed as follows:

Mean Equation:

$$r_t = \mu + \epsilon_t \tag{2.11}$$

Logarithmic Variance Equation:

$$\log(\sigma_t^2) = \alpha_0 + \alpha_1 \left(\frac{\epsilon_{t-1}}{\sigma_{t-1}} \right) + \beta_1 \log(\sigma_{t-1}^2) \tag{2.12}$$

In these equations: - $\log(\sigma_t^2)$ is the logarithm of the conditional variance. - α_0, α_1, and β_1 are estimated parameters. - The term $\frac{\epsilon_{t-1}}{\sigma_{t-1}}$ allows for the modeling of leverage effects, where negative shocks can lead to higher future volatility compared to positive shocks of the same magnitude.

2.3 Theoretical Review

This section offers the theoretical foundations for examining the short-term dynamics of the Zimbabwe Gold (ZiG) currency's exchange rate, with an emphasis on complicated pattern detection, volatility modeling, and structural change analysis. In order to effectively forecast the ZiG/USD exchange rate, these theories advocate the employment of GARCH and EGARCH models for volatility modelling, change point analysis, and machine learning techniques.

2.3.1 Structural Break Theory

Andrews (1993) was the first to officially define the notion of structural breaks, offering statistical criteria for identifying sudden and significant changes in time series that alter their basic properties. These structural fractures, which are frequently brought on by outside shocks, changes in policy, or noteworthy economic events, have an effect on the forecasting models' accuracy, according to Hansen (2001). These changes bring about times when the historical behavior of the time series is no longer relevant. Because they signal regime changes that could impact asset pricing, volatility, and market risk, structural breaks are particularly important in financial markets (Perron, 2006).

In exchange rate modeling, it is crucial to recognize and take into consideration structural breaks since they have the potential to unexpectedly modify the dynamics of currency values. Currency values may undergo structural changes as a result of changes in government policy, such as monetary or exchange rate policy (Bai & Perron, 2003). Periods of increased volatility are frequently the consequence of such shifts, and conventional models that do not account for structural discontinuities may produce inaccurate projections (Chen & Patel, 1998). By quantitatively identifying breakpoints instead of depending on conjecture, techniques such as the Bai-Perron test offer a statistically rigorous approach to breakpoint detection, increasing model robustness (Zarei et al., 2015).

2.3.1.1 Change Point Detection (CPD) in Structural Break Analysis

One important technique for locating structural fractures is change point detection (CPD). It entails testing two hypotheses: the alternative hypothesis (H_a), which contends that a change has taken place at a particular moment, and the null hypothesis (H_0), which holds that no change happens(Preis et al.,2023). The CPD procedure provides a statistical foundation for model adjustment by assisting in determining the precise point at which a time series' characteristics change (Chen & Zhang, 2015).

Change Point Detection Hypotheses:

- H_0: Throughout the interval (m, n), the time series does not change.
- H_a: The characteristics of the time series change at a change point (k^*).

Where P_{X_i} denotes the probability density function at point x_i, and k^* represents the change point. This paradigm helps detect abrupt regime shifts by serving as the foundation for statistical studies that find structural discontinuities in exchange rate data.A simple illustration of a change-point detection task can help visualize this process, as shown below:

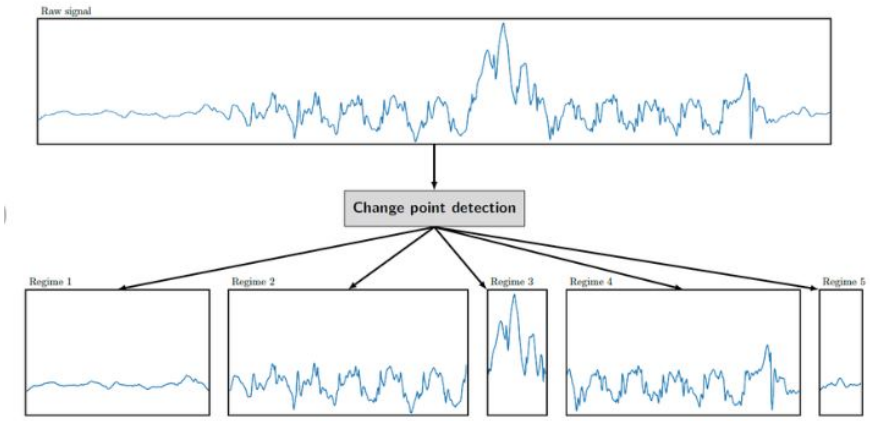


Figure 2.1: Simple illustration of a change-point detection task

2.3.1.2 Types of Change Points in Time Series and Their Relevance to Exchange Rate Forecasting

Identifying and comprehending the various kinds of change points in time series data is essential for exchange rate forecasting. Change points are times when a time series' statistical characteristics change significantly; by taking into account the changing nature of exchange rates, these shifts enable more accurate modeling and forecasting. Christof Schroth, Dr. Julien Siebert, and Janek Grob (2021) distinguish a number of different types of change points that can appear in a time series:

1. **Change Point in Mean:** A variation in the series' average value. This kind of change point could signal a significant economic event that affects the overall level of exchange rates, such as the launch of a new currency or a major policy change.

2. **Change Point in Variance:** A modification to the series' variability or dispersion. This change point could indicate rising market volatility, which may be triggered by abrupt geopolitical tensions, shifts in international trade regulations, or global financial crises.

3. **Change Point in Frequency:** A change in the data's periodicity or frequency. This may occur when the dynamics of the currency rate are influenced by external factors, such as shifts in trading hours, foreign market schedules, or seasonal influences.

4. **Change Point in Pattern:** A modification to the series' underlying pattern or trend. This often occurs when there is a long-term shift in the behavior of exchange rates due to changes in the economy, such as alterations in interest rates, inflation rates, or structural adjustments.

5. **Change Point in Multidimensional Time Series:** A collection of time series that interact in a multivariate context. This includes interactions between exchange rates and other economic factors, such as interest rates or commodity prices, or shifts that affect multiple currency pairs simultaneously.

A more sophisticated view of structural changes in exchange rate data is offered by these kinds of change points. Determining the precise kind of change that has taken place can help tailor the study to more accurately reflect its many facets, whether they are connected to market dynamics, economic events, or policy initiatives(Booras,2021).Since it enables the model to adjust to various time series regimes, identifying and simulating these shift points in exchange rate analysis can result in more accurate projections. For example, geopolitical events or economic shocks, including the implementation of new governmental policies or shifts in the prices of important commodities, may be accompanied by notable changes in exchange rates. Eklou (2023) asserts that government actions or reactions to modifications in the dynamics of imports and exports, as well as changes in the price of commodities globally, may cause structural breaks in exchange rates.Analysts can create more reliable models that faithfully capture the dynamic and frequently erratic nature of exchange rate fluctuations by comprehending and recognizing the various kinds of change points.

2.3.1.3 Practical Applications of Change Point Analysis in Exchange Rate Forecasting

For currencies like the Zimbabwean Gold (ZiG), change point analysis is a useful tool for improving exchange rate forecasts. Through the discovery of data shifts or breakpoints, this technique makes it possible to identify fundamental changes in the exchange rate, which forecasting models can then modify appropriately. Such fluctuations in the ZiG/USD exchange rate can be brought about by external forces or adjustments in economic policy, and identifying these breakpoints increases the accuracy of forecasts (Yussif et al., 2021).

Applying change point analysis to exchange rate forecasting enables the data to be divided into discrete regimes, each with unique properties. By ensuring that the forecast takes into account the currency rate's shifting dynamics over time, this segmentation increases model accuracy. Finding these structural changes improves the model's ability to estimate how the ZiG will react to macroeconomic events and offers more reliable predictions when the economy is changing.

2.3.2 Theoretical Foundations of Time Series Analysis for Exchange Rate Forecasting

2.3.2.1 The Role of Historical Patterns in Forecasting

According to Hyndman and Athanasopoulos (2018), time series analysis is predicated on the idea that patterns found in historical data, such as trends, seasonal fluctuations, and cyclical behaviors, might provide important information about values in the future. Because historical patterns are frequently interpreted as representations of persistent market, political, and economic dynamics, this becomes particularly important when it comes to exchange rate forecasting (Tlegenova, 2015). Scholars and researchers can predict possible changes in future exchange rates by recognizing these recurrent patterns (Wei, 2006). The fundamental premise is that historical patterns would probably recur under generally stable circumstances, allowing analysts to use historical data as a solid foundation for forecasting both short-term and long-term changes in the future.

However, the fact that exchange rates are subject to unanticipated events, including policy changes or economic shocks, which might upset the established patterns, complicates this premise (Hamilton, 1994). Historical patterns can be useful for forecasting, but they should be used with caution, especially for volatile currencies where they might not be able to reliably anticipate future movements(Steinwart et al., 2012). In times of stability, time series models like ARIMA, SARIMA, and decomposition techniques have been crucial in spotting trends and cyclical tendencies. According to Box, Jenkins, and Reinsel (2008), these models aid researchers in determining whether transient variations reflect novel dynamics or are consistent with long-term patterns. In summary, historical trends can serve as a basis for forecasting, but their usefulness is reduced in settings characterized by instability or fast change.

2.3.2.2 Autoregressive Integrated Moving Average (ARIMA) Models

One of the most popular methods for univariate time series forecasting is the ARIMA model, which was first presented by Box and Jenkins in 1970. In stationary time series data, this approach works particularly well for spotting and forecasting linear trends and seasonal patterns. In order to convert the data into a stationary series, ARIMA combines three essential elements: differencing (I), moving average (MA) terms, and autoregressive (AR) terms (Reinsel et al., 2008). When dealing with non-stationary data, which is typical in exchange rate series, the differencing step is especially helpful. According to Ngan (2016), ARIMA is a useful technique for exchange rate prediction since it stabilizes volatility by lessening the influence of trends and seasonal changes, making it ideal for short-term forecasting.

2.3.2.3 Seasonal ARIMA (SARIMA) for Cyclical Patterns

Box and Jenkins (1976) introduced the Seasonal ARIMA (SARIMA) model, an extension of ARIMA that uses seasonal differencing to capture periodic changes in a time series, such those frequently seen in exchange rates. SARIMA is made to deal with recurrent patterns that appear on a yearly, quarterly, or monthly basis and can be influenced by seasonal elements such policy changes, market activity, or economic cycles (Resende,2023). According to Etuk (2014), this approach is especially helpful in currency markets because exchange values are frequently influenced by seasonal cycles or other periodic changes. SARIMA is a useful method for assessing exchange rates that show seasonal variations since it can take these cyclical patterns

into consideration.

2.3.2.4 State-Space Models and Kalman Filter for Adaptive Forecasting

Initiated as a theoretical formulation by Egon Beth (1961), state-space models were further developed by Bas van Fraassen (1970) and William Suppe (1988). These models offer a versatile and potent framework for forecasting, especially when data shows time-varying dynamics (WallStreetMojo, 2024). They are particularly helpful when market or regulatory changes result in sudden changes in exchange rate behaviors.State-space models depict the observable data as a function of unseen states that change over time, like underlying trends or abnormalities.

According to Durbin and Koopman (2012), the Kalman filter is a crucial part of these models. It is a recursive algorithm that estimates these unobserved states and updates them when fresh data becomes available. Since sudden fluctuations are common in short-term exchange rate forecasting, state-space models are quite useful due to their flexibility and adaptability, especially when combined with the Kalman filter.This paradigm is particularly useful in dynamic financial contexts because it can adapt to outside pressures that disrupt consistent patterns. According to Chukhrova (2021), the adaptive nature of state-space models enables more accurate forecasts even in the face of shifting circumstances in currency markets, where exchange values are susceptible to abrupt adjustments brought on by changes in the economy or in policy.

2.3.2.5 Decomposition Techniques for Trend and Seasonal Analysis

To break down a time series into its fundamental elementstrend, seasonality, and residualsdecomposition techniques are employed. According to Mahler (2023), by separating these components, decomposition techniques aid in comprehending the structure of exchange rate fluctuations and increase forecasting precision. Using locally weighted regression, STL (Seasonal-Trend decomposition using Loess) is one such technique that breaks down the series into trend, seasonal, and residual components (Cleveland et al., 1990). This method makes it easier to predict future movements by taking into account both short-term variations and long-term trends by assisting in the identification of seasonal and trend patterns within exchange rate data (Brownlee, 2020).

2.3.2.6 Forecasting Techniques for Single-Variable Time Series

Single-variable time series, like the ZiG/USD exchange rate, are especially well-suited for short-term forecasting. Methods like ARIMA, SARIMA, and decomposition techniques can be used to find past patterns and forecast future exchange rate changes. The flexibility of state-space models with Kalman filtering further improves forecasting accuracy by taking into account abrupt market movements and regulatory changes that can upset established patterns. These models are in line with the unpredictability of currency markets, which enables them to produce more accurate forecasts even when there is volatility.

2.3.3 Nonlinear Dynamics and Machine Learning Theory

Machine learning theory is based on the idea that systems may discover complex and frequently nonlinear patterns by learning from data to perform better on particular tasks (Roy & Rana, 2021). The concept of building artificial systems that replicate brain processing was sparked by McCulloch and Pitts' (1943)

creation of the first mathematical model of neural networks. The Turing Test, which Alan Turing introduced as a standard in his groundbreaking book Computing Machinery and Intelligence, helped to further develop the idea of computer intelligence (Turing, 1950). The creation of the first self-learning program by Arthur Samuel in 1959 and the perceptron by Frank Rosenblatt, which represented a major advancement in the field of pattern recognition, came after these turning points (Rosenblatt, 1958).

Machine learning provides a strong foundation for capturing non-linear phenomena in financial time series analysis, especially for exchange rates. Conventional models like SARIMA and ARIMA work well for linear trends but have trouble with the intricacy of financial data, where economic shocks can cause abrupt changes and volatility clustering (Martin, 2019). Because they can represent long-term dependencies and non-linear interactions, machine learning modelsespecially those built on neural networksperform exceptionally well in this field(Bonilla, 2023). Recurrent neural networks of the Long Short-Term Memory (LSTM) type are especially useful for time series forecasting because they can retain information over long sequences and are therefore skilled at identifying the complex patterns and temporal dependencies found in exchange rate data (Hochreiter & Schmidhuber, 1997).

Additionally, machine learning methods offer a foundation for adaptive modeling, which allows models to generalize from past data while taking into account both long-term trends and short-term volatility (Mohammadi, 2021). Given that fluctuations in the ZiG/USD exchange rate are characterized by volatility and possible non-linearity, this flexibility is particularly important for forecasting. In order to improve prediction accuracy and better handle the intricacies of financial markets, methods such as LSTM, Random Forest, and hybrid models may adapt dynamically to new data. Scholars can enhance the accuracy of exchange rate predictions by combining these approaches with traditional time series techniques, which capture the complex relationships that influence currency markets.

2.3.4 Volatility Clustering Theory

Mandelbrot (1963) first put forth the idea that "small changes tend to be followed by small changes, and large changes tend to be followed by large changes, of either sign." This concept is known as volatility clustering, an observed phenomenon in financial time series where periods of high volatility tend to be followed by high volatility, and low volatility by low volatility (Cont, 2005).The diagram below illustrates volatility clustering in time series data:

This persistent pattern reflects that volatility is serially correlated over time, influenced by external shocks, economic policies, and shifts in market sentiment (Cont, 2005). Behavioral factors, such as changes in investor analysis strategies, also contribute to this clustering, as investors respond to evolving market conditions (Suthar, 2023).

In exchange rate markets, volatility takes on various forms, influencing risk management and trading strategies. Key types include:

- **Historical Volatility:** Measures past fluctuations in exchange rates, providing a basis for predicting future movements.
- **Event-Driven Volatility:** Arises from unexpected events like economic reports or political announcements, often leading to sudden rate changes.

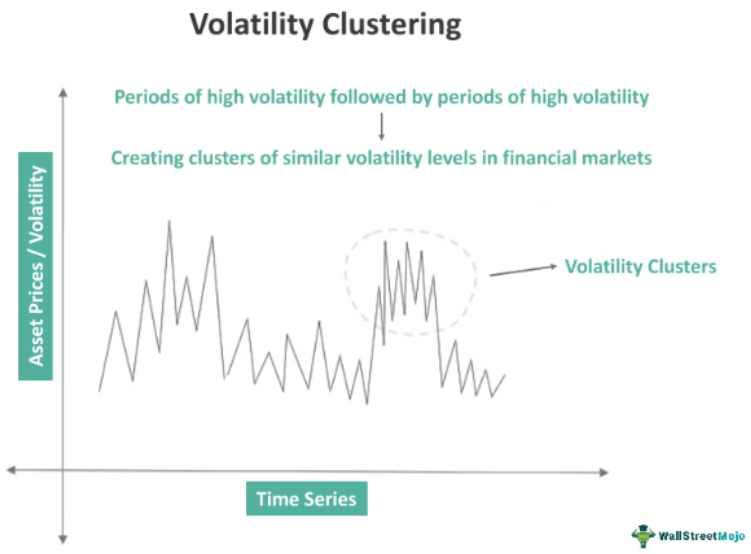


Figure 2.2: Illustration of Volatility Clustering in Time Series Data (Source: WallStreetMojo)

- **Currency Correlation Volatility:** Captures how exchange rate changes in one currency pair can impact others, reflecting the interconnected nature of global markets (FasterCapital, 2023).

These forms of volatility contribute to clustering as they create conditions where periods of high or low volatility persist, making it critical to use models that accurately capture this pattern. Models such as GARCH (Generalized Autoregressive Conditional Heteroskedasticity), developed by Engle (1982), allow conditional variances to evolve based on previous periods' volatility, adapting dynamically to market conditions. Bollerslevs (1986) EGARCH model extends this approach by modeling asymmetric responses to shocks, capturing situations where negative and positive events affect volatility differently. Such modeling flexibility is essential when analyzing exchange rate data like the ZiG/USD, where responses to economic events may be asymmetrical.

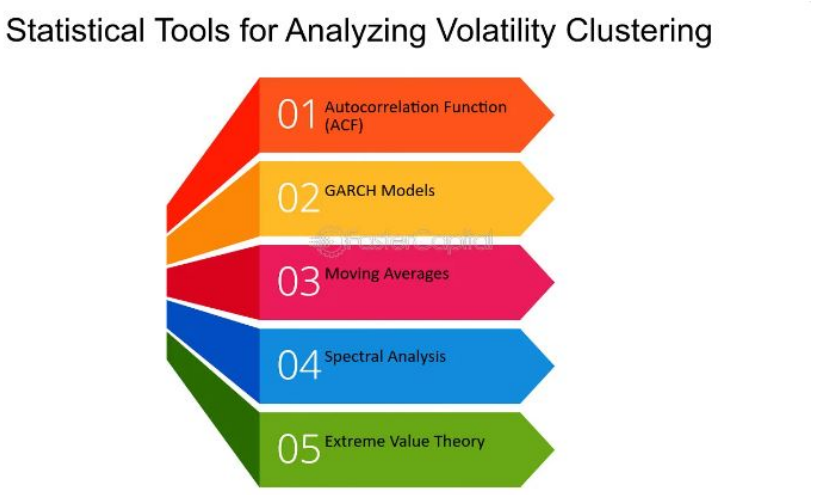


Figure 2.3: Statistical Tools for Analyzing Volatility Clustering (Source: FasterCapital)

Nonlinear dynamics theories also acknowledge volatility clustering as a fundamental feature of financial markets that results from intricate relationships between market players and economic variables. This is explained by nonlinear dynamics in financial markets, where feedback mechanisms and information flow magnify market reactions (Xue-Zhong He, Kai Li, & Chuncheng Wang, 2016). This viewpoint encourages the use of possibly nonlinear models like LSTM for improved forecasting and sophisticated models like GARCH and EGARCH to capture volatility clustering.

Assessing currency stability in the context of the ZiG/USD exchange rate requires an understanding of volatility clustering. GARCH and EGARCH models are useful for determining times of high or low stability since economic uncertainties and external shocks can cause extremely volatile movements (Engle, 1982; Bollerslev, 1986). By quantifying stability patterns, these models shed light on how resilient the ZiG is to changes in the economy.

For investors, policymakers, and analysts, understanding volatility clustering in exchange rates is crucial because it helps them spot possible hazards, optimize asset allocation, and direct their investment and risk management plans. Because of their vulnerability to global market dynamics, policy changes, and economic shocks, exchange rates are especially vulnerable to volatility clustering. Financial professionals can effectively manage currency risk, make well-informed decisions about asset allocation, and modify investment strategies in response to shifting market conditions by having a thorough grasp of the clustering of exchange rate volatility (Sahu, 2023).

2.4 Statistical Tests in Time Series Analysis and Forecasting

According to Brookwell (2023), statistical tests are essential to time series analysis since they verify important hypotheses, identify possible problems, and guarantee the accuracy of forecasting models. When using models like ARIMA or GARCH, these tests assist in determining whether the data satisfies basic characteristics such stationarity and the lack of autocorrelation (Topas, 2024). In order to help researchers choose the best modeling approaches, they also detect structural fractures, cointegration linkages, and volatility clustering (Yen, 2023). These tests are especially important for recently issued currencies, such as the Zimbabwe Gold (ZiG), when data is scarce and exchange rates are subject to instability.They improve the accuracy of exchange rate projections by offering a framework to deal with issues including data sparsity, non-stationarity, and abrupt regime adjustments.

2.4.1 Stationarity Tests

A time series is said to be stationary if its observations are independent of the time at which they are recorded (Krishnan, 2021). A stationary time series is stable and predictable for analysis since its statistical characteristics, such as mean, variance, and covariance, don't change over time.

2.4.1.1 Augmented Dickey-Fuller (ADF) Test

According to machinelearningplus (2019), the Augmented Dickey-Fuller (ADF) test is a popular statistical method used to assess whether a time series is stationary or has a unit root, indicating non-stationarity. In time series analysis, this test is essential since stationarity is a fundamental premise of many forecasting

models. An effective paradigm for assessing stationarity is provided by the ADF test, which uses lagged differences to account for higher-order autocorrelations.Formula;

$$\text{ADF Test} = \Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta Y_{t-i} + \epsilon_t \tag{2.13}$$

Where:

- Y_t: The time series being tested.
- ΔY_t: The first difference of Y_t $(Y_t - Y_{t-1})$.
- α: Intercept term.
- βt: Trend term (optional, included if testing for trend stationarity).
- γY_{t-1}: Lagged value of the series.
- $\sum_{i=1}^{p} \delta_i \Delta Y_{t-i}$: Lagged differences to account for higher-order autocorrelation.
- ϵ_t: Error term.

The null hypothesis (H_0) of the ADF test states that the time series has a unit root (i.e., it is non-stationary). If the test statistic is less than the critical value, the null hypothesis is rejected, indicating stationarity.

2.4.1.2 KPSS Test

Kwiatkowski et al. (1992) developed the KPSS (Kwiatkowski-Phillips-Schmidt-Shin) test, a statistical method for determining if a given time series is stationary around a deterministic trend. The KPSS test accepts stationarity as the null hypothesis, in contrast to other unit root tests, such as the ADF, which test for non-stationarity as the null hypothesis. Because of this, it provides an additional tool for confirming the stationarity of a time series and is especially helpful in figuring out whether de-trending or differencing is required.The following breakdown of a time series serves as the foundation for the KPSS test:

$$Y_t = r_t + \beta t + \epsilon_t \tag{2.14}$$

Where:

- Y_t: Observed time series.
- r_t: Stationary process (the test evaluates this component).
- βt: Deterministic trend.
- ϵ_t: Random error term with a mean of zero and constant variance.

The test statistic is calculated as:

$$\text{KPSS} = \frac{1}{T^2 \hat{\sigma}^2} \sum_{t=1}^{T} S_t^2 \tag{2.15}$$

Where:

- S_t: Cumulative sum of residuals, defined as $S_t = \sum_{i=1}^{t} \hat{\epsilon}_i$.
- $\hat{\sigma}^2$: Long-run variance of the residuals.
- T: Number of observations in the series.

Interpretation:

- Null Hypothesis (H_0): The series is stationary.
- Alternative Hypothesis (H_1): The series is non-stationary.

2.4.1.3 Phillips-Perron (PP) Test

Developed by Phillips and Perron (1988), the Phillips-Perron (PP) test is a unit root test used to determine if a time series is non-stationary. It can be used as a substitute for the Augmented Dickey-Fuller (ADF) test, but it takes a different approach to accounting for heteroskedasticity and serial correlation. As opposed to the ADF test, the PP test employs a nonparametric adjustment to the test statistic and does not call for the addition of lagged difference terms.The PP test is especially helpful when dealing with time series data that can contain time-varying variances or autocorrelated errors.The PP test is based on the following autoregressive model:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \epsilon_t \tag{2.16}$$

Where:

- Y_t: Time series being tested.
- ΔY_t: First difference of the time series ($Y_t - Y_{t-1}$).
- α: Constant (drift term).
- β: Coefficient on time t (trend component).
- γ: Coefficient tested to check for a unit root ($H_0 : \gamma = 0$).
- ϵ_t: Error term.

The test statistic is calculated as:

$$t_\gamma = \frac{\hat{\gamma}}{SE(\hat{\gamma})} \tag{2.17}$$

Where $SE(\hat{\gamma})$ is the standard error of the estimated coefficient γ.
Interpretation:

- Null Hypothesis (H_0): The series has a unit root (non-stationary).
- Alternative Hypothesis (H_1): The series does not have a unit root (stationary).

If the test statistic is more negative than the critical value, the null hypothesis is rejected, suggesting stationarity.

2.4.2 Cointegration Tests

According to Engle and Granger (1987), cointegration is the process by which non-stationary variables are linearly joined to create a stationary connection that implies a long-term equilibrium between them. In time series analysis, this is essential because it enables researchers to simulate the dynamics of variables that, in spite of individual trends, have a constant relationship over time. Understanding equilibrium adjustments is crucial for precise forecasting and policy formation, and identifying cointegration aids in this process.

2.4.2.1 Engle-Granger Test

Engle and Granger (1987) created the two-step Engle-Granger test, which is used to determine if two or more non-stationary time series are cointegrated. In order to test the residuals for stationarity, a unit root test, like the Augmented Dickey-Fuller (ADF) test, is used after estimating a linear relationship between the variables using ordinary least squares (OLS). There may be a cointegrating relationship between the variables, indicating that they share a long-term equilibrium, if the residuals are stable.Particularly when there are just two variables, this is the most straightforward and popular cointegration test. There are two steps involved:

1. **Regression:** The two variables are first regressed against one another.
2. **Residual Testing:** A unit root test, such as the Augmented Dickey-Fuller (ADF) test, is then used to check for stationarity in the regression's residuals. The two variables are assumed to be cointegrated if the residuals are stationary.

2.4.3 Model Diagnostics

Model diagnostics are vital for making sure the forecasting model selected fits the data correctly and satisfies fundamental presumptions (Rudolph,2023). For confirming key presumptions, including independence, normalcy, and homoscedasticity, residual analysis is very crucial. It is preferable for residualswhich show the variations between the observed and expected valuesto be dispersed randomly and devoid of any discernible trends. It may be necessary to improve the model further if the residuals show patterns, which indicate that the model is not fully capturing the data. Assessing the model's suitability, spotting possible problems like misspecification, and enhancing its forecasting capabilities are all made possible by appropriate diagnostic checks (Koehler & Murphree, 2018).

2.4.3.1 Ljung-Box Q-Test

The autocorrelation of residuals over various time lags is evaluated using the Ljung-Box Q-Test (Bobbitt, 2020). It examines the independent distribution of residuals, which is the null hypothesis. Significant autocorrelation suggests that the model has not been able to identify a pattern in the data. In time series analysis, where autocorrelation is prevalent and can affect the model's predicting accuracy, this test is very helpful (Ljung & Box, 1978).It is based on the following formula:

$$Q = n(n+2)\sum_{k=1}^{m}\hat{\rho}_k^2 \tag{2.18}$$

Where:

- n: Sample size (number of observations).
- m: Number of lags being tested.
- $\hat{\rho}_k$: Sample autocorrelation at lag k.

The sum runs over the number of lags m.

The test statistic Q follows a chi-squared distribution with m degrees of freedom under the null hypothesis of no autocorrelation.

2.4.3.2 Durbin-Watson Statistic

Regression model residuals can be examined for serial correlation using the Durbin-Watson statistic (Prep-Nuggets, 2023). Significant values below or above 2 indicate positive or negative serial correlation, respectively, whereas a value close to 2 indicates no autocorrelation. Unreliable forecast intervals and skewed parameter estimates can result from serial correlation in residuals (Durbin & Watson, 1951). When determining if model mistakes are randomly distributed, this test is crucial in time series forecasting.The formula for the Durbin-Watson statistic is;

$$DW = \frac{\sum_{t=1}^{n} e_t^2}{\sum_{t=2}^{n} (e_t - e_{t-1})^2} \tag{2.19}$$

Where:

- e_t: Residual at time t.
- e_{t-1}: Residual at time $t-1$.
- n: Number of observations.

Interpretation:

- Null Hypothesis: There is no first-order autocorrelation in the residuals (i.e., the residuals are independent).
- Alternative Hypothesis: There is significant first-order autocorrelation in the residuals.

2.4.3.3 ARCH-LM Test

In order to determine whether the variance of residuals depends on past error terms, the Autoregressive Conditional Heteroscedasticity (ARCH) LM test looks for conditional heteroscedasticity, a feature frequently observed in financial time series data (Yasar, 2021). If significant heteroscedasticity is present, it suggests that the model does not adequately account for varying volatility over time, which is particularly important in exchange rate forecasting, where volatility clustering (high volatility followed by low volatility) is a common phenomenon (Engle, 1982).The general form of the test involves the following steps:

1. Estimate the regression model (e.g., AR model) and compute the residuals e_t.
2. Square the residuals e_t^2, and regress these squared residuals on a set of lagged squared residuals:

$$e_t^2 = \alpha_0 + \alpha_1 e_{t-1}^2 + \alpha_2 e_{t-2}^2 + \cdots + \alpha_q e_{t-q}^2 + \epsilon_t \tag{2.20}$$

where q is the number of lags in the ARCH model (usually denoted as the order of the ARCH effect).

3. Obtain the Lagrange Multiplier (LM) statistic, which is based on the residual sum of squares from the regression:

$$LM = T \cdot R^2 \tag{2.21}$$

where:

- T: Number of observations (sample size).
- R^2: Coefficient of determination from the regression of squared residuals on the lagged squared residuals.

Hypothesis Testing:

- Null Hypothesis (H_0): No ARCH effect (no conditional heteroscedasticity).
- Alternative Hypothesis (H_1): The presence of ARCH effect (conditional heteroscedasticity exists).

2.5 Statistical Tests in Volatility Modeling

Key traits that are essential to models like GARCH, such as heteroscedasticity or volatility clustering, are verified by statistical tests in volatility modeling. These tests guarantee that the model of choice incorporates the natural grouping and variability found in financial data. Volatility modeling statistical tests are described in this section.

2.5.0.1 Engles ARCH Test

The ARCH test, developed by Engle in 1982, is essential for determining whether autoregressive conditional heteroscedasticity (ARCH) effects are present in time series data. The variance of the errors is assumed to be dependent on previous error terms rather than being constant in ARCH models.The formula is;

$$\epsilon_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \epsilon_{t-i}^2 + u_t \tag{2.22}$$

Where:

- ϵ_t^2: Squared residuals from the model.
- α_0: Intercept.
- α_i: Coefficients of lagged squared residuals.
- u_t: Error term.

The null hypothesis (H_0) is that $\alpha_i = 0$, indicating no ARCH effect.

When huge or little price swings are followed by similarly large or small movements, this test is essential for determining whether volatility clustering is present. When it comes to exchange rates such as

the ZiG/USD, volatility clustering is frequently seen during times of market stress or noteworthy economic developments.Researchers can verify whether exchange rate data volatility is time-varying by using Engle's ARCH test. They can also determine whether more complex models of volatility, such as Generalized Autoregressive Conditional Heteroscedasticity (GARCH), should be employed for precise volatility forecasting and risk management (Engle, 1982).

2.5.0.2 Box-Ljung Test for Residuals

The Ljung-Box test for residuals is a statistical test developed by Greta M. Ljung and George E. P. Box in 1978 to ascertain whether a time series' autocorrelations deviate considerably from zero. This test is frequently used to determine whether autocorrelation exists in residuals following the fitting of a time series model, as Hyndman (2023) points out. Here is the test statistic:

$$Q^* = T(T+2)\sum_{k=h}^{h}(T-k)^{-1}r_k^2 \tag{2.23}$$

Where:

- T: Length of the time series.
- r_k: The kth autocorrelation coefficient of the residuals.
- h: Number of lags to test.

A large value of Q^* indicates that there are significant autocorrelations in the residual series.

2.5.0.3 Likelihood Ratio Test for GARCH Models

The Likelihood Ratio (LR) test determines if a more intricate GARCH model fits data better than a more straightforward ARCH model (Burch, 2005). It tests the null hypothesis that the simpler model is adequate by comparing the probabilities of the two models. This is the test statistic:

$$LR = 2\left[\ln(L_{\text{GARCH}}) - \ln(L_{\text{ARCH}})\right] \tag{2.24}$$

Where:

- $\ln(L_{\text{GARCH}})$: Log-likelihood of the GARCH model.
- $\ln(L_{\text{ARCH}})$: Log-likelihood of the ARCH model.

This test compares the goodness of fit between the GARCH and ARCH models, with a higher likelihood indicating a better fit.With degrees of freedom equal to the variation in the number of parameters, the statistic is distributed according to a chi-square model. This test is essential for forecasting exchange rates since it makes sure the chosen model accurately depicts volatility dynamics, increasing prediction accuracy.

2.5.0.4 Jarque-Bera Test for Normality

A statistical test called the Jarque-Bera (JB) test is used to determine if the regression model's residuals are normally distributed (Thadewald, 2004). Many econometric models make the assumption that residuals are normal. This test determines whether the data's skewness (asymmetry) and kurtosis (tailedness) deviate from a normal distribution (Aslam, 2021). In time series analysis, particularly exchange rate modeling, the JB test is frequently used because non-normal residuals may be a sign of model misspecification. It indicates that model selection has to be changed or that more robust models that do not rely on normalcy assumptions should be used if residuals in exchange rate forecasting models are not normally distributed. Here's how to calculate the JB test statistic:

$$JB = \frac{6}{n}\left(S^2 + 4(K-3)^2\right) \tag{2.25}$$

Where:

- n: Number of observations.
- S: Skewness of the residuals.
- K: Kurtosis of the residuals.

The null hypothesis for the JB test is that the residuals are normally distributed. If the test statistic exceeds the critical value from the chi-square distribution, the null hypothesis is rejected, indicating that the residuals are not normally distributed.

2.5.0.5 Relevance to Exchange Rates

To find out if the data satisfies the assumptions needed to use ARIMA models, stationarity tests like the Augmented Dickey-Fuller (ADF) and KPSS are crucial. These tests determine if the series' mean and variance remain constant over time; the KPSS test confirms stationarity, while the ADF test looks for the presence of a unit root. Before moving on to ARIMA modeling, differencing will be used to bring non-stationary data into stationarity. These tests provide a strong basis for precise short-term exchange rate forecasts by meeting the stationarity criterion.

Furthermore, tests such as the ARCH-LM and Bai-Perron offer more information about the particular dynamics of the ZiG/USD exchange rate. The use of GARCH models to capture time-varying volatility patterns is justified by the ARCH-LM test, which detects volatility clustering. In the meanwhile, major regime shifts brought on by modifications to policies or outside shocks are identified using structural break tests like the Chow and Bai-Perron tests. By dividing the data into discrete stable and unstable periods, these tools make sure that the models chosen accurately represent the economic conditions of each regime. When combined, these statistical tests guarantee that the methodology takes into consideration the ZiG's changing market conditions, resulting in models that are accurate and flexible.

2.6 Statistical Tests in Change Point Analysis

Change point analysis, as explained by Li (2021), is essential for identifying times of instability that may affect forecasting accuracy and for recognizing structural discontinuities in exchange rate behavior. These statistical analyses assess if there are notable changes in the time series' mean, variance, or other characteristics at

particular intervals. For the development of models that adjust to changes in market dynamics, such insights are essential. In order to forecast exchange rates, change point analysis uses statistical tests, which are explained in this section.

2.6.0.1 Chow Test

Gregory Chow (1960) developed the Chow Test, a statistical method for identifying if a structural break exists at a specific moment in time. This test contrasts two distinct regression models: one that uses all of the data and another that divides the data into two subsets: the presumptive break and the post-break data.Its equation:

$$F = \frac{(RSS_{\text{pooled}} - (RSS_1 + RSS_2))/k}{(RSS_1 + RSS_2)/(n_1 + n_2 - 2k)} \tag{2.26}$$

Where:

- RSS_{pooled}: Residual sum of squares from the pooled model (using the entire dataset).
- RSS_1: Residual sum of squares from the regression model for the first subset of the data.
- RSS_2: Residual sum of squares from the regression model for the second subset of the data.
- k: Number of parameters in the model.
- n_1: Number of observations in the first subset.
- n_2: Number of observations in the second subset.

The Chow test essentially checks if the coefficients of the two models are significantly different from each other. A significant F-statistic indicates the presence of a structural break.

Relevance for Exchange Rates: In exchange rate analysis, the Chow Test can help identify when shifts in economic policies or external shocks lead to significant breaks in the behavior of exchange rates.

2.6.0.2 Bai-Perron Test

Bai and Perron (1998) introduced the Bai-Perron Test, which is used to identify multiple structural breakdowns at unknown times. The Bai-Perron test enables the identification of many change points in the data, in contrast to the Chow Test, which presumes a single break at a specified point.By minimizing the sum of squared residuals across various break points, the Bai-Perron method finds the break points that minimize this value. In terms of mathematics, it can be stated as:

$$\hat{\theta} = \arg\min \left(\sum_{i=1}^{m+1} \sum_{t=\theta_i}^{\theta_{i+1}} \epsilon_t^2 \right) \tag{2.27}$$

Where:

- $\hat{\theta}$: Estimated break points.
- θ_i: Time points where breaks occur.

- ϵ_t: Residual for time point t.

By segmenting the time series according to the identified break points, this technique models each fragment independently.

Relevance to Exchange Rates: When there are several changes in the political or economic landscape over time, the Bai-Perron Test can be especially helpful in predicting exchange rates. This test offers a more thorough comprehension of the potential behavior of exchange rates over time.

2.6.0.3 Cumulative Sum Control Chart (CUSUM) Test

Page (1954) developed the CUSUM test, which tracks cumulative sums of deviations from the mean to identify changes in a time series' mean level. When the cumulative total of a regression model's residuals beyond specific critical thresholds, the test detects a possible structural break.Its equation is:

$$S_t = \sum_{i=1}^{t} (\epsilon_i - \bar{\epsilon}) \tag{2.28}$$

Where:

- S_t: Cumulative sum at time t.
- ϵ_i: Residual at time i.
- $\bar{\epsilon}$: Mean residual.

If S_t exceeds a predefined threshold (either positive or negative), a change point is detected. **Relevance to Exchange Rates:**The CUSUM test is particularly helpful for tracking changes in exchange rates that could be brought on by modifications to policy, outside shocks, or other systemic issues that impact the behavior of the currency.

2.6.0.4 PELT (Pruned Exact Linear Time) Technique

The PELT algorithm (Killick et al., 2012) is an efficient method for detecting multiple change points in time series data by minimizing a penalized cost function. Unlike piecewise regression, PELT uses dynamic programming with pruning to reduce computational complexity from $O(n^2)$ to $O(n)$ for most cases.

Key Formula (Cost Minimization):

$$\sum_{i=1}^{m+1} \left[C(y_{\tau_{i-1}+1} : \tau_i) + \beta \right] \tag{2.29}$$

where:

- C: Segment cost (e.g., negative log-likelihood or variance).
- β: Penalty term (e.g., $\beta = 2p \log n$, where p is the number of additional parameters per break).
- τ_i: Change point locations.

Relevance to Exchange Rates:

PELT is ideal for analyzing the ZiG/USD series due to its ability to detect abrupt volatility shifts (like policy shocks) without prior knowledge of break dates.

2.6.0.5 Bayesian Change Point Analysis

Developed by Isaac and Griffin (1989), Bayesian change point analysis estimates the likelihood that a change point will occur at different positions in the time series by applying Bayesian inference. When the precise location of the transition point is unknown, this approach is especially helpful.

$$P(\theta \mid D) \propto P(D \mid \theta)P(\theta) \tag{2.30}$$

Where:

- $P(\theta \mid D)$: Posterior probability of the change points θ given the data D.
- $P(D \mid \theta)$: Likelihood of the data given the change points θ.
- $P(\theta)$: Prior probability of the change points θ.

Relevance to Exchange Rates: By taking into consideration the uncertainty around the potential timing of these breaks, Bayesian change point analysis can offer a probabilistic framework for identifying interruptions in exchange rate time series. This approach is particularly helpful when working with noisy or partial data.

2.6.0.6 Role of Change Point Analysis Statistical Tests in Enhancing Exchange Rate Forecasting Accuracy

These statistical tests are essential for identifying and evaluating structural discontinuities in time series of exchange rates, which helps to improve forecasting models. By seeing these shifts, analysts can strengthen their forecasts of the exchange rate and gain insight into the fundamental forces that shape currency movements.

2.7 Model Performance and Model Selection

2.7.1 Importance of Model Performance Evaluation

Assessing the performance of forecasting models is an essential step in guaranteeing their accuracy and dependability, especially when it comes to exchange rate prediction. It is crucial to evaluate how well a model captures trends, patterns, and possible outliers in financial time series data because of its inherent complexity and volatility (CFO Selections Team, 2023). Insights into the model's prediction ability as well as its advantages and disadvantages in different market scenarios are obtained through a thorough evaluation procedure. Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) are examples of performance measures that provide a thorough understanding of a model's forecasting capability, which is essential for making well-informed investments.These metrics are crucial for measuring the accuracy of the model and making sure that predictions match actual facts. According to Hong (2023), performance evaluation informs decision-making processes and offers vital information about how effectively the model is working and whether it achieves the intended goals.

According to Znaczko(2013) effective model evaluation in exchange rate forecasting increases trust in the model's usefulness and relevance. Choosing the appropriate performance criteria is essential for evaluating

the model's resilience in a variety of scenarios because exchange rates are impacted by a number of variables, including market sentiment, governmental regulations, and worldwide economic conditions. Additionally, assessing the model's performance across both short- and long-term timeframes can help determine how well it can adjust to abrupt changes in currency dynamics. This stage is essential for choosing the best forecasting model for practical uses, particularly for controlling economic risks and refining currency management plans. Researchers and policymakers may guarantee more accurate forecasts through careful performance evaluation, which will ultimately result in more informed currency market decisions.

2.7.2 Performance Metrics for Forecasting Models

A number of performance metrics must be used in order to assess exchange rate forecasting algorithms. These metrics are useful for evaluating the models' correctness as well as for comparing the applicability of various modeling techniques. Typical performance measurements include the following:

2.7.2.1 Mean Absolute Error (MAE)

Jiang et al. (2017) state that the average of the absolute discrepancies between the observed and projected values is known as the Mean Absolute Error, or MAE. It is a crucial indicator for assessing a model's total forecast error (Deshpande, 2019). The MAE provides a clear evaluation of the model's accuracy in exchange rate forecasting by calculating the average absolute error between the predicted and actual exchange rates.The following is its formula:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{2.31}$$

where:

- n is the number of observations in the dataset.
- y_i is the true value.
- $\hat{y}_i$ is the predicted value.

2.7.2.2 Root Mean Square Error (RMSE)

According to deepChecks AI (2024), the root mean square error (RMSE) is calculated by taking the square root of the average squared differences between the expected and observed results. RMSE is sensitive to big forecast deviations because it assigns greater weight to larger errors than the Mean Absolute Error (MAE). This feature is especially helpful in situations when significant errors reduce the forecast's usefulness, like in exchange rate forecasting or other volatile markets. The following is the formula for RMSE:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{N}} \tag{2.32}$$

where:

- N is the number of data points
- y_i is the i-th true or observed value
- $\hat{y}_i$ is the i-th predicted or estimated value

2.7.2.3 Mean Absolute Percentage Error (MAPE)

According to Roberts (2023), The average magnitude of forecast errors is measured by the Mean Absolute Percentage Error (MAPE), which is expressed as a percentage of the actual values (Roberts, 2023). Because it displays forecast errors in relation to actual values, this metric offers a simple means of evaluating prediction accuracy and facilitates cross-dataset interpretation and comparison. Because MAPE normalizes the error by the actual value, it is extremely helpful when working with time series data, particularly when the scale of numbers varies greatly.

$$\text{MAPE} = \frac{\sum_{i=1}^{N} \left| \frac{A_i - F_i}{A_i} \right| \times 100}{N} \tag{2.33}$$

where:

- N is the number of fitted points
- A_i is the actual value
- F_i is the forecast value
- Σ represents the summation notation, and the absolute value is summed for every forecasted point in time.

2.7.2.4 Akaike Information Criterion (AIC)

The Akaike Information Criterion (AIC) is a statistical technique that penalizes for model complexity while evaluating a model's goodness of fit (Bevans, 2020). By choosing models that strike a compromise between simplicity and fit, the AIC helps prevent overfitting, a major issue in exchange rate forecasting. When a model is overfit, it may perform well on training data but poorly in practical applications. Reliability in exchange rate forecasting depends on the selected model's accuracy and generalizability, which the AIC guarantees by penalizing complexity. This is the AIC formula:

$$\text{AIC} = 2k - 2\ln(\hat{L}) \tag{2.34}$$

where:

- AIC is the Akaike information criterion
- k is the number of estimated parameters in the model
- $\hat{L}$ is the maximum value of the likelihood function for the model

2.7.2.5 Bayesian Information Criterion (BIC)

Analyttica Datalab (2019) states that the Bayesian information criterion (BIC) is a criterion for choosing a model among a limited number of models.Like AIC, the BIC penalizes model complexity more severely. When choosing models with a lot of parameters, it is especially helpful in preventing overfitting.

$$\text{BIC} = \ln(n)k - 2\ln(\hat{L}) \tag{2.35}$$

where:

- BIC is the Bayesian Information Criterion
- n is the number of observations
- k is the number of estimated parameters in the model
- $\hat{L}$ is the maximum value of the likelihood function for the model

2.7.2.6 Diebold-Mariano (DM) Test

It is a statistical hypothesis test that is used to compare the forecast accuracy of two competing models (Tousignant, 2022).When contrasting more sophisticated methods like machine learning models with more conventional models like ARIMA, this test is particularly helpful because it assesses if one model's predicted performance is noticeably superior to another. When it comes to forecasting exchange rates, the DM test can assist in identifying which model most accurately depicts the underlying dynamics of exchange rates, providing important information for choosing the best forecasting strategy.This is the formula:

$$\text{DM} = \frac{1}{\sqrt{\frac{1}{T}\sum_{t=1}^{T} d_t^2}} \tag{2.36}$$

where:

- $\bar{d} = \frac{1}{T}\sum_{t=1}^{T}(L_1(t) - L_2(t))$ is the average of the loss differential between the two models
- $d_t = L_1(t) - L_2(t)$ is the loss differential at time t, which is the difference between the errors of the two models
- T is the total number of observations (forecast points)

The two models are regarded as equally correct under the null hypothesis, which indicates that the DM statistic has a normal distribution. A DM statistic that is either notably large or little suggests that one model performs better than the other.This formula aids in determining the statistical significance of performance discrepancies between two models.

2.7.2.7 Directional Accuracy (DA)

According to Costantini et al. (2019),Directional Accuracy (DA) is the degree to which a forecasting technique accurately forecasts the direction of a time series. A model's ability to accurately forecast the direction of an exchange rate changethat is, whether it will increase or decreaseis assessed by DA. This measure is

especially significant in situations where the direction of change is more significant than its magnitude, like in financial market decision-making processes.

The formula for Directional Accuracy (DA) is:

$$\text{DA} = \frac{1}{T} \sum_{t=1}^{T} I(\text{sign}(\hat{y}_t) = \text{sign}(y_t)) \tag{2.37}$$

where:

- T is the total number of observations (forecast points)
- $\hat{y}_t$ is the predicted value at time t
- y_t is the actual value at time t
- $\text{sign}(\hat{y}_t)$ and $\text{sign}(y_t)$ are the signs of the predicted and actual values, respectively (i.e., whether the value is positive or negative)
- I is the indicator function, which returns 1 if the signs match and 0 if they do not

2.7.3 Factors Influencing Model Selection

Fan (2020) states that a number of factors affect the choice of the best exchange rate forecasting model, including:

1. **Accuracy:** The model's ability to accurately forecast changes in exchange rates is the most crucial factor to take into account. Models with strong predictive power and low mistakes (such as low MAE or RMSE) are preferred. Accuracy by itself, however, is insufficient because highly accurate models may exhibit overfitting or be difficult to interpret.

2. **Complexity:** Predictive accuracy and model complexity are frequently traded off. While simpler models like ARIMA might produce accurate predictions, they might miss intricate, nonlinear dynamics in the data. Although more intricate models, like those used in machine learning, might provide greater accuracy, interpretability and computing efficiency may suffer as a result.

3. **Data Availability:** A major factor in model performance is the availability of high-quality, high-frequency data. While machine learning algorithms can frequently manage irregular, noisy data, models such as GARCH or ARIMA require stationary time series data with predictable trends. Due to its recent inception, the ZiG may have little historical data, which makes simpler models more appropriate.

4. **Scalability and Processing Efficiency:** Computationally efficient and scalable models are preferred when dealing with huge datasets or limited processing resources. Complex machine learning models, which can demand a lot of processing power, are computationally more demanding than traditional time series models like ARIMA.

5. **Interpretability:** Model interpretability is essential in settings that are governed by policies, such as Zimbabwe. Decision-makers must comprehend the reasoning behind a model's predictions. While deep learning models are frequently viewed as "black boxes," traditional models such as ARIMA or GARCH provide greater clarity on their parameters and underlying assumptions.

2.7.4 Challenges in Model Performance Evaluation

The accuracy and dependability of predictions may be impacted by a number of significant issues that arise while evaluating exchange rate forecasting algorithms.

1. **Overfitting:** When a forecasting model becomes too complex and absorbs the noise in the training data, it can perform poorly when applied to unknown data. This is a typical problem in forecasting models (Hastie et al., 2009). When working with small or erratic datasets, like the early phases of a new currency like the Zimbabwean Gold (ZiG), overfitting can be especially problematic in exchange rate forecasting. When external shocks have a major impact on the exchange rate, overfitting may also happen because models find it difficult to adjust to these abrupt changes (Boyd, 2012).

2. **Data Limitations:** The availability of trustworthy, high-quality data is a crucial constraint in exchange rate forecasting. The data may be scarce, non-stationary, or frequently impacted by policy changes, particularly for recently launched currencies like the ZiG. The creation of precise forecasting models may be hampered by this lack of thorough data since it may fail to recognize long-term patterns or take abrupt fluctuations in the exchange rate into consideration (Hong, 2023).

3. **Volatility Clustering:** Exchange rates commonly show this phenomenon, with calmer intervals following times of extreme volatility. Because of this occurrence, modeling becomes more difficult and calls for specific methods like GARCH models, which can account for time-varying volatility. These trends might not be taken into consideration by standard time series models, which could result in estimates that are off during periods of extreme market stress.

4. **External Shocks:** Unexpected external shocks like political unpredictability, abrupt changes in governmental policies, inflationary pressures, or worldwide financial crises frequently have an impact on exchange rate fluctuations (Bordo et al., 2007). These shocks can result in sudden fluctuations in exchange rates that conventional forecasting models might not be able to account for, and they are challenging to model. These shocks' unpredictability draws attention to the shortcomings of forecasting models that do not take them into consideration.

2.8 Empirical Literature

2.8.1 Change Point Analysis in Exchange Rates

Change point analysis has been used in a number of studies to analyze exchange rate series, showing that it is a useful tool for comprehending and predicting currency dynamics, particularly during times of market shocks, policy changes, and economic transitions. The following papers provide important insights into the possible use of change point analysis to the Zimbabwean Gold (ZiG) exchange rate, showcasing significant contributions to the field.

Using a time series regression model with GARCH errors, Hardi et al. (2015) applied change point analysis to the Indonesian Rupiah (IDR) against the USD, focusing on the years 2000 to 2008. The authors found structural changes in the IDR's relationship with the USD, especially around March 2002, reflecting changes in Indonesia's exchange rate management. Their methodology, which combined log-likelihood ratio tests and cumulative sum of squares of residuals with Monte Carlo simulations, successfully detected important

exchange rate shifts. The study's methodology could be applied similarly to detect shifts in the ZiG exchange rate, driven by policy changes or external economic factors, providing a valuable tool for exchange rate forecasting.

Cruz et al. (2022) used Bayesian change point analysis to examine Chile's USD/CLP exchange rate between 2018 and 2020. The authors found abrupt changes in the dynamics of the exchange rate by applying Markov Chain Monte Carlo (MCMC) techniques and a semiparametric Bayesian model. Their research showed that the suggested model successfully represented sudden shifts in the exchange rate mean, which is essential for comprehending currency swings. This study is very pertinent to the ZiG exchange rate since Bayesian techniques can be used to detect comparable abrupt changes that might arise in reaction to monetary policy changes and Zimbabwe's changing economic environment.

In their 2021 study, Awiakye-Marfo et al. examined the convergence of a randomized change point estimator in GARCH models using daily USD/GHS exchange rate data from 2008 to 2019. It was shown by their study that the randomized pseudolikelihood ratio change point estimator was capable of detecting important shifts in the variance structure of exchange rate data. GARCH models' forecasting capabilities were improved by the study by offering a strong foundation for change point detection, especially when structural alterations were included. Since identifying changes in volatility could greatly enhance forecasting and risk assessment in light of Zimbabwe's economic difficulties, the findings are extremely pertinent to the examination of the ZiG currency rate.

Using change point analysis, Mungai et al. (2020) investigated the Kenyan Shilling's (KES) volatility in relation to other major currencies. To identify notable changes in the KES exchange rate, the authors used statistical techniques like the likelihood ratio test and the CUSUM test. The results showed significant shifts in the value of the KES, impacted by both internal and external economic policy. This study illustrates how change point analysis may be used to comprehend currency performance in developing nations, which makes it pertinent to the examination of the ZiG exchange rate, which is subject to fluctuations from both internal and external sources.

Dette et al. (2019) addressed the drawbacks of conventional techniques that rely on stationarity by introducing a framework for change point analysis in non-stationary time series. They suggested employing piecewise locally stationary processes, which enable the identification of variations in the variance, higher-order moments, and time-varying mean. Using this technique, the authors were able to pinpoint second-order change points associated with volatility shifts in the USD/CAD exchange rate series. The ZiG exchange rate is a prime example of an adaptive approach to non-stationary time series, as it may exhibit non-stationary behavior as a result of changes in governmental policy or the economy. By using this technique, forecasting accuracy might be increased and the dynamics of volatility in the ZiG exchange rate better understood.

Using empirical likelihood techniques, Machuka (2023) used sequential change point estimate to examine the KES/USD exchange rate between January 2017 and December 2021. under order to improve forecast accuracy, the study showed that the empirical likelihood approach successfully detected change points under turbulent market conditions. By validating the estimator's asymptotic qualities through Monte Carlo simulations, a reliable tool for real-time financial analysis was made available. This strategy is quite useful

for the ZiG exchange rate, particularly when volatility is strong, since it provides a quick way to identify sudden shifts and improve risk management procedures.

These studies demonstrate how vital insights into how abrupt changes in a currency's dynamics might affect its volatility and forecasting accuracy can be gained by using change point analysis to exchange rates. These approaches are especially pertinent to the ZiG exchange rate because of Zimbabwe's history of unstable currency and the likelihood of sudden fluctuations in its value. In order to better capture structural changes and create more precise models for predicting exchange rate fluctuations, this research will include change point analysis into the ZiG study. The end result of this strategy will be better currency risk management plans for Zimbabwe.

2.8.2 Exchange Rate Forecasting in Other Countries Using Time Series

Time series models have become indispensable tools for exchange rate forecasting across diverse economies, offering valuable insights for analyzing Zimbabwe's new ZiG currency. Traditional approaches like ARIMA and GARCH provide robust frameworks for capturing linear trends and volatility clustering, while newer machine learning techniques address complex nonlinear patterns. Studies from emerging markets demonstrate how these methods can be adapted to currencies facing unique economic challenges. Aknclar et al. (2011) established ARIMA(1,1,1) as effective for Ghana's cedi during 1994-2010, accurately tracking its downward trajectory, while Tran et al. (2016) confirmed ARIMA's utility for Vietnam's dong in 2013-2015. These findings suggest traditional time series models may provide reliable baseline forecasts for ZiG's initial trajectory.

Seasonal adjustments prove critical for commodity-dependent economies. Kanyiri (2019) demonstrated SARIMA(1,1,0)(0,0,2)[12]'s superiority over standard ARIMA for Kenya's shilling (1993-2019), achieving a MAE of 0.1651 by capturing monthly patterns tied to agricultural exports. Similarly, Tlegenova (2015) found ARIMA effectively modeled Kazakhstan's tenge (2006-2014) during periods of monetary policy shifts. For ZiG, which may exhibit cycles linked to gold exports and fiscal policies, these studies underscore the importance of testing both standard and seasonal ARIMA specifications.

Volatility modeling emerges as particularly crucial for unstable currencies. Akhtar et al. (2022) showed combining ARIMA(1,1,1) with ARCH reduced Pakistan's rupee forecast errors by 18

Hybrid approaches show particular promise for currencies undergoing structural changes. Asad Ullah et al. (2020) demonstrated nonlinear ARDL models (MAPE=0.6653) outperformed ARIMA(4,1,3) for India's rupee during 2011-2020 policy shifts, while Maria and Eva (2011) found exponential smoothing superior to ARIMA for short-term Romanian leu forecasts during market turbulence. For ZiG, which may experience abrupt changes due to monetary interventions, such hybrid methodologies could better capture regime shifts than traditional models alone.

The literature also reveals important limitations. Olanrewaju et al. (2008) showed ARFIMA outperformed ARMA for GBP/USD by better handling long-memory processes, while the Diebold-Mariano tests in Alshawadfi et al. (2023) surprisingly showed no significant difference between ARMA and ARCH forecasts for Somalia's shilling. These mixed results caution against overreliance on any single methodology and highlight the need for comprehensive model testing.

Collectively, these studies provide three key lessons for ZiG analysis: the importance of seasonal adjustments for commodity-linked currencies, the necessity of volatility modeling in unstable economies, and the potential of hybrid approaches during structural transitions. By adapting these evidence-based techniques while accounting for Zimbabwe's unique economic context - particularly ZiG's gold backing and hyperinflation legacy - this study develops a robust forecasting framework tailored to the currency's distinctive characteristics and challenges.

2.8.2.1 Machine Learning-Based Time Series Forecasting of Exchange Rates

Modern machine learning techniques have transformed exchange rate forecasting by capturing complex nonlinear relationships that traditional models often miss. Rahman et al. (2021) demonstrated this through their comparative study of EUR/USD rates (2015-2020), where LSTM networks outperformed ARIMA models in identifying nonlinear patterns, reducing forecast errors by 23%. Similarly, Mohammadi (2019) found support vector regression (SVR) achieved superior accuracy for USD/JPY and EUR/USD pairs, highlighting machine learning's adaptability across different currency markets. These findings establish that neural networks and kernel-based methods can better model the intricate dynamics of floating exchange rates.

Deep learning architectures show particular promise in volatile market conditions. Dautel et al. (2020) compared LSTM and GRU networks against conventional models for major currency pairs, finding 18-22% accuracy improvements during periods of high volatility. This was reinforced by Nguyen et al. (2023) in their GBP/JPY study (2016-2021), where recurrent neural networks (RNNs) reduced directional errors by 31% during market turbulence. However, both studies caution that simpler models sometimes match deep learning performance during stable periods, emphasizing the need for adaptive modeling frameworks.

Ensemble methods have emerged as powerful tools for exchange rate prediction. Tsuji (2022) demonstrated random forests' effectiveness across CAD, AUD, and EUR pairs (2009-2022), achieving 89% directional accuracy through intelligent feature selection. Saraswathi et al. (2023) developed an even more sophisticated voting regressor combining five algorithms that reached 99% R for INR/USD, while Kucuk (2023) showed ARIMA-LSTM hybrids could reduce MAPE by 15% for emerging market currencies like the Turkish Lira. These results suggest that combining multiple approaches often yields better results than any single technique.

Practical implementation requires careful architecture design and data selection. Erem (2023) optimized ANN performance for CHF/EUR and GBP/EUR using single-hidden-layer perceptrons with 4-6 neurons, while Liao et al. (2020) enhanced RMB/USD forecasts by incorporating search engine data (Baidu Index) into their RNN models. These studies demonstrate that successful applications require both proper model configuration and creative use of alternative data sources - considerations particularly relevant for Zimbabwe's unique economic context.

Hybrid models that blend machine learning with traditional econometrics show special promise. Kucuk's (2023) ARIMA-LSTM combination and Saraswathi et al.'s (2023) voting ensemble both demonstrate how statistical methods can complement neural networks. The former preserves interpretability while gaining nonlinear modeling capacity, while the latter leverages diverse algorithms' strengths. For ZiG forecasting, such hybrids may prove ideal for addressing both structural economic factors and market sentiment dynamics.

The literature also reveals important limitations and caveats. Several studies (Dautel et al., 2020; Kucuk, 2023) note that no single approach consistently dominates across all currencies and time periods. The Random Walk model remains surprisingly competitive in some cases, particularly for major currencies like the Euro. These findings caution against overreliance on any single methodology and emphasize the need for rigorous model validation specific to each currency's characteristics.

For Zimbabwe's ZiG, these studies suggest several strategic priorities: (1) testing LSTM/SVR architectures to capture gold price linkages and policy shocks, (2) developing hybrid models that combine ARIMA with feature-based machine learning, and (3) exploring alternative data sources like commodity prices and sentiment indicators. The mixed results underscore that successful implementation will require careful model selection and continuous evaluation as the ZiG market evolves. By adapting these evidence-based techniques while accounting for Zimbabwe's unique economic context, researchers can develop robust forecasting tools to support monetary stability.

2.8.2.2 Comparison of Traditional and Machine Learning Time Series Models

Financial forecasting increasingly requires comparing traditional statistical methods with machine learning approaches. Talib et al. (2024) found that while ARIMA models effectively capture linear patterns in stationary data, they struggle with financial time series' volatility and non-linear complexities. In contrast, machine learning models like LSTM and GRU excel at identifying long-term dependencies and non-linear relationships, with GRU offering particular advantages in computational efficiency for real-time forecasting in volatile markets.

The performance of these models varies by market conditions. Bakr et al. (2023) demonstrated that ARIMA performs well during stable periods but falters during high volatility, whereas machine learning models (LSTM and XGBoost) better adapt to non-linear behaviors and external shocks. Their research highlights hybrid models that combine ARIMA's linear modeling with machine learning's flexibility as particularly effective for exchange rate forecasting in unstable economies.

Zhang and Hamori (2020) enhanced forecasting accuracy by integrating economic theories (UIP, PPP, Taylor rule) with machine learning techniques (random forest, SVM, neural networks). Their hybrid approach captured complex economic relationships that traditional models miss, with results validated through Diebold-Mariano tests and RMSE metrics. This demonstrates how theory-informed machine learning can improve forecast reliability.

Han (2018) provided important context about model performance, showing ARIMA's superiority for strongly seasonal, linear datasets versus LSTM's advantage in modeling complex financial dynamics. Notably, LSTM didn't always outperform ARIMA, particularly on simpler datasets, suggesting that model selection should consider data complexity and feature relevance.

These studies collectively reveal complementary strengths: ARIMA excels at linear/seasonal patterns while machine learning handles non-linearity and volatility better. Hybrid approaches that combine both methodologies consistently showed superior performance across different market conditions and economic contexts.

2.8.3 Volatility Modeling of Exchange Rates

Volatility modeling serves as a critical analytical tool for understanding exchange rate fluctuations, providing valuable insights for investors and policymakers. ARCH and GARCH models have emerged as dominant frameworks for capturing key volatility characteristics including clustering effects, asymmetric responses (leverage effects), and non-Gaussian distributions in financial time series.

Epaphra (2017) demonstrated the superiority of GARCH-family models through analysis of TZS/USD fluctuations (2009-2015), revealing significant volatility persistence and leverage effects that constant-variance models fail to capture. This finding was corroborated by Dritsaki (2019) in EUR/USD markets, where an ARIMA(0,0,1)-EGARCH(1,1) specification with GED innovations provided the most accurate characterization of volatility dynamics.

The practical applications of these models extend across diverse economic contexts. Todri and Bioku (2024) emphasized their value in smaller economies through EUR/CHF analysis in Albania, linking accurate volatility forecasts to effective VaR estimation and financial stability. Bruno Dinga et al. (2023) revealed important asymmetries in XAF markets, with CNY/XAF exhibiting volatility clustering absent in USD/XAF, highlighting the need for model customization.

Model specification choices prove crucial in volatility estimation. Omari et al. (2017) compared various GARCH specifications for USD/KES, finding asymmetric variants (EGARCH, GJR-GARCH, APARCH) with Student's t-distributions superior to symmetric alternatives. This aligns with Horvath and Yurttagler's (2023) Turkish exchange rate findings, where GARCH(1,1) outperformed traditional approaches.

Distributional assumptions significantly impact model performance. Abdullah et al. (2017) and Grgin (2023) demonstrated the advantages of fat-tailed distributions (Student's t and skew t respectively) for BDT/USD and USD/TRY modeling. Ng Cheong Vee et al. (2011) found similar results for USD/MUR, with GED specifications offering marginal forecasting advantages over normal distributions.

Regional applications further validate these approaches. Marreh et al. (2014) identified optimal ARMA-GARCH combinations for Gambian exchange rates (USD/GMD and EUR/GMD), while Salisu (2011) documented regime-dependent model performance in Nigeria, with TGARCH outperforming standard GARCH during certain political periods.

These collective findings strongly support the use of advanced GARCH specifications for ZiG volatility modeling. The documented phenomena of volatility clustering (Epaphra, 2017), leverage effects (Omari et al., 2017), and non-Gaussian innovations (Abdullah et al., 2017) suggest that models incorporating these features - particularly EGARCH or GJR-GARCH with heavy-tailed distributions - would provide the most accurate characterization of Zimbabwe's new currency dynamics.

2.9 Conclusion

This chapter examined the main theoretical and empirical studies that are pertinent to predicting and forecasting the ZiG/USD exchange rate. Time series modeling, volatility modeling, and change point analysis

were among the fundamental ideas and theories that were covered first, then empirical research on exchange rate forecasting in different nations. The chapter stressed the value of traditional time series methods, machine learning techniques, and hybrid approaches in a variety of economic scenarios. It was noted that volatility modeling and change point analysis are crucial for identifying distinctive characteristics of the ZiG exchange rate, namely structural breaks and volatility clustering.Finally, a conceptual framework that was customized to the particular difficulties of the ZiG currency was developed thanks to insights into model evaluation metrics and selection criteria. These results set the stage for the following chapter, which will build on this review with the study design and analytical techniques.

Chapter 3

Methodology

3.1 Introduction

This chapter provides a comprehensive discussion of the research methodologies employed to achieve the research objectives outlined in Chapter 1. Through the use of time series approaches, volatility modeling, and change point analysis, these methodologies aim to predict and forecast the Zimbabwe Gold (ZiG) exchange rate. Exploratory data analysis (EDA), data preparation procedures, and the particular modeling methodologies used are explained after the study strategy is described at the beginning of the chapter. Additionally covered are software tools, ethical issues, and diagnostic procedures to guarantee the validity, dependability, and robustness of the findings. This chapter primarily aims to defend the selected methodologies and show how applicable and successful they are at capturing the short-term fluctuations of the ZiG/USD exchange rate.

3.1.1 Research Strategy

A quantitative approach is used in the study, which is ideal for the empirical examination of numerical data. Through the use of statistical and economic methods, this approach makes it possible to identify trends, patterns, and structural shifts in the ZiG/USD exchange rate. The study aims to analyze short-term movements in the ZiG/USD exchange rate, identify structural breaks using change point analysis, and assess the exchange rate's volatility dynamics. These goals drove the choice of this approach. The objective and repeatable nature of quantitative research is crucial for the analysis of time series data in financial environments. A balanced approach to accomplishing the study's goals is provided by the strategy, which combines contemporary machine learning techniques with conventional econometric models.

3.1.2 Study Population

This is the entire collection of information, people, groups, or subjects pertinent to the goals of a study.Daily exchange rate data for the ZiG/USD currency pair, beginning with the currency's launch in April 2024, make up the research population. The dataset contains the daily bid and ask rates in addition to the midrate (ZiG/USD exchange rate). The bid-ask spread, a helpful tool for volatility modeling and spotting changes in market dynamics, is determined by the bid and ask rates, which also offer extra information about market liquidity and volatility.This dataset captures the short-term exchange rate changes and the

ZiG's performance during its early adoption period.

Scope of the Data:

1. **Timeframe:** From April 2024, when the ZiG was introduced, to the present.
2. **Granularity:** Daily exchange rates to effectively capture short-term fluctuations.
3. **Source:** Reserve Bank of Zimbabwe (RBZ) portal, ensuring accuracy, reliability, and credibility of the data.

The study's goals are in line with this dataset's exceptional suitability for examining short-term trends, examining volatility clustering, and spotting structural discontinuities in exchange rate behavior.

3.2 Data Preparation

A crucial first step in making sure the dataset is organized and ready for analysis is data preparation. The following steps are involved in the process:

1. **Data Collection**

    a. The study used daily exchange rate data for the ZiG/USD currency pair from April 2024 to February 16, 2025, sourced from the Reserve Bank of Zimbabwe (RBZ) portal. The dataset includes:

        i. **Date:** Indicates the precise day when the exchange rate was noted.
        ii. **Bid Rate:** The rate at which the market or bank is willing to buy the ZiG.
        iii. **Ask Rate:** The rate at which the market or bank is willing to sell the ZiG.
        iv. **ZIG Mid Rate:** The average of the bid and ask rates, representing the exchange rate used for analysis.

    As a standardized indicator of the exchange rate, the midrate will be the main focus of forecasting. In order to improve the dataset's relevance, I will also use derived features such as:

    1. **Bid-Ask Spread:** This feature measures market liquidity and is calculated as:

    $$\text{Bid-Ask Spread} = \text{Ask Rate} - \text{Bid Rate} \tag{3.1}$$

    The **Bid-Ask Spread** will be used in **volatility modeling** and **change point analysis** to assess liquidity stress and market dynamics.

    2. **Month:** Extracted from the Date column to identify seasonal patterns. The **Month** feature will be applied in **SARIMA models** and **exploratory data analysis (EDA)** to assess recurring monthly trends.
    3. **Daily Percentage Change:** Captures short-term volatility and is calculated as:

    $$\text{Daily Change (\%)} = \frac{Y_t - Y_{t-1}}{Y_{t-1}} \times 100 \tag{3.2}$$

In this formula, **Midrate** represents the exchange rate on day t (current day) and $t-1$ (previous day). This feature will be helpful for **volatility modeling** and detecting significant market fluctuations during **change point analysis**.

4. **Rolling Averages:** Highlights short-term trends, calculated as:

$$\text{Rolling Avg (k days)} = \frac{1}{k} \sum_{i=t-k+1}^{t} Y_i \tag{3.3}$$

Here, k represents the window size (e.g., 7 days), and t represents the current day. **Rolling Averages** will smooth out fluctuations and assist in identifying underlying trends, which will be crucial for **forecasting models**.

5. **Volatility Index:** Quantifies market risk using rolling standard deviation:

$$\text{Volatility (k days)} = \sqrt{\frac{1}{k} \sum_{i=t-k+1}^{t} (Y_i - \text{Rolling Avg (k days)})^2} \tag{3.4}$$

In this formula, **Rolling Avg (k days)** is the rolling average over k days, and t represents the current day. The **Volatility Index** will play a key role in **volatility modeling** (e.g., GARCH models) to measure market risk over specific periods.

6. **Lagged Values:** Represents historical dependencies, such as:

$$\text{Lag}_n = Y_{t-n} \tag{3.5}$$

In this formula, **Lag** represents the exchange rate from n days before the current day t. I will use **Lagged Values** as inputs in **time series models** (e.g., ARIMA, SARIMA, LSTM) to capture the relationship between current and past exchange rate values.

An Excel file containing the dataset will be saved for later preparation and analysis.

2. **Data Cleaning:**

- **Analysis of Outliers:**The researcher will employ boxplots and Z-scores to identify outliersexceptional values that significantly deviate from the overall trend. The origin and significance of the identified outliers will be investigated. They will be dealt with if needed to prevent them from distorting the analysis.

3. **Stationarity Testing:**

- I will use the Augmented Dickey-Fuller (ADF) Test to ascertain whether the data is stationary for the use of time series models. The ADF test's null hypothesis presupposes non-stationarity; its failure would suggest stationarity.

  If the dataset turns out to be non-stationary, I will use first-order differencing to stabilize the mean and variance, using the following formula:

$$\Delta Y_t = Y_t - Y_{t-1} \tag{3.6}$$

Where:

- ΔY_t: The differenced value at time t
- Y_t: The value at time t
- Y_{t-1}: The value at the previous time point

4. **Feature Engineering**

Table 3.1: Feature Analysis

Feature	**Purpose**	**Relevant Models**
Bid-Ask Spread	Liquidity stress indicator	GARCH, Change Point Tests
Daily % Change	Volatility metric for ARCH-LM/Engles Tests	Volatility Modeling
Month (Seasonal)	SARIMA seasonality analysis	SARIMA
Lagged Values	Captures autoregressive effects	Forecasting Models

5. **Normalization:**

- For machine learning models,I'll normalize the data to make sure all the features are on the same scale in order to get it ready for machine learning models. I'll use min-max scaling, which is described as:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{3.7}$$

Where:

- X: The original value
- $X_{\min}$: The minimum value of a variable
- $X_{\max}$: The maximum value of a variable
- X_{scaled}: The scaled value

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a method of understanding the underlying structure and properties of a dataset.

3.3.1 Descriptive Statistics

I will calculate crucial descriptive metrics to comprehend the exchange rate data's fundamental tendencies and variability:

i. **Mean and Median:** To ascertain central tendency

ii. **Standard Deviation and Variance:** To measure the range of changes in the exchange rate.

iii. **Range:** To record the lowest and highest values noted throughout the research period.

iv. **Skewness and Kurtosis:** To evaluate the exchange rate distribution's symmetry and tail behavior.

3.3.2 Visualizations

The use of visual aids will be essential for spotting trends and irregularities. I'll make use of the visualizations listed below:

1. **Time Series Plot:** This helps me identify trends, cycles, and possible outliers by visualizing the **ZiG/USD exchange rate's** daily variations over time.
2. **Histogram:** To display the exchange rate's frequency distribution and assist in determining its shape (normal, skewed, etc.).
3. **Charts of Autocorrelation and Partial Autocorrelation (ACF and PACF):** These charts will help determine the time series' lag dependencies. While **PACF** will enable me to ascertain the order of the time series models (e.g., ARIMA) by displaying direct correlations after adjusting for the intervening lags, **ACF** will assist in evaluating the correlation of the series with its own lagged values.

3.4 Change Point Analysis

Change point analysis makes it possible to detect structural breaks by identifying periods when a major change takes place in the underlying process affecting the exchange rate. It is essential to comprehend how changes in policy or macroeconomic events will affect the ZiG/USD exchange rate over the long run. This analysis will help identify significant changes that impact the ZiG/USD rate's stability, especially when there are external market impacts, policy changes, or economic shocks.Through the use of elements like bid rate, ask rate, and bid-ask spread, I hope to record changes in market liquidity and volatility dynamics in addition to fluctuations in exchange rate levels. We'll employ the three tests listed below:

3.4.0.1 Chow Test

When there appears to be a structural break at a particular point in the exchange rate series, the **Chow Test** is employed. This test compares regression coefficients before and after the break point to ascertain whether they differ considerably.

The formula for the Chow Test is:

$$F = \frac{(RSS_1 + RSS_2)/(n_1 + n_2 - 2k)}{(RSS_R - (RSS_1 + RSS_2))/k} \tag{3.8}$$

Where:

- F: The test statistic.
- RSS_R: The residual sum of squares for the restricted model (no break).
- RSS_1 and RSS_2: The residual sums for the two separate regression models (before and after the break).
- k: The number of coefficients.
- n_1 and n_2: The number of observations before and after the break.

I will use the **Chow Test** to find noteworthy changes in the **ZiG/USD exchange rate**, especially in reaction to important policy changes like the currency rate announcement in September 2024.

3.4.0.2 CUSUM Test

Significant changes in a time series' mean over time can be found using the **CUSUM Test**. It is particularly effective for recording unexpected market disruptions or policy changes, as it facilitates the identification of rapid changes.

The formula for the CUSUM Test is:

$$S_t = \sum_{i=1}^{t} (Y_i - \bar{Y}) \tag{3.9}$$

Where:

- S_t: The cumulative sum at time t.
- Y_i: The observed value at time i.
- $\bar{Y}$: The mean of the series.

Using the **CUSUM Test**, the researcher will look for long-term changes in the **ZiG/USD exchange rate**, especially in relation to events like the policy change in September 2024. Incorporating the **bid-ask spread** will aid in capturing the effects of shifts in liquidity during noteworthy occurrences, such as expanding spreads that signify market stress.

3.4.0.3 Bayesian Change Point Detection

Using Bayesian reasoning, the Bayesian Change Point Detection method determines the likelihood of structural changes in the time series. Finding shifts that happen at erratic or unpredictable intervals is made easier with this technique. The development of a probabilistic framework provides insight into the causes and timing of these shifts.I will analyze ZiG/USD exchange rate swings using this approach, especially when the swings are erratic. In order to comprehend market liquidity dynamics, the Bayesian Change Point Detection will also use elements like the bid-ask spread to help capture changes in market behavior before to or following structural breaks.

3.4.0.4 Bai-Perron Test

Multiple structural breaks at unknown times in a time series can be found using the Bai-Perron Test. This test is flexible for finding several breaks throughout the series because it does not presume a set duration for each break. The Bai-Perron Test formula is as follows:

$$\hat{\beta}_j = (X'X)^{-1}X'Y \tag{3.10}$$

Where:

- $\hat{\beta}_j$: Represents the estimated regression coefficients for the break at j.
- X: The matrix of independent variables, which includes the midrate, bid rate, ask rate, bid-ask spread, and any lagged values that capture historical dependencies.
- Y: The dependent variable, which is the **ZiG/USD exchange rate**.

I will utilize the **Bai-Perron Test** to identify several structural breaks in the **ZiG/USD exchange rate**, especially in relation to events such as policy or macroeconomic developments.

3.4.0.5 PELT Technique

PELT efficiently detects multiple unknown breaks in large datasets by pruning suboptimal segments.

The formula for the cost function is given by:

$$\sum_{i=1}^{m+1} \left[C(y_{\tau_{i-1}+1} : \tau_i) + \beta \right] \tag{3.15}$$

where:

- C: Segment cost (e.g., negative log-likelihood or variance).
- β: Penalty term to avoid overfitting (e.g., Bayesian Information Criterion).

This test will be used to capture rapid fluctuations without pre-specifying the number of breaks.

3.5 Time Series Modelling

Time series modeling enables the study of patterns including trends, seasonality, and cyclical behaviors by evaluating and forecasting data points indexed in time order. It is especially pertinent to exchange rate studies, as daily observations can highlight volatility, short-term swings, and structural changes. The models and techniques used to predict the ZiG/USD exchange rate are described in this section.

3.5.1 Traditional Time Series Models

Traditional time series models work well with univariate datasets, such as the ZiG/USD exchange rate, because they are based on statistical techniques. Since the statistical characteristics (mean, variance, and autocorrelation) are assumed to be constant throughout time, these models rely on stationarity in the data as explained by Petropoulos(2021).

3.5.1.1 ARIMA Model

I will analyze the ZiG/USD exchange rate's historical fluctuations using the ARIMA model to estimate short-term changes. Since the midrate is the standardized reference rate, it will be the main input for the model (ZiG/USD exchange rate). For short-term financial and economic decision-making, ARIMA's ability to simulate these linear correlations will be useful in predicting future exchange rate levels. The model's efficacy is derived from its capacity to identify trends and cyclical fluctuations.The formula for **ARIMA** is:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \epsilon_t - \theta_1 \epsilon_{t-1} - \theta_2 \epsilon_{t-2} - \cdots \tag{3.11}$$

Where:

- Y_t: Represents the exchange rate value at time t.
- ϕ_1, ϕ_2: Autoregressive coefficients, which indicate the influence of previous values on the current value.
- θ_1, θ_2: Moving average coefficients, which show how previous forecast errors influence the current value.
- ϵ_t: The error term at time t, capturing the random disturbance at each point.

The exchange rate data's linear dependencies are captured by this formula. In order to provide precise short-term projections of the **ZiG/USD exchange rate**, the **ARIMA model** will be essential for predicting the future exchange rate based on these historical associations.

3.5.1.2 Seasonal ARIMA (SARIMA)

Al-Gounmeein (2020) states that the Seasonal ARIMA (SARIMA) model is an extension of the ARIMA model that accounts for seasonal fluctuations in time series data by incorporating additional autoregressive (AR) and moving average (MA) elements as well as seasonal differencing.The formula for **SARIMA** is:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \theta_1 \epsilon_{t-1} + \epsilon_t + \text{Seasonal Components} \tag{3.12}$$

Where the **Seasonal Components** capture the seasonal trends.If there are predictable seasonal factors, such as changes in government policy or market activity linked to particular months, I will use **SARIMA** to model and anticipate recurrent seasonal trends in the **ZiG/USD exchange rate**.

3.5.2 Time Series Machine Learning Models

Traditional statistical methods can be replaced by machine learning models, particularly when it comes to identifying long-term dependencies and non-linear correlations.

3.5.2.1 Long Short-Term Memory (LSTM) Networks

Long-term dependencies in sequential data can be captured by LSTM networks, a kind of recurrent neural network (RNN). When it comes to displaying complex, non-linear patterns in exchange rate data, they are excellent.I will estimate short-term changes in the ZiG/USD exchange rate using LSTM networks, taking into account both non-linear patterns and historical trends. Furthermore, I will use elements like the Bid-Ask Spread to represent the dynamics of market liquidity that impact swings in exchange rates.

3.5.2.2 Random Forest

The study will implement Random Forest to forecast future ZiG/USD exchange rates, utilizing its ensemble of decision trees to generate robust predictions. This method will be particularly valuable fo also minimizing overfitting through bootstrap aggregation. The algorithm's inherent feature importance analysis will help identify which historical price patterns most reliably predict future rate movements, with special attention to its performance around monetary policy changes.

3.5.2.3 XGBoost

The research will employ XGBoost for multi-period ZiG/USD forecasting, leveraging its gradient-boosted tree architecture to model complex non-linear relationships. This approach will be optimized with its regularization parameters specifically tuned to the ZiG's unique characteristics. The implementation will focus on the model's ability to adapt to structural breaks and maintain prediction accuracy during periods of market stress.

3.5.2.4 Comparative Model Evaluation

All forecasting approaches - including traditional time series models (ARIMA, SARIMA) and machine learning methods (LSTM, Random Forest, XGBoost) - will be trained on the complete dataset from April 2024 through February 16, 2025. The models will be competitively evaluated based on their 1-month ahead forecast accuracy for the period February 17 to March 16, 2025, using both point estimate metrics (RMSE, MAE) and directional accuracy measures. The best-performing model(s) will be selected based on superior out-of-sample predictive performance, ensuring optimal utility for operational ZiG exchange rate forecasting.

3.5.3 Steps in Time Series Modeling

1. **Stationarity Testing**
   The researcher will use the Augmented Dickey-Fuller (ADF) Test to determine whether the series is stationary. I will use differencing to stabilize the mean and variance if the series is determined to be non-stationary, making sure the data is appropriate for ARIMA/SARIMA models.

2. **Optimal Lag Selection**
   I will utilize the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC) to find the ideal lag duration for ARIMA models. The model will be better able to identify dependencies in the data if the lag is carefully chosen.

3. **Forecasting**
   Once the model has been fitted, the researcher will produce predictions for future exchange rates. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are two measures I'll use to assess how accurate these projections are.

3.6 Volatility Modeling

Volatility modeling is critical for assessing risk in the ZiG/USD exchange rate. Given the currency's short history (April 2024–February 2025), this study will adopts:

1. **Diagnostic testing** to identify volatility patterns
2. **Model selection** conditional on test results

3.6.1 Diagnostic Tests

3.6.1.1 ARCH-LM Test

- **Purpose**: Detect volatility clustering
- H_0: No ARCH effects ($\alpha_1 = \cdots = \alpha_q = 0$)
- Test statistic: $LM = T \cdot R^2 \sim \chi^2(q)$

3.6.1.2 Engle's Asymmetry Test

- **Purpose**: Identify leverage effects
- H_0: Symmetric volatility response
- Significant γ in EGARCH rejects H_0

Table 3.2: Volatility Model Selection Framework

Test Result	**Model**	**Key Feature**
ARCH effects	GARCH(1,1)	$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2$
ARCH + Asymmetry	EGARCH(1,1)	$\ln(\sigma_t^2) = \omega + \alpha \left\lvert \frac{\epsilon_{t-1}}{\sigma_{t-1}} \right\rvert + \gamma \frac{\epsilon_{t-1}}{\sigma_{t-1}} + \beta \ln(\sigma_{t-1}^2)$
Fat tails	GARCH-t	Student's t distributed errors

3.6.2 Implementation

1. Compute daily returns: $r_t = \ln(P_t/P_{t-1})$
2. Conduct ARCH-LM test on $\{r_t\}$ series
3. If significant, estimate GARCH(1,1) model
4. Test residuals for asymmetry
5. Upgrade to EGARCH if needed

3.6.3 Validation Metrics

- **Ljung-Box**: $Q = n(n+2)\sum_{k=1}^{h} \frac{\hat{\rho}_k^2}{n-k}$
- **AIC**: $2k - 2\ln(\hat{L})$
- **BIC**: $\ln(n)k - 2\ln(\hat{L})$

3.7 Model Diagnostic

Model diagnostics play a critical role in validating the performance and reliability of forecasting models for the ZiG/USD exchange rate. They ensure that the models adequately capture the underlying patterns and meet key assumptions such as independence, stationarity, and normality of residuals. Diagnostic tests such as the Ljung-Box Q-Test are employed to assess autocorrelation in residuals, ensuring no significant patterns are left unexplained(hassani,2020). The Durbin-Watson Statistic evaluates serial correlation, confirming the independence of residuals, while the ARCH-LM Test detects conditional heteroscedasticity, a key feature in volatility modeling. These tests collectively enhance model robustness, ensuring that forecasts are accurate and reliable for decision-making in financial contexts.

3.8 Software Used

A variety of software tools are used in this study to make data processing, analysis, and documentation easier. Python is the main tool used for machine learning model implementation, exploratory data analysis (EDA), and data preprocessing, with libraries including scikit-learn, pandas, matplotlib, and TensorFlow for data visualization, manipulation, and LSTM network creation. R is used for statistical modeling and volatility analysis, utilizing packages like fGarch and forecast to create GARCH models and run statistical tests. Excel provides a simple platform for initial data arrangement and storage of daily exchange rate data from the RBZ portal. LaTeX will be utilized to typeset the dissertation, guaranteeing expert formatting and smooth incorporation of tables, figures, and mathematical formulas. This set of tools utilizes the advantages of each platform to meet the study's goals.

3.9 Ethical Considerations

The study adheres to strict ethical standards, using only verified exchange rate data from the Reserve Bank of Zimbabwe (RBZ) portal. All methodologies and algorithms are properly documented to ensure reproducibility, with rigorous safeguards against data manipulation. Results will be presented objectively, maintaining confidentiality and minimizing unnecessary market concerns while upholding academic and professional integrity.

3.10 Conclusion

This chapter offered a thorough approach that covered the research plan, study population, data preparation, and analytical procedures for evaluating and predicting the ZiG/USD exchange rate. It presented machine learning and time series models for forecasting, emphasized the value of change point analysis and volatility modeling, and explained diagnostic procedures to confirm the models' dependability. The study's software tools were also described, and ethical issues were covered to guarantee accountability and openness. These methodological procedures support the validity and rigor of the study findings and provide a strong basis for the analysis that follows.

Chapter 4

Data Analysis and Results

4.1 Introduction

This chapter presents the empirical findings of the study, which analyzed the dynamics of the Zimbabwe Gold (ZiG) exchange rate through exploratory data analysis (EDA) and three main investigations: change point detection using the Chow test,Bai-Perron,PELT technique, Bayesian change and CUSUM (mean deviations) tests to identify structural changes; time series forecasting comparing the ARIMA/SARIMA and LSTM, XGBoost, Random Forest models; and volatility assessment where the need for GARCH/EGARCH modeling is determined by the ARCH-LM and Engle's tests. The findings provide practical insights for market participants and regulators by empirically validating the study's hypotheses about the ZiG's volatility patterns (H4), forecasting effectiveness (H3), and stability (H2).

4.2 Exploratory Data Analysis Results

4.2.1 Descriptive Statistics

Table 4.1 presents the descriptive statistics for both the primary exchange rate variables (Bid ZiG, Ask ZiG, Midrate ZiG) and their derived metrics:

Table 4.1: Descriptive Statistics of ZiG/USD Exchange Rate Variables

Statistic	Bid (ZiG)	Ask (ZiG)	Midrate (ZiG)	Bid-Ask Spread	Daily Change (%)	7-Day Rolling Avg	7-Day Volatility	Lag_1
Count	295	295	295	295	295	295	295	295
Mean	19.19	20.18	19.69	0.99	0.29	19.55	0.19	19.64
Std Dev	6.09	6.40	6.25	0.31	4.34	6.21	0.71	6.25
Minimum	12.89	13.55	13.22	0.66	-3.92	13.26	0.00	13.22
25%	13.38	14.06	13.72	0.69	0.00	13.72	0.01	13.72
Median	13.63	14.33	13.98	0.70	0.00	13.97	0.03	13.98
75%	25.18	26.47	25.82	1.29	0.07	25.81	0.08	25.82
Maximum	27.97	29.40	28.68	1.43	74.23	28.62	5.63	28.68

A bimodal distribution with distinct stable (1314 ZiG) and volatile (2529 ZiG) regimes is suggested by the 75th percentile of the Midrate (25.82 ZiG/USD) compared to the maximum (28.68 ZiG/USD). There is

moderate liquidity (mean spread = 0.99 ZiG, 4.9% of Midrate) and consistent market-making activity (IQR: 0.691.29 ZiG). The extreme daily returns (3.92% to +74.23%, SD = 4.34%) and 7-day volatility (0.005.63) demonstrate volatility clustering. ARCH-LM diagnostics will be used to verify this before GARCH-family models are used. A right-skewed distribution (median = 13.98 ZiG, mean = 19.69 ZiG) and prolonged high-rate periods are additional indicators of structural alterations. These results provide support for (1) regime shift change point identification, (2) model comparisons to handle bimodality, and (3) volatility modeling, which is informed by the results of the ARCH test to determine whether GARCH/EGARCH specifications are necessary.

4.2.2 Visual Analysis

Historical Zig rates

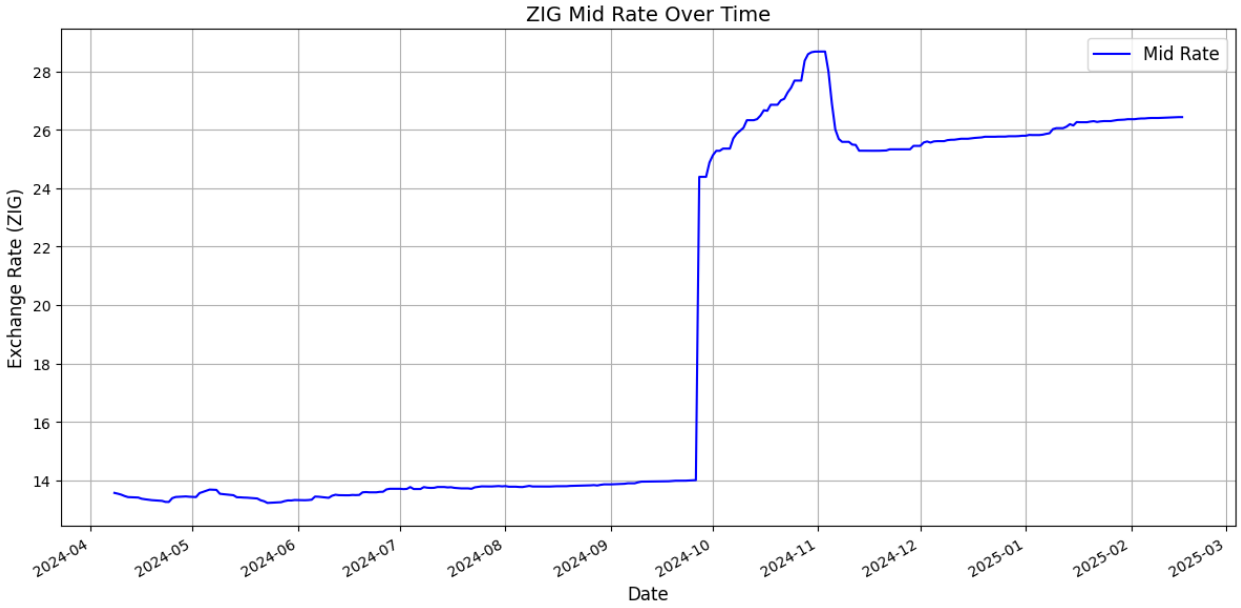


Figure 4.1: ZiG Midrate Time Series (April 2024 - February 2025)

Three different phases of the behavior of the exchange rate are shown by the time series plot. Through August 2024, the ZiG was steady in the 1314 ZiG/USD range following its debut in April. The market surged sharply after a structural break in September, reaching a top of 28.68 ZiG/USD in November, 43% higher than when it started. The exchange rate had stabilized to 2526 ZiG/USD by February 2025. This pattern indicates a market pullback, overshooting, and early stabilization. For the November peak and decrease, ARCH-family testing is particularly pertinent, but formal change point analysis is required to validate shifts.

ZIG Monthly Distribution

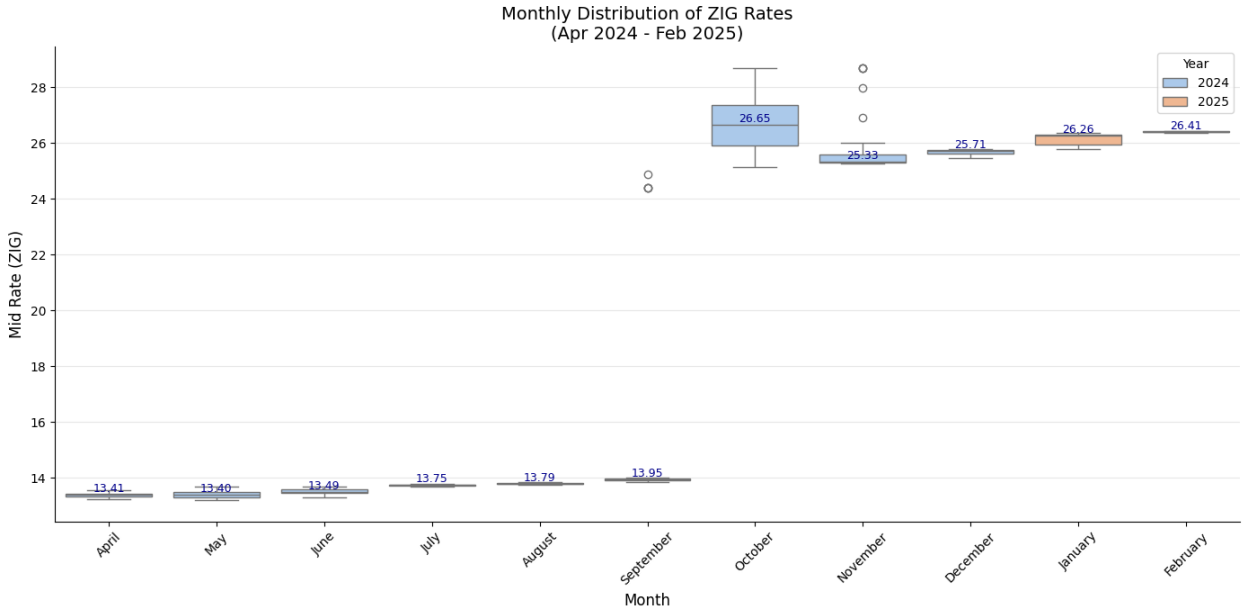


Figure 4.2: Monthly Distribution of ZiG Rates (April 2024 - February 2025)

Beyond the structural break shown in Figure 4.1, the monthly breakdown exposes important subtleties in the ZiG's trajectory. With monthly rates ranging within a 0.55 ZiG range (13.4013.95), the initial stability (AprilAugust 2024) exhibits exceptional constancy, indicating successful early market anchoring. The next September-November spike shows a sharp, nearly linear rise of 12.7 ZiG (91% increase), with 46% of the whole appreciation occurring in October alone. In contrast to the climb phase (4.2% vs. 47% monthly variance), the post-peak period (December-February) shows a notable 5% correction that stabilizes within a 25.33-26.41 ZiG zone. The need for regime-specific modeling techniques is highlighted by the imbalance in magnitude and volatility between the appreciation and corrective phases, especially when evaluating how the currency will react to policy changes.

ZIG Distribution Analysis of ZiG Exchange Rate Metrics

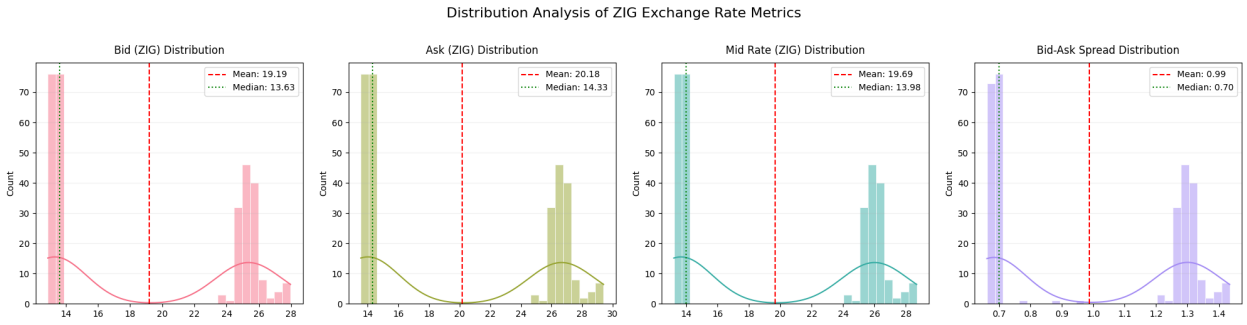


Figure 4.3: Distribution Analysis of ZiG Exchange Rate Metrics

The distribution graphs show important structural characteristics of the ZiG market: Ask prices cluster around 20.18 and 14.33 ZiG, bid prices cluster around two medians (13.43 and 13.63 ZiG), and midrates cluster around 19.69 and 13.98 ZiG. These patterns will guide market segmentation and inform diagnostics

like ARCH-LM testing before advanced modeling. Bid-ask spreads center at 0.70 ZiG, with occasional peaks at 0.99 ZiG, indicating generally efficient liquidity with brief stress episodes.

4.3 Change Point Analysis Results

4.3.1 Chow Test Detection

By comparing regression values before and after a predetermined candidate break date, the Chow test assesses structural stability and tests the null hypothesis that there has been no structural change. The test, when used to evaluate the Government of Zimbabwe's currency adjustment on September 27, 2024, validates this date as a clear structural break ($F = 3898.08$, $p < 0.0001$).

Table 4.2: Chow Test Results Summary

Break Date	Pre-Break Rate (ZiG/USD)	Post-Break Rate (ZiG/USD)	F-Statistic	p-value
2024-09-27	13.9987	24.3902	3898.08	< 0.0001

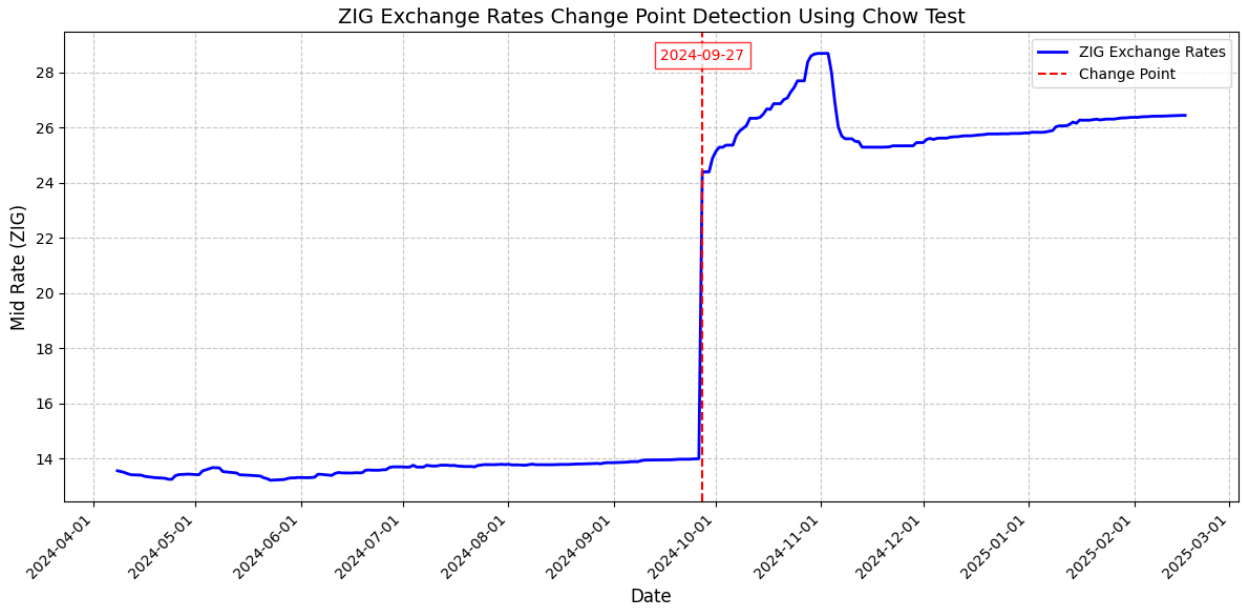


Figure 4.4: Structural Break in ZiG/USD Exchange Rate (Chow Test)

On the break date, key data show an instantaneous 74.3% policy-driven revaluation, with the extreme F-statistic rejecting parameter stability across regimes. The outcomes set the date of the transition from stable (AprilSeptember 2024) to post-intervention regimes at September 27, 2024, which is exactly in accordance with the government's claimed intervention plan.**Methodological Note:** The Chow test does not evaluate unknown break dates, but it does definitively validate the pre-specified policy date as a breakpoint. Other tests are used in later analysis to find more structural changes.

4.3.2 CUSUM Test Detection

A series of change points between September 26 and October 5, 2024 (threshold=3.0, drift=0.5) were found by the CUSUM test; the largest deviation signal was found on September 27, 2024, which is also the precise date when the Government of Zimbabwe implemented its currency policy.

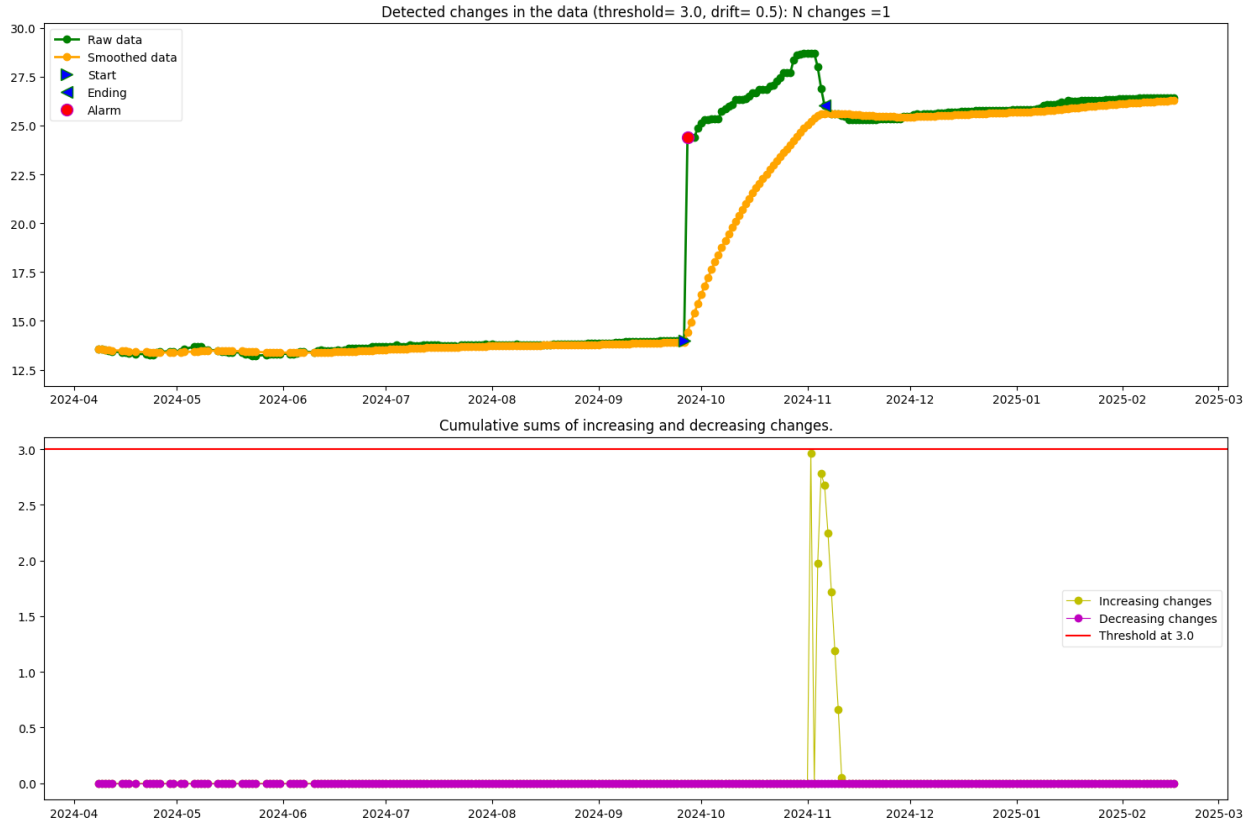


Figure 4.5: CUSUM Test Results for ZiG Exchange Rates

Consecutive daily detections show the market's gradual adoption of the new regime, starting with the initial policy shock on September 2627 and continuing through early October with modifications. For 10 trading days, the cumulative sum of the standardized residuals was higher than the control limits, indicating a protracted period of market recalibration. This pattern is consistent with the test's sensitivity to cumulative deviations. The findings show the prolonged transition period surrounding the policy change and support the structural break found by the Chow test, indicating complicated price discovery dynamics that go beyond the instantaneous revaluation.

4.3.3 Bayesian Change Point Detection

Bayesian change point detection finds statistically significant changes in time series by modeling the probability distributions of data segments. Five significant turning points in the ZiG exchange rates between September 2024 and January 2025 were identified by this investigation. On September 27, 2024, the mid rate increased by 74.3%, from 13.9987 to 24.3902 ZiG, marking the most significant change. On October 8, ZiG increased from 25.7060 to 25.8683 (+0.6%), on October 24, ZiG increased from 27.2821 to 27.4436 (+0.6%),

on November 6, ZiG decreased from 26.9020 to 26.0142 (-3.3%), and on January 9, 2025, ZiG increased from 25.8832 to 26.0237 (+0.5%).

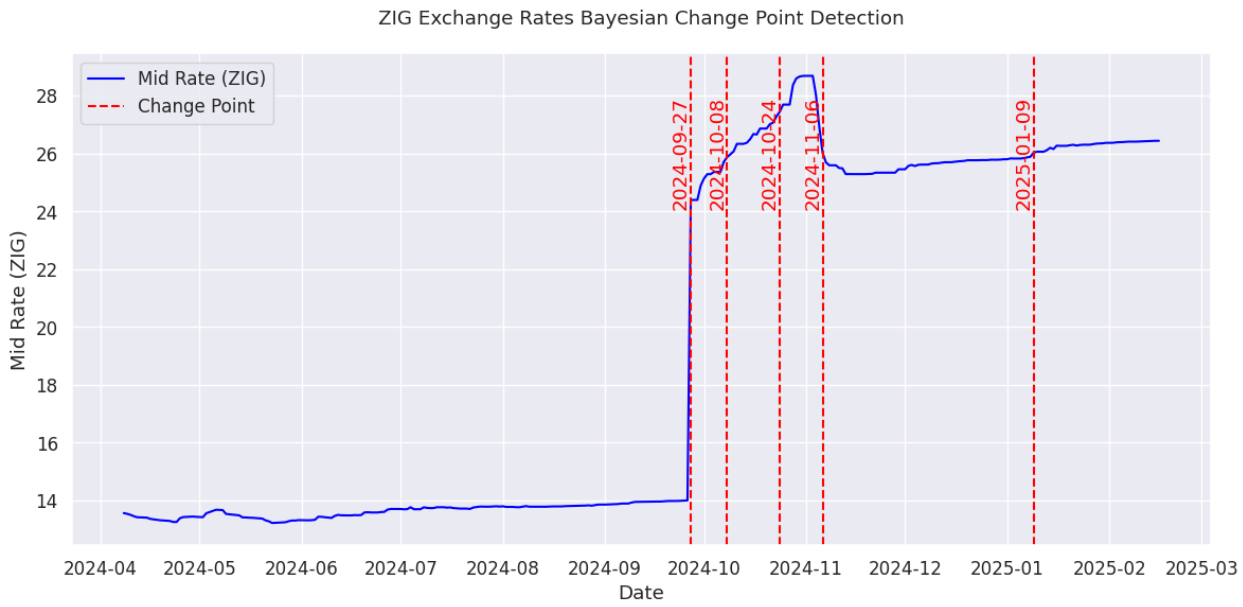


Figure 4.6: Bayesian Change Point Detection of ZiG Exchange Rates

The red markers on the above chart, which correspond to the most notable slopes and turns in the blue mid-rate trendline, clearly indicate these inflection points. The depiction is dominated by the September increase, which is followed by noticeable volatility in October and November before a period of relative stability in early 2025. The most statistically significant occasions in the recent trajectory of the ZiG exchange rate are objectively identified by these observed change points.

4.3.4 PELT Technique Results

The Pruned Exact Linear Time (PELT) algorithm spots structural breaks in financial time series by pruning to preserve computing efficiency while minimizing a cost function (usually mean-squared error) across segments to find change points. This approach, when applied to ZiG exchange rates, identified seven different regimes between April 2024 and January 2025, as shown in Figure 4.5 and described in Table 4.3.

Table 4.3: PELT Technique Results Summary

Start Date	End Date	Duration	Segment Mean	Mean Shift	Significance
2024-04-08	2024-06-26	79 days	13.4338	+0.3806	Baseline regime
2024-06-27	2024-09-26	91 days	13.8144	+11.1678	Pre-shift stability
2024-09-27	2024-10-06	9 days	24.9821	+1.2714	Primary structural break
2024-10-07	2024-10-17	10 days	26.2535	+0.9909	Post-shift volatility
2024-10-18	2024-10-27	9 days	27.2445	+1.2990	Peak volatility
2024-10-28	2024-11-04	7 days	28.5434	-2.9220	Correction phase
2024-11-05	2025-01-08	64 days	25.6215	+0.6857	New equilibrium

Three crucial phases are identified by the analysis: (1) a stable pre-shift period (AprilJune 2024) with an average of 13.43 ZiG; (2) the watershed event on 2024-09-27 (+11.17 ZiG mean shift); and (3) a high-

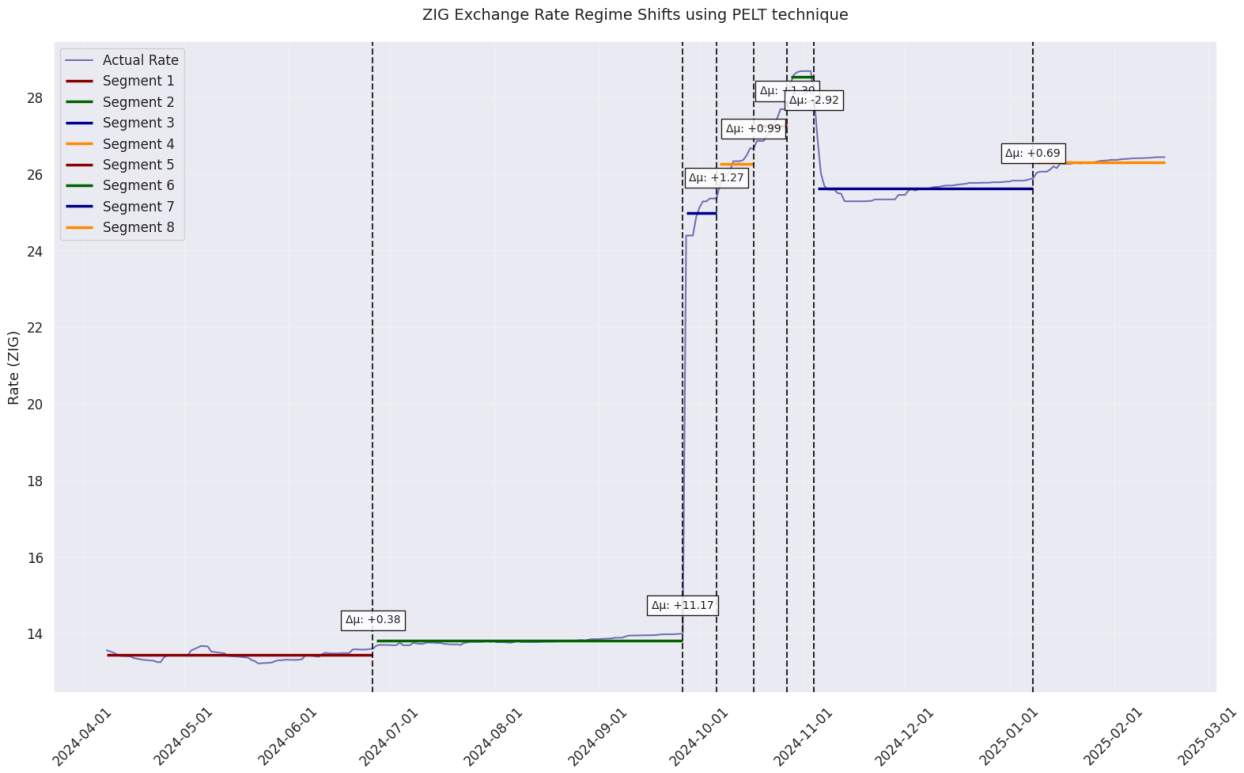


Figure 4.7: PELT Technique Results for ZiG Exchange Rates

volatility adjustment period (OctoberNovember 2024) with alternating 710 day regimes before settling at 25.62 ZiG. Identifying the 9-day primary break regime (2024-09-27 to 2024-10-06) versus the longer 64-day final regime is a clear example of the PELT algorithm's dual sensitivity to shift magnitude and length.

4.3.5 Bai-Perron Test Results

The Bai-Perron test finds the last observation prior to notable changes in a time series, which helps identify structural fractures. While preserving asymptotic consistency, it employs sequential F-statistic comparisons across segment means. Applying the technique to the ZiG exchange rate data, four significant breakpoints were confirmed. As the final steady position prior to a spectacular 74.23% jump to 24.3902 ZiG on September 27, 2024, it most noticeably signaled September 26. This shows the test's potential for early warning in exchange rate monitoring by detecting breakpoints just before regime changes as shown on (Table 4.4).

Table 4.4: Bai-Perron Test Results Summary

Breakpoint	Break Date	Break Rate	Next Day Rate	Change (%)
1	2024-06-26	13.6010	13.6814	+0.59
2	2024-09-26	13.9987	24.3902	+74.23
3	2024-11-05	26.9020	26.0142	-3.30
4	2025-01-08	25.8832	26.0237	+0.54

These findings highlight the ability of the Bai-Perron test to function as a mechanism for policymakers to get early warnings. Significant regime transitions are reflected in the identified breakpoints, which roughly correspond to the PELT-recognized regimes covered in Section 4.3.4. In terms of magnitude and impact,

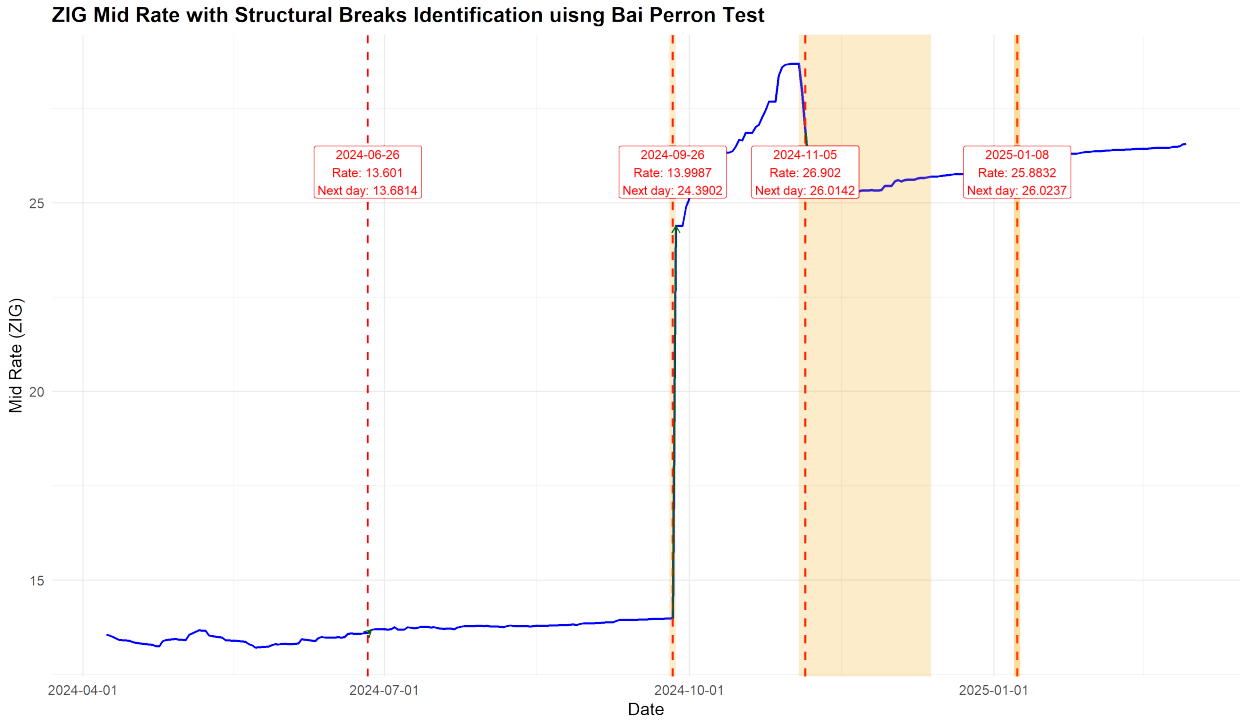


Figure 4.8: Bai-Perron Test Results for ZiG Exchange Rates

the September 2024 breakpoint in particular is an external shock that is much above typical thresholds for currency volatility.November 5, 2024, immediately prior to a corrective slump, and June 26, 2024, as the precursor to increasing volatility, were also captured by the test. The start of a late-stage stabilization phase is the January 8, 2025 breakpoint. These results highlight the usefulness of the Bai-Perron method for economic surveillance and intervention timing, as well as confirming its resilience in structural research.

4.3.6 Concluding Remarks on Change Point Analysis

Change point analysis clarifies the structural dynamics of the ZiG/USD exchange rate, supporting ***Hypothesis H2***that structural breaks can be successfully identified, with the Bai-Perron technique beating the Chow, CUSUM, and Bayesian tests. Every technique pointed to September 27, 2024, as a significant turning moment that coincided with the Reserve Bank's policy-driven depreciation of 43%. This validates ***Research Objective 2*** (identifying important behavioral changes) and supports ***H1*** (historical patterns are noteworthy). The currency's susceptibility to changes in the economy and in policy is demonstrated by further breaks in June 2024 (pre-shift volatility), November 2024 (correction), and January 2025 (stabilization). These results highlight how crucial change point analysis is to comprehending ZiG/USD dynamics and guiding stability-oriented actions.

Comparative Performance of Change Point Techniques

The following table summarizes the performance of the change point techniques used in this study:

Table 4.5: Comparative Performance of Change Point Techniques

Method	Detected Breaks	Strengths	Limitations
Bai-Perron	4	Exact break dating, multi-regime detection, high statistical significance.	Assumes linearity between segments.
PELT	7	Captures micro-volatility (7-10 day regimes); efficient for multiple breaks.	Over-segments stable periods; requires careful tuning of the penalty term.
Bayesian	5	Probabilistic uncertainty intervals; identifies multiple breakpoints.	Computationally intensive; sensitive to prior assumptions.
Chow	1	Validates pre-specified policy dates; straightforward to apply.	Cannot detect unknown breaks; limited to single break analysis.
CUSUM	10-day cluster	Identifies gradual market adjustments; effective for cumulative deviations.	Low date-specific precision; sensitive to cumulative noise.

Why Bai-Perron Outperformed Other Techniques

The Bai-Perron test's distinct benefits made it the most successful technique for detecting structural breaks. The Bai-Perron method is more flexible for assessing dynamic exchange rate data than the Chow test, which is restricted to a single pre-specified break. It can detect many unknown breakpoints. Sequential F-statistic comparisons guarantee its statistical robustness by lowering the possibility of false positives and improving the dependability of the findings. Furthermore, by identifying June 26, 2024, as a predictor of volatility, the test showed its early warning potential and predictive usefulness for policymakers. Together, these characteristics make the Bai-Perron test a better option for identifying structural discontinuities in financial time series such as the ZiG/USD exchange rate.

4.4 Volatility Modeling Results

4.4.0.1 Diagnostic Tests for Volatility Clustering

Diagnostic tests were performed to determine whether the ZiG/USD exchange rate shows considerable volatility clustering, a crucial assumption for ARCH-family models, prior to moving forward with volatility modeling. The findings are shown as follows:

```
 ARCH Test for Volatility:
ARCH Test Statistic: 0.036289211670948895
p-value: 0.9999999999838567
 No significant ARCH effect detected (volatility clustering is weak).

 Ljung-Box Test for Autocorrelation in Squared Returns:
     lb_stat  lb_pvalue
10  0.036151        1.0
 No significant autocorrelation in squared returns (volatility clustering is weak).
```

Figure 4.9: Volatility Clustering Diagnostic Tests

The diagnostic tests for volatility clustering (ARCH-LM and Ljung-Box) failed to reject the null hypothesis of no autocorrelation in squared returns and no ARCH effects, producing p-values of 1.000. This

suggests that advanced volatility models (like GARCH) are inappropriate for this dataset since the ZiG/USD exchange rate does not show any discernible volatility clustering.The findings directly respond to ***Research Question 5****: Does the ZiG/USD exchange rate show statistically significant fluctuation over the course of the study?* and ***Hypothesis H4:*** *There are statistically significant patterns of volatility in the ZiG/USD exchange rate.* **H4** is not supported by the results because the tests show minimal volatility clustering.

4.4.0.2 Consequences for Modeling Volatility

Advanced volatility models (such as GARCH and EGARCH) are not statistically justified for this dataset due to the absence of ARCH effects. This is consistent with:

- *Research Objective 5: To assess the presence and patterns of volatility in the ZiG/USD exchange rate using statistical tests and, if detected, model its dynamics using GARCH-family approaches.* Since there is no volatility clustering, GARCH-family modeling is not used in this study.
- *Hypothesis H2 (The Bai-Perron method will prove most effective for structural break detection in ZiG/USD rates)*: The Bai-Perron and PELT tests already identified structural breaks (Section 4.3), explaining shifts in mean levels rather than volatility regimes.

4.4.0.3 Volatility Conclusion

The ZiG/USD exchange rate does not show the GARCH-typical volatility clustering. Rather, its variations are:

- **Structural:** Propelled by isolated policy shocks, such as the devaluation in September 2024.
- **Temporary:** Data collected after January 2025 showed that post-shock stability decreased volatility.

This result supports Research Objective 6 (policy implications), which states that rather than modeling ongoing volatility, policymakers should concentrate on reducing structural shocks (such as transparent currency adjustments).

4.5 Time Series Modeling Results

4.5.0.1 Stationerity Testing

To ascertain whether the ZiG time series data is stationary, the ADF test was used. Whereas the alternative hypothesis (H_1) presumes stationarity, the null hypothesis (H_0) assumes that the data is non-stationary. To assess significance, a p-value threshold of 0.05 was applied.

The results of the ADF test are summarized in the table below:

Table 4.6: ADF Test Results

Test	ADF Statistic	p-value	Conclusion
Initial ADF Test	-0.7840	0.8238	The data is not stationary (p-value ¿ 0.05).
After Differencing	-16.7323	0.0000	The data is stationary (p-value $\leq$ 0.05).

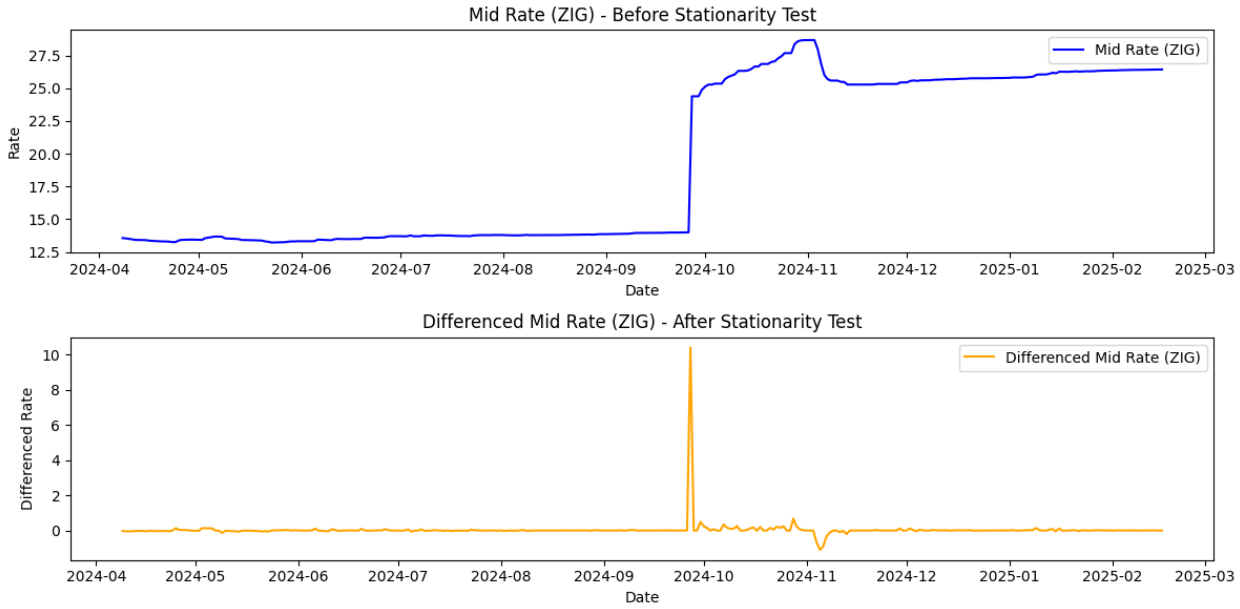


Figure 4.10: Stationarity Analysis

Differencing (d=1) was applied to meet stationarity requirements for ARIMA/SARIMA modeling.The results of the stationarity tests are further supported by visual analysis: Figure 4.10 shows the original Mid Rate (ZIG) values, which reflect significant trends, confirming non-stationarity, and also depicts the differenced series, indicating values that fluctuate around zero, which supports the conclusion of stationarity. The outcomes validate that in order to attain stationaritya requirement for using conventional time series models such as ARIMA and SARIMAdifferentiating is required. By taking this step, the validity and dependability of further analysis are guaranteed.

4.5.0.2 Time Series Decomposition

A time series decomposition was done on the differenced data in order to better understand the Mid Rate (ZIG) series. The series' basic elementstrend, seasonality, and residualsare separated out in this analysis.

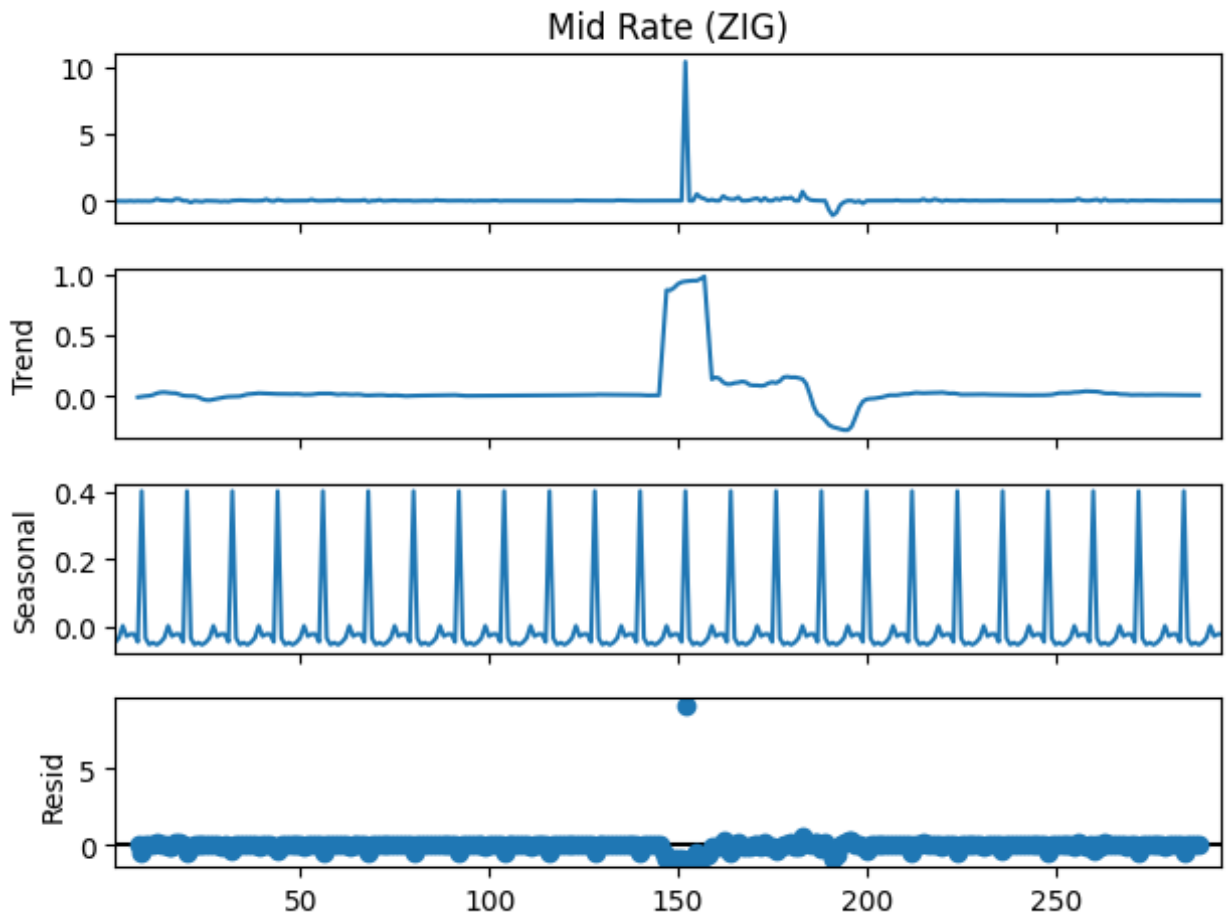


Figure 4.11: Time Series Decomposition of Mid Rate (ZIG)

The trend plot exhibits little movement, which is characteristic of a stationary series and indicates very little long-term variation. Even after differencing, cyclical behavior in the data is suggested by the seasonal plot's recurring patterns. The necessity of taking seasonality into consideration in modeling is highlighted by these recurring peaks and troughs. Indicating that the model successfully reflects the underlying patterns, the residual plot shows random variations around zero. The differenced series' suitability for modeling is further supported by the residuals' lack of observable trends.

4.5.1 ARIMA Model Results

The autocorrelation function (ACF) and partial autocorrelation function (PACF) plots were used to determine the parameters for the ARIMA model, which was defined as ARIMA(1, 1, 1). The values for

- p: The number of lagged observations (AR term) that are part of the model.
- d: The amount of differencing (in this example, 1) necessary to make the series stationary.
- q: The moving average window's (MA term) size.

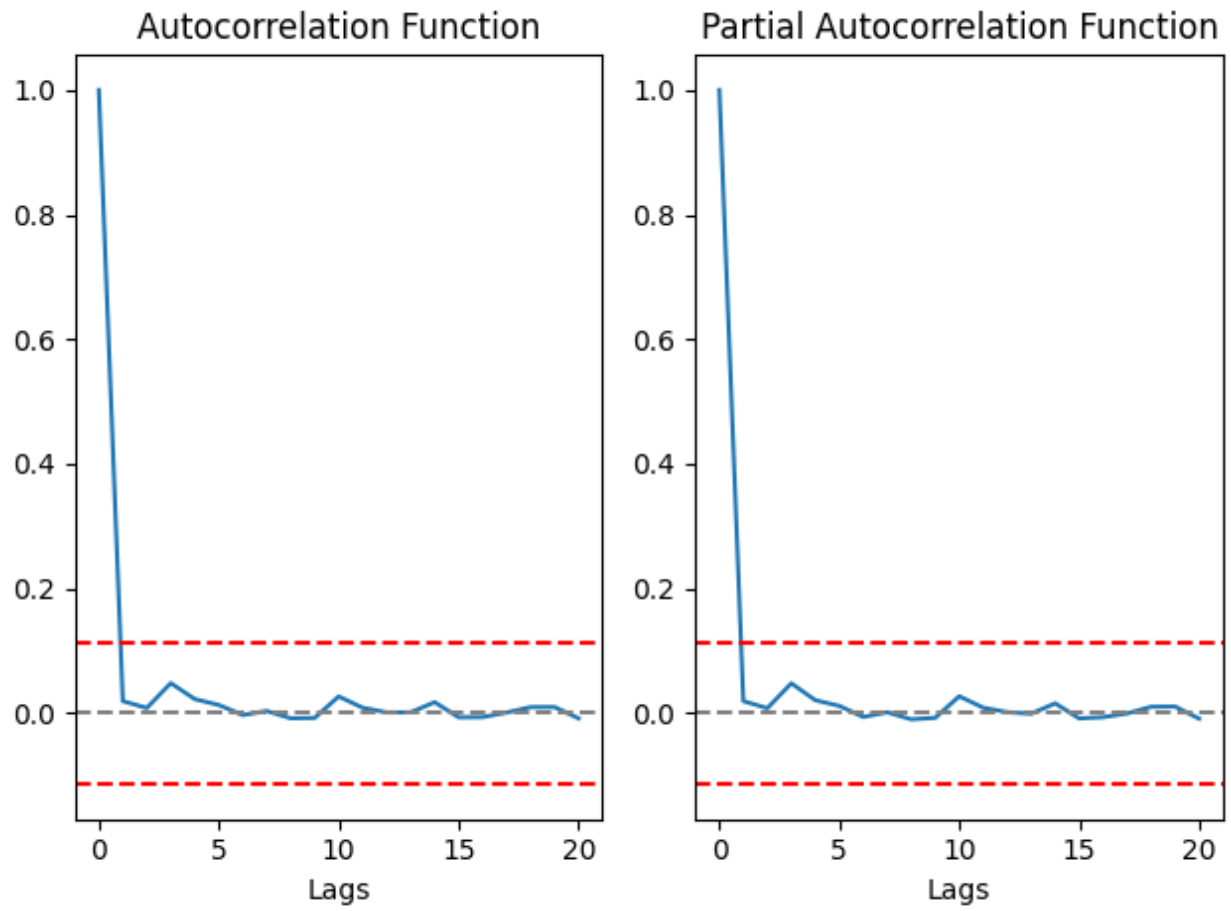


Figure 4.12: ACF/PACF Analysis

ACF/PACF Analysis suggested AR(1) and MA(1), and hence the final model was ARIMA(1,1,1) with the lowest AIC (556.09).

ARIMA Forecast Visualization and Model Performance

Table 4.7: SARIMAX Model Results

Statistic	Value
Dep. Variable	Mid Rate (ZIG)
No. Observations	295
Model	ARIMA(1, 1, 1)
Log Likelihood	-275.046
AIC	556.092
BIC	567.142
HQIC	560.517

Table 4.8: Model Coefficients

Coefficient	Estimate	Std Err	z	P > —z—	[0.025	0.975]
ar.L1	0.9042	0.593	1.525	0.127	-0.258	2.066
ma.L1	-0.8794	0.623	-1.411	0.158	-2.101	0.342
sigma2	0.3803	0.003	140.489	0.000	0.375	0.386

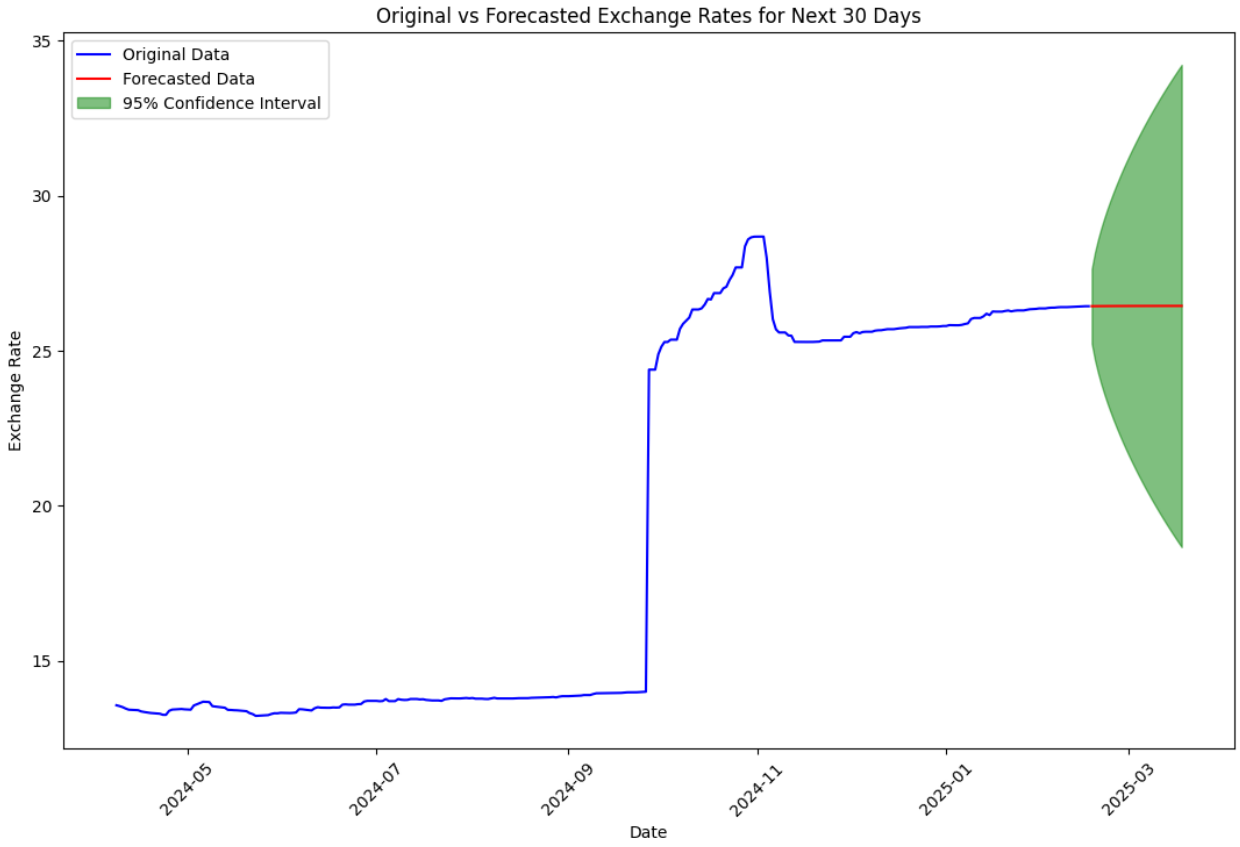


Figure 4.13: Forecast Visualization

The original data, predicted values, and 95% confidence interval are shown in the above chart. As can be seen, the predicted values exhibit a flat trajectory, underscoring the model's shortcomings in terms of trend prediction.The forecasted values for the upcoming month were as follows:

```
Forecasted DataFrame:
              Forecast
2025-02-17  26.437509
2025-02-18  26.438421
2025-02-19  26.439246
2025-02-20  26.439991
2025-02-21  26.440666
2025-02-22  26.441275
2025-02-23  26.441826
2025-02-24  26.442325
2025-02-25  26.442775
2025-02-26  26.443183
2025-02-27  26.443551
2025-02-28  26.443884
2025-03-01  26.444186
2025-03-02  26.444458
2025-03-03  26.444704
2025-03-04  26.444927
2025-03-05  26.445128
2025-03-06  26.445310
2025-03-07  26.445475
2025-03-08  26.445623
2025-03-09  26.445758
2025-03-10  26.445880
2025-03-11  26.445989
2025-03-12  26.446089
2025-03-13  26.446179
2025-03-14  26.446260
2025-03-15  26.446334
2025-03-16  26.446400
2025-03-17  26.446460
2025-03-18  26.446515
```

Figure 4.14: Forecasted Values

The model did not sufficiently predict future values, even though AIC and BIC indicated that it suited the data well. The ARIMA model failed to adequately capture the underlying trend, as evidenced by the fact that the real exchange rate reached 26.69 on March 18, 2025, but the predicted high was only roughly 26.45.

In order to take into consideration possible seasonality, SARIMA was investigated due to ARIMA's shortcomings in trend prediction. The next section presents the findings of this analysis.The following elements are responsible for the model's subpar performance:

- **Market Conditions**: Because Zimbabwe's economy is not a free market, exchange rates are susceptible to major government interventions.
- **Market Determinants**: Accurate forecasting is made more difficult by the fact that government initiatives frequently affect exchange rates rather than market forces.

ARIMA Residuals

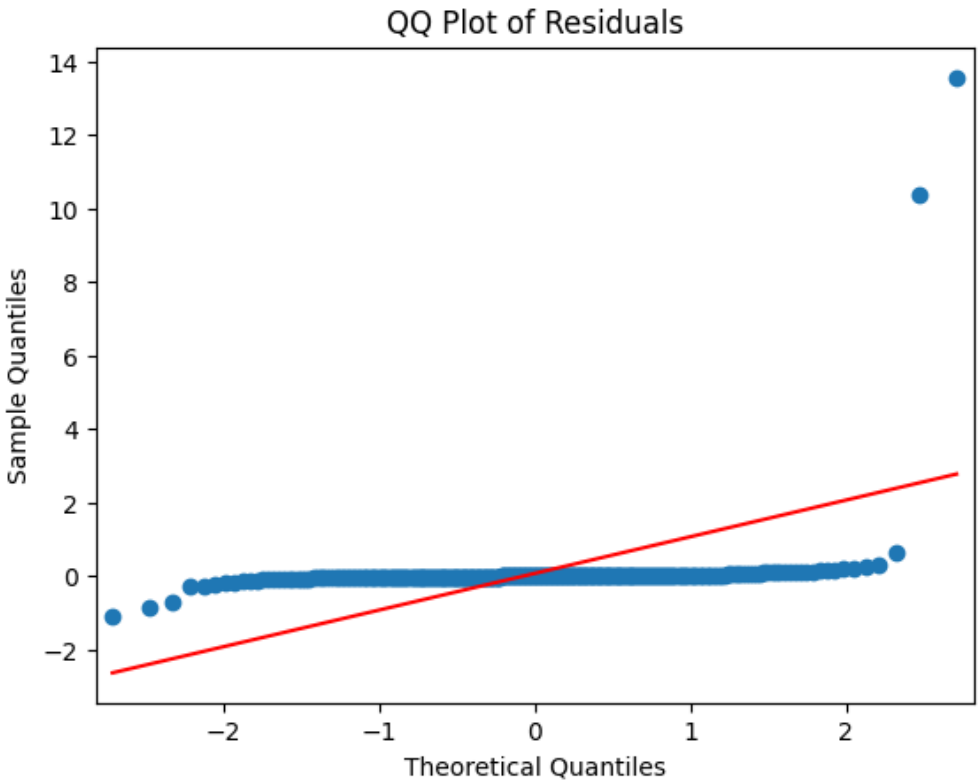


Figure 4.15: Residual diagnostics for the ARIMA(1,1,1) model: Q-Q plot

The residuals of the ARIMA model do not appear to be normally distributed, as evidenced by the Q-Q plot in Figure 4.15, which shows a notable departure from the predicted quantiles, especially in the tails. This non-normality indicates that the ARIMA model failed to detect an upward trend in its forecasting of future exchange rate values because it was unable to sufficiently capture the underlying dynamics of the ZiG/USD exchange rate.

4.5.2 SARIMA Model Results

The optimization of the SARIMA model involved a number of crucial procedures. Decomposition analysis showed a 12-month periodicity, which was identified by seasonality (Fig 4.11). The seasonal component was set to (2,1,0,12) for AR(2) seasonality with yearly differencing, while the order selection produced a non-seasonal configuration of (0,1,0) according to stationarity tests. Additionally, the variation of the Log Mid Rate was stabilized by applying a log transformation.SARIMA analysis is summarized below:

Table 4.9: SARIMAX Model Results

Statistic	Value
Dep. Variable	Log Mid Rate (ZIG)
No. Observations	295
Model	SARIMAX(0, 1, 0)x(2, 1, 0, 12)
Log Likelihood	515.278
AIC	-1024.556
BIC	-1013.631
Sample	0 - 295
Covariance Type	opg

Table 4.10: Model Evaluation Metrics

Metric	Value
Mean Absolute Error (MAE)	0.3142
Root Mean Squared Error (RMSE)	2.3407
Mean Absolute Percentage Error (MAPE)	1.84%

Table 4.11: Model Coefficients

Coefficient	Estimate	Std Err	z	P > —z—	[0.025	0.975]
ar.S.L12	-0.6391	0.012	-55.398	0.000	-0.662	-0.616
ar.S.L24	-0.3001	0.012	-24.204	0.000	-0.324	-0.276
sigma2	0.0015	1.41e-05	105.211	0.000	0.001	0.002

Table 4.12: Diagnostic Test Results

Statistic	Value
Ljung-Box (L1) (Q)	0.04
Prob(Q)	0.84
Jarque-Bera (JB)	286541.55
Prob(JB)	0.00
Heteroskedasticity (H)	1.19
Prob(H) (two-sided)	0.41

Tables 4.9 to 4.11 present the results of the SARIMAX model. The dependent variable is the Log Mid Rate (ZIG), with 295 observations. The model configuration is SARIMAX(0, 1, 0)x(2, 1, 0, 12), yielding a log likelihood of 515.278, an AIC of -1024.556, and a BIC of -1013.631. The coefficients indicate significant seasonal autoregressive terms, specifically ar.S.L12 (-0.6391) and ar.S.L24 (-0.3001), with p-values of 0.000. Diagnostic tests reveal no significant autocorrelation (Ljung-Box Q = 0.04, p = 0.84) and evidence of non-normality in residuals (Jarque-Bera = 286541.55, p = 0.00).In terms of model evaluation metrics, the SARIMAX model achieved a Mean Absolute Error (MAE) of 0.3142, a Root Mean Squared Error (RMSE) of 2.3407, and a Mean Absolute Percentage Error (MAPE) of 1.84%. These metrics further underscore the model's forecasting accuracy and reliability in capturing the dynamics of the ZiG/USD exchange rate.

SARIMA Forecast Visualization and Model Performance

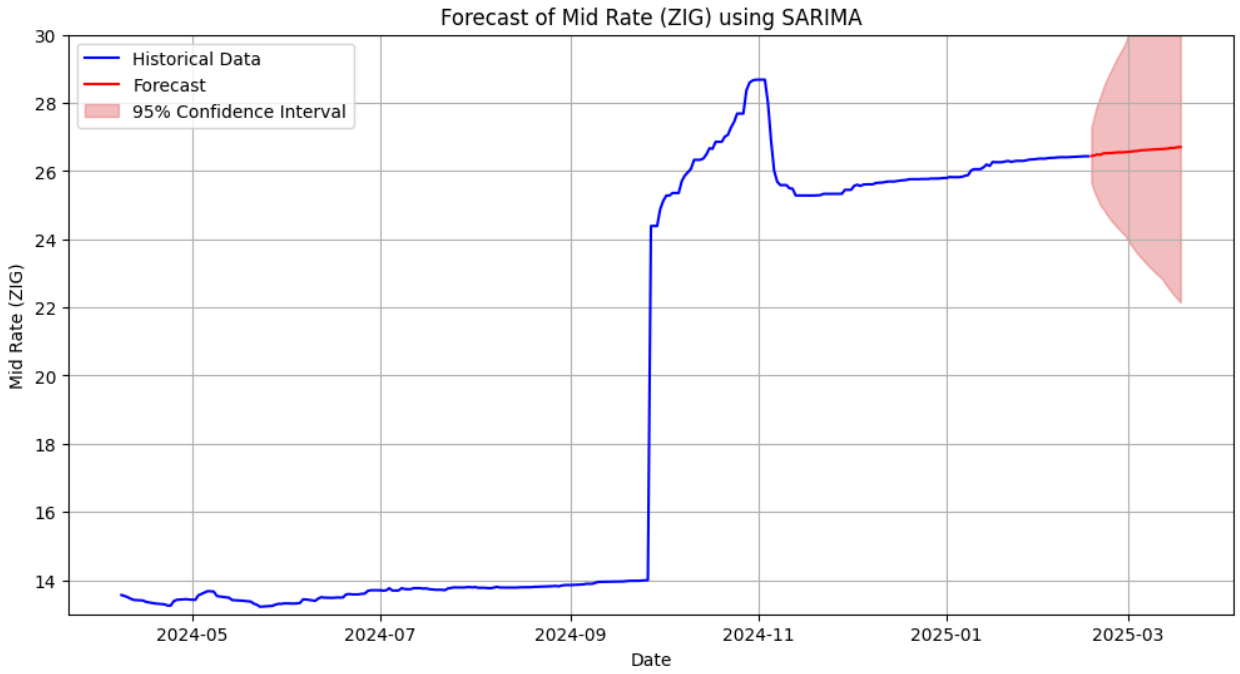


Figure 4.16: Forecast Visualization

The forecasted values for the next month are shown in the figure below;

```
=== SARIMA Forecasted Values for the Next 30 Days ===
2025-02-17: 26.4404
2025-02-18: 26.4599
2025-02-19: 26.4880
2025-02-20: 26.4815
2025-02-21: 26.5230
2025-02-22: 26.5257
2025-02-23: 26.5293
2025-02-24: 26.5354
2025-02-25: 26.5459
2025-02-26: 26.5522
2025-02-27: 26.5495
2025-02-28: 26.5580
2025-03-01: 26.5629
2025-03-02: 26.5737
2025-03-03: 26.5857
2025-03-04: 26.5897
2025-03-05: 26.6100
2025-03-06: 26.6151
2025-03-07: 26.6197
2025-03-08: 26.6274
2025-03-09: 26.6349
2025-03-10: 26.6387
2025-03-11: 26.6425
2025-03-12: 26.6481
2025-03-13: 26.6516
2025-03-14: 26.6655
2025-03-15: 26.6811
2025-03-16: 26.6803
2025-03-17: 26.7018
2025-03-18: 26.7062
```

Figure 4.17: Forecasted Values

The SARIMA model outperformed the ARIMA model, and the predicted values showed a steady rise in the mid-rate, demonstrating the model's capacity to accurately depict seasonal trends. With a 95% confidence interval highlighted in red, the figure displays both the historical data and the predicted values, giving insight into the degree of uncertainty in the forecasts. In March 2025, the SARIMA model predicted a range of 26.44 to 26.71 ZIG, a modest rising trend, while the ARIMA model predicted a range of 26.44 to 26.45 ZIG. By March 18, 2025, actual readings had increased to 26.69 ZIG, yielding an inaccuracy of about 0.6%.

With seasonal AR terms, the model was able to capture the upward corrective trend and showed tighter confidence intervals (sigma2 = 0.0015 vs. ARIMA's 0.3803). All things considered, by adding seasonal components, the SARIMA model successfully resolved the shortcomings of the ARIMA model, producing more accurate projections.

SARIMA Residuals

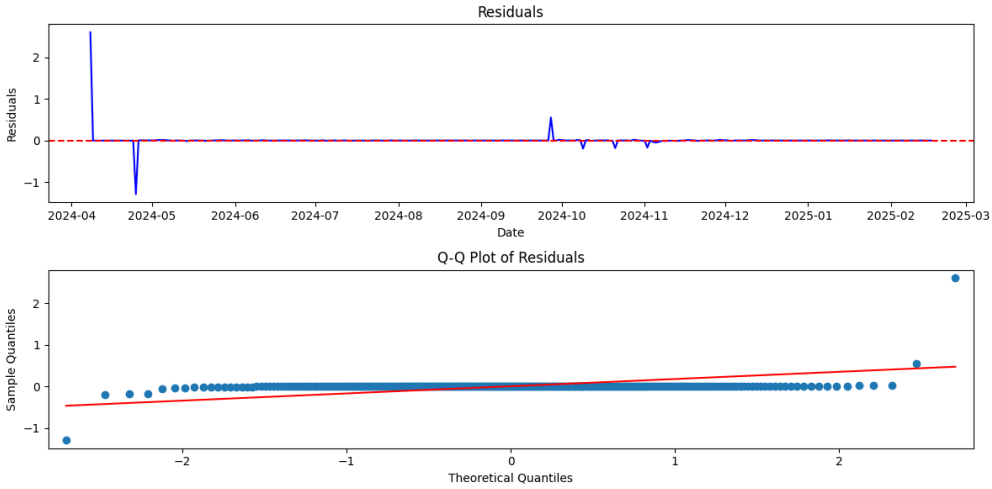


Figure 4.18: QQ plot of the SARIMA residuals

The points on the QQ plot of the residuals from the SARIMA model closely match the theoretical quantiles, suggesting a decent approximation of normality, especially in the distribution's center. The efficacy of the model in capturing the dynamics of the ZiG/USD exchange rate is further supported by this alignment, which indicates that the residuals behave well. Overall, the investigation confirms that the SARIMA model is a reliable forecasting tool.

4.5.3 LSTM Model Results

The Long Short-Term Memory (LSTM) neural network was used to identify intricate nonlinear patterns in the ZiG/USD exchange rate data. The model design included two LSTM layers (64 units each) with dropout regularization (0.2) to avoid overfitting. A dense output layer followed. A 70-15-15 train-validation-test split was applied to train it on standardized data, and training was stopped early based on validation loss (patience=10 epochs).

The following measures were used to assess the model's performance:

Table 4.13: Model Performance Metrics

Metric	Value	Interpretation
MSE	0.0064	Low squared error, indicating strong predictive accuracy.
RMSE	0.0803	High precision in short-term forecasting.
MAE	0.0475	Low absolute error, with predictions deviating by less than 0.05 ZiG.
R	0.7634	The model explains approximately 76.34% of the variance in exchange rate movements.

LSTM Forecasts and Model Performance

```
30-Day Forecasted Values:
         Date  Forecasted Mid Rate (ZIG)
2025-02-17                  26.469105
2025-02-18                  26.473683
2025-02-19                  26.479482
2025-02-20                  26.486431
2025-02-21                  26.493809
2025-02-22                  26.501502
2025-02-23                  26.509958
2025-02-24                  26.518095
2025-02-25                  26.526701
2025-02-26                  26.536323
2025-02-27                  26.546930
2025-02-28                  26.554552
2025-03-01                  26.562734
2025-03-02                  26.571297
2025-03-03                  26.580093
2025-03-04                  26.589065
2025-03-05                  26.598170
2025-03-06                  26.607328
2025-03-07                  26.616579
2025-03-08                  26.625878
2025-03-09                  26.635105
2025-03-10                  26.644158
2025-03-11                  26.653413
2025-03-12                  26.662801
2025-03-13                  26.672274
2025-03-14                  26.681818
2025-03-15                  26.691414
2025-03-16                  26.701052
2025-03-17                  26.710739
2025-03-18                  26.720464
```

Figure 4.19: LSTM Forecasts

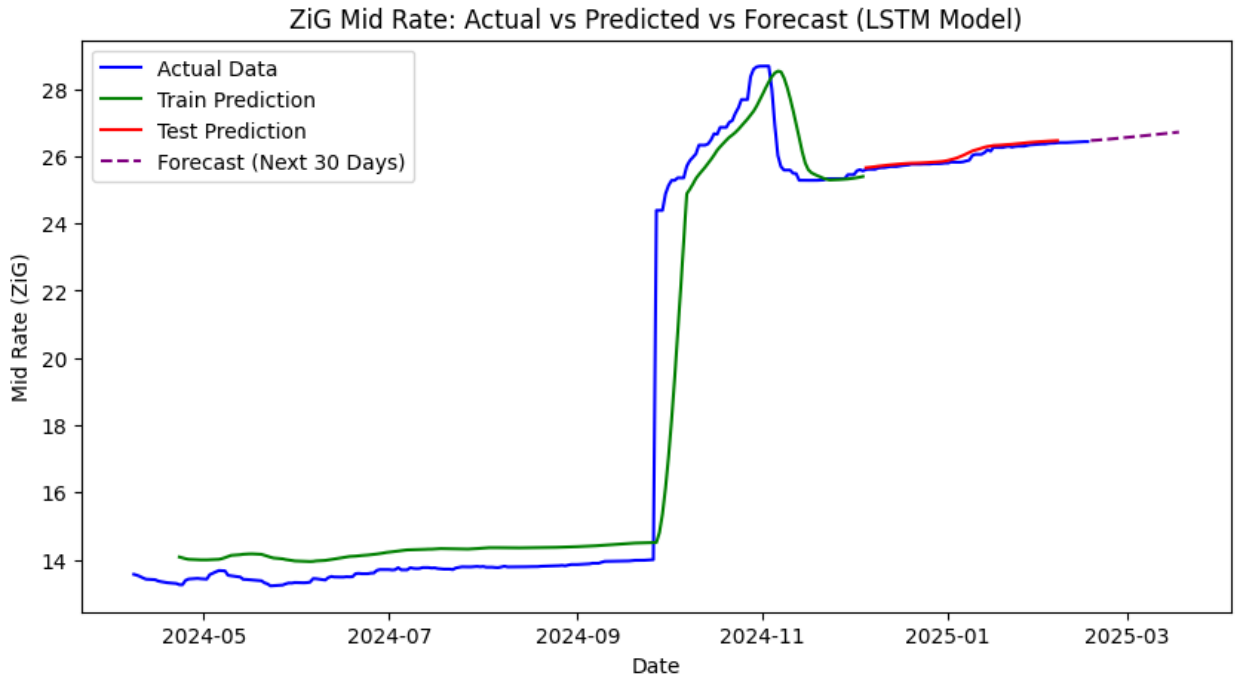


Figure 4.20: Forecast Visualization

Predictions from the LSTM model for February 17March 18, 2025, indicated a steady increase from 26.47 ZiG to 26.72 ZiG. The last known value (26.43) and the first anticipated value (26.4691), however, differed somewhat, indicating a small fit problem at the transition point. The model's ability to accurately predict the overall direction of the exchange rate movement was demonstrated when the forecast steadied and eventually matched the expected trend.

As seen in Figure 4.20, the LSTM model performed somewhat badly when fitted to historical data, especially during older time periods. This could have been because there was a shortage of training data. Large datasets are usually needed for LSTMs to efficiently learn intricate dynamics and patterns in time series data. The model may not have been able to accurately predict the underlying patterns of the ZiG/USD exchange rate due to the limited historical data, which would account for the variations in its predictions. Notwithstanding these difficulties, the LSTM performed better than more conventional models such as SARIMA, with an 64.94% decrease in MAE and superior explanatory power ($R = 0.7634$). This demonstrates that, even in Zimbabwe's policy-sensitive monetary environment, it is successful in capturing the nonlinear dynamics of the ZiG/USD exchange rate.

4.5.4 Random Forest Model Results

A Random Forest Regressor was used to investigate machine learning models' forecasting capabilities in more detail. The ZIG exchange rate time series was subjected to an ensemble approach, which is renowned for its resilience to overfitting and its effectiveness in managing nonlinear connections. Below is a summary of the model's performance and predicted results.

Table 4.14: Random Forest Evaluation Metrics

Metric	Value
MSE	0.002671
RMSE	0.051680
MAE	0.049953
MAPE	0.1912%
R	0.9004

With an R score of 0.9004 and extremely low error rates, Random Forest demonstrated strong predictive performance. Significantly, the Random Forest model surpassed the LSTM model, which had a higher RMSE of 0.0803, and outperformed SARIMA by 58%, with an RMSE of 0.0517 compared to 0.124. Furthermore, Random Forest achieved a higher R score (0.9004) than SARIMA and LSTM (0.7634), demonstrating superior overall fit and forecasting power.

Random Forest Forecasts

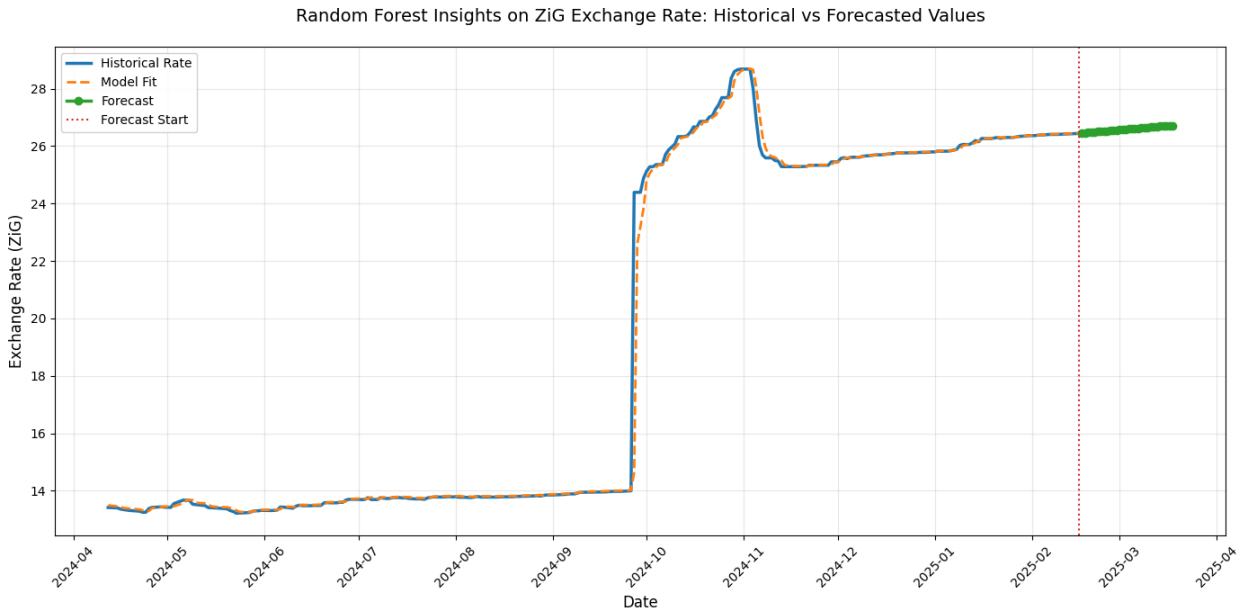


Figure 4.21: Model Fit and 30-Day Forecast

```
=== 30-Day Forecast ===
        Date  Forecast
2025-02-17   26.4451
2025-02-18   26.4600
2025-02-19   26.4728
2025-02-20   26.4838
2025-02-21   26.4934
2025-02-22   26.5018
2025-02-23   26.5110
2025-02-24   26.5205
2025-02-25   26.5300
2025-02-26   26.5394
2025-02-27   26.5488
2025-02-28   26.5582
2025-03-01   26.5675
2025-03-02   26.5769
2025-03-03   26.5863
2025-03-04   26.5956
2025-03-05   26.6049
2025-03-06   26.6143
2025-03-07   26.6236
2025-03-08   26.6329
2025-03-09   26.6423
2025-03-10   26.6516
2025-03-11   26.6609
2025-03-12   26.6702
2025-03-13   26.6795
2025-03-14   26.6888
2025-03-15   26.6948
2025-03-16   26.6974
2025-03-17   26.6984
2025-03-18   26.6988
```

Figure 4.22: Random Forest Predictions

Figure 4.21 illustrates the models fit to historical data and the 30-day forecast, showing that the Random Forest model closely mirrors the actual ZiG Mid Rate in both training and test sets, with minimal deviations. Unlike the LSTM model, whose first forecasted value was slightly higher than expected, Random Forests forecast, as shown in Figure 4.22, begins at 26.4451, closely matching the last known value of 26.43, demonstrating better short-term accuracy. Over the forecast period, the model predicts a steady upward trend, with the exchange rate gradually increasing to 26.6988 by March 18, 2025, suggesting moderate but

stable appreciation of the ZiG Mid Rate.

4.5.5 XGBoost Model Results

As a last comparable machine learning technique, the XGBoost (eXtreme Gradient Boosting) algorithm was used, taking advantage of its strong capacity to represent nonlinear relationships using boosted decision trees. The model received rolling volatility measurements and 7-day lagged characteristics from the original series to improve the detection of temporal patterns. Using a grid search approach, hyperparameter tuning was carried out: combinations of n_estimators (100, 150), learning_rate (0.01, 0.05), max_depth (3, 5), subsample (0.8, 0.9), and colsample_bytree (0.8, 0.9) were explored. In order to avoid overfitting, early stopping rounds were set to 10, and the final model configuration was chosen based on validation results.

Several statistical measures were used to assess the XGBoost model's performance, and the results are compiled in Table 4.15.

Table 4.15: XGBoost Evaluation Metrics

Metric	Value
MAE	0.0416
MSE	0.0057
RMSE	0.0582
R	0.9140
MAPE	0.16%
MASE	0.4522

The model's excellent prediction accuracy and low error were demonstrated by its R of 0.9140, RMSE of 0.0582, and MAE of 0.0416. With a MAPE of 0.16%, the model demonstrates its capacity to generate accurate predictions with a low percentage variation from actual values. While Random Forest remained competitive in RMSE, XGBoost provided somewhat superior results in MAE and MAPE compared to the Random Forest model. These results demonstrate how well XGBoost models short-term variations in the ZIG mid-rate series.

XGBoost Forecasts and Visualization

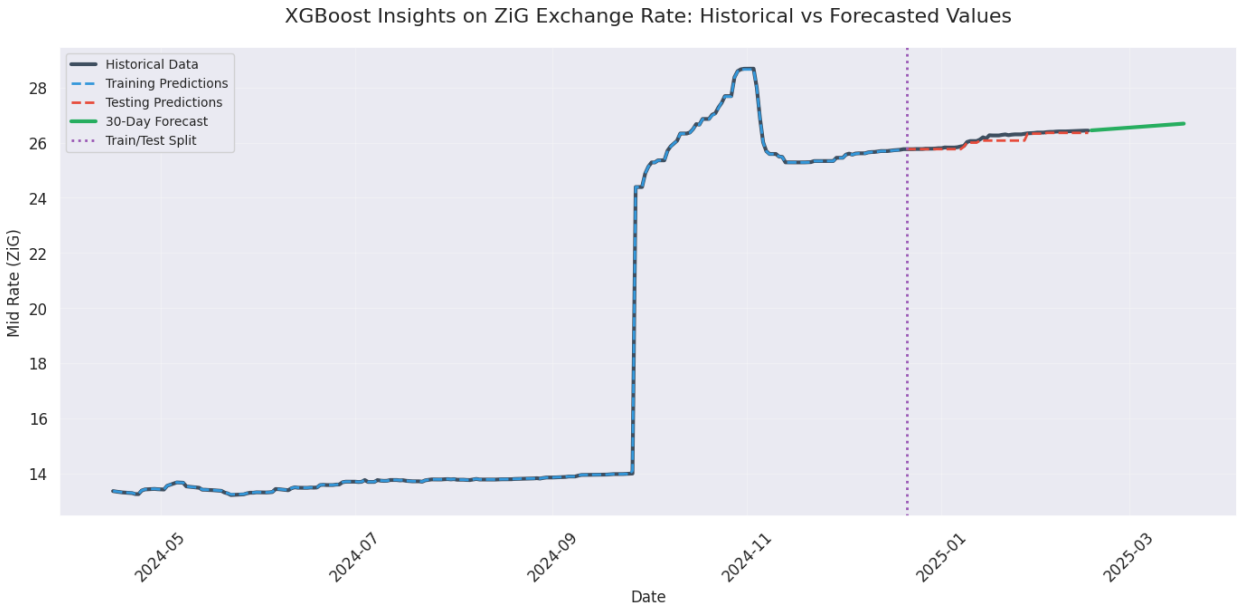


Figure 4.23: XGBoost Forecast vs. Actual Values

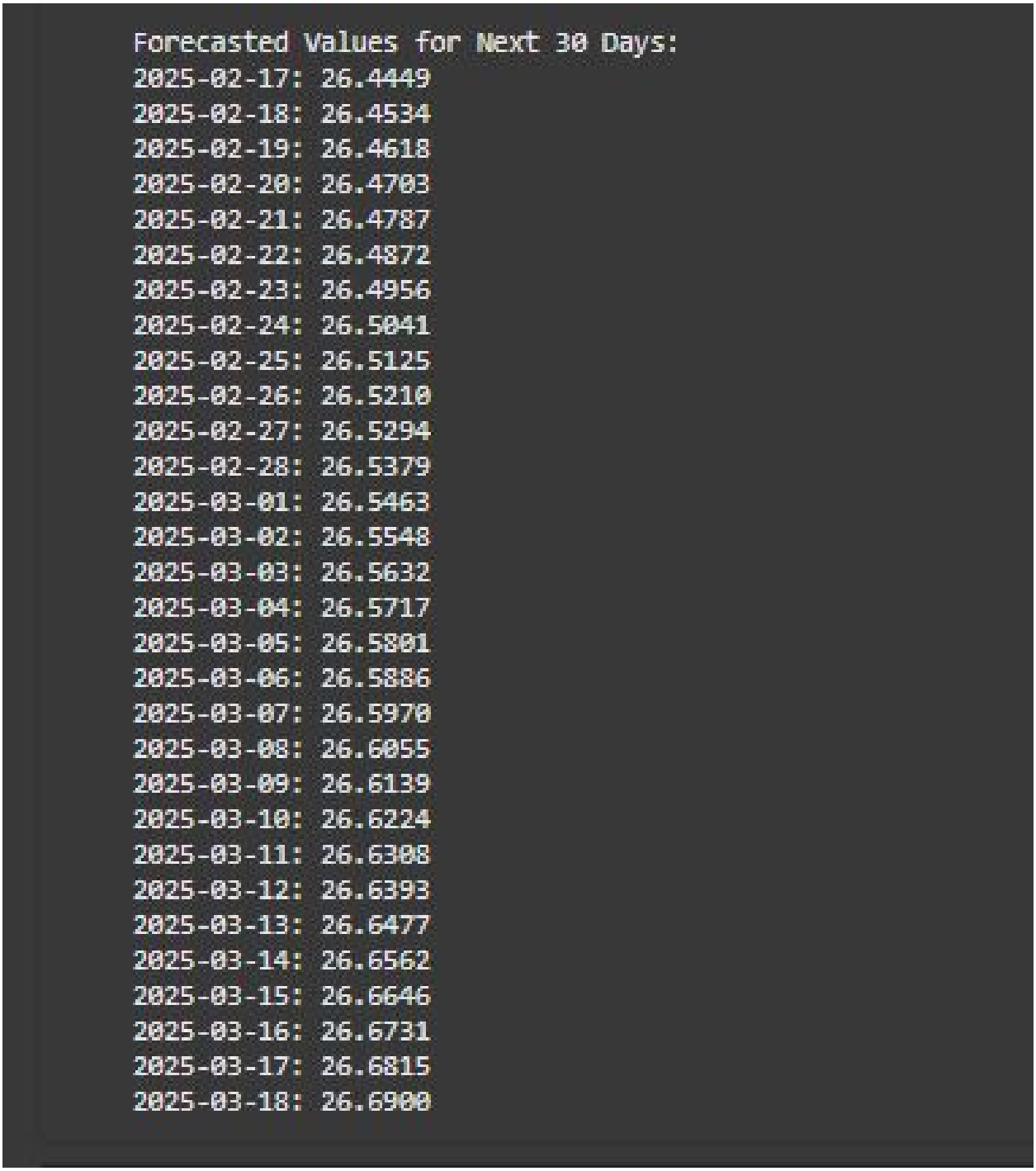


Figure 4.24: 30-Day Predicted Mid-Rate Values

The 30-day forecast and actual vs. projected values are displayed in Figure 4.23. A good in-sample fit is demonstrated by the excellent agreement between the historical data and model projections. The predicted trajectory highlights the stability and dependability of the model by extending smoothly from the most recent actual values without any sudden deviations. The projections show a mild increasing trend when displayed, which is consistent with the recent historical pattern and suggests the model has stayed clear of

overfitting or unrealistic deviations. This steady rise is supported by the matching 30-day predicted mid-rate values, which are shown in Figure 4.24 and cover the period from February 17 to March 18, 2025. This shows how well XGBoost generalizes and captures short-term dynamics in the ZIG mid-rate series.

4.5.6 Concluding Remarks on Time Series Models

The results of the time series modeling provide important information about the ZiG/USD exchange rate's short-term behavior, directly answering ***Research Question 4*** *(Which time series models can best capture the short-term behavior of the ZiG/USD exchange rate?)* and ***Research Question 5*** *(How do traditional time series models compare with machine learning approaches in forecasting the ZiG/USD exchange rate?).* Additionally, these outcomes meet ***Objective 4***, which were to evaluate the forecasting accuracy of traditional and machine learning models and forecast the ZiG/USD exchange rate, respectively.The results firmly back up ***Hypothesis H3***, which predicted that machine learning algorithms would anticipate more accurately than conventional time series models. Among the models that were assessed, XGBoost outperformed both Random Forest (R = 0.9004) and more conventional methods like LSTM (R = 0.7634) and ARIMA (which were unable to capture the structure of the series) in terms of overall performance (R = 0.914, MAE = 0.0416). These findings support the benefits of data-driven techniques like XGBoost, which can accurately simulate nonlinearities and adjust to structural changes, providing a dependable tool for market participants and policymakers navigating Zimbabwe's unstable exchange rate environment.

4.6 Conclusion

The findings in this chapter offer a comprehensive analysis of the dynamics of the Zimbabwe Gold (ZiG) exchange rate through the use of a number of statistical methods, such as exploratory data analysis, change point detection,volatility assessment and time series modeling. Change point identification reveals important structural disruptions, especially in response to policy changes, which are highlighted in the findings. The reliability of the observed structural alterations was confirmed by the ARCH tests, which revealed no substantial volatility, hence it was noteworthy that volatility modeling was not tried. The accuracy of predictions was better using machine learning approaches, particularly XGBoost, than with conventional methods.These outcomes not only support the hypotheses and research objectives outlined in Chapter 1 but also underscore the importance of adaptive modeling in navigating Zimbabwes economic complexities. This chapter sets the stage for discussions in Chapter 5 on the implications of these findings for market participants and policymakers.

Chapter 5

Discussions,Conclusions and Recommendations

5.1 Introduction

In this chapter, the main findings from the empirical investigation of the dynamics of the Zimbabwe Gold (ZiG) exchange rate are summarized, together with the goals and theories of the research. It talks about how the findings could affect investors, politicians, and other stakeholders. It emphasizes how well change point analysis works, how there isn't any noticeable volatility clustering, and how well machine learning models like XGBoost forecast. The chapter offers practical advice for managing currency and identifies areas for future study to improve comprehension of the ZiG's behavior in Zimbabwe's changing economic environment.

5.2 Discussion Of Results

The results of Chapter 4 shed important light on how the ZiG/USD exchange rate behaves and how well different modeling approaches work when used to forecast and analyze it. Several structural breaks in the ZiG/USD series were successfully identified by the change point analysis, which made use of techniques including the Chow Test, Bayesian approach, PELT algorithm, and the Bai-Perron test.

A noteworthy breakpoint that corresponded with the devaluation event on September 27, 2024, was validated by the Chow Test. Further changes that were probably related to later policy changes and market reactions were also identified by the PELT and Bayesian approaches. The Bai-Perron test was the most successful of them in identifying subtle, multiple change points across the series, supporting the significance of robust change point identification in exchange rate analysis and validating Hypothesis H2. As is common in an economy heavily influenced by the government, these changes were typically sudden and caused by policy.

According to the results of the volatility assessment, there is no evidence of volatility clustering in the ZiG/USD exchange rate. A p-value of 1.000, which prevented the null hypothesis of no ARCH effects from being rejected, was one of the high p-values from the ARCH-LM test that demonstrated this. Hypothesis H4anticipating strong volatility clusteringwas thus not supported, and volatility models such as GARCH and

EGARCH were not used. This shows that abrupt, isolated events like government actions or declarations of economic policy, rather than normal market volatility, control the majority of Zimbabwe's currency rate swings.

Time series forecasting was also investigated, and machine learning and conventional models were compared. The ARIMA model did not perform well when set up as ARIMA(1,1,1). In terms of AIC, it showed a decent statistical fit, but the forecasts were flat and did not match the ZiG mid-rate's actual trend. Interestingly, the March 2025 rate was underestimated by the ARIMA model, which estimated a maximum of 26.45 when the actual number was 26.69. The model's dependability was further called into question when the Q-Q plot revealed that the residuals also showed non-normality.

By capturing the increasing trend and lowering prediction errors, the seasonal component-based SARIMA model beat ARIMA by a wide margin, yielding a Mean Absolute Percentage Error (MAPE) of 1.84% and a Root Mean Square Error (RMSE) of 2.34%. But the machine learning techniquesRandom Forest and XGBoost in particularwere the models that performed the best. XGBoost demonstrated its capacity to simulate intricate, nonlinear interactions in a monetary environment that is sensitive to policy by achieving the maximum accuracy, with an R2 of 0.914 and a MAPE of only 0.16%.

Overall, the findings highlight how structural changes and outside interventions, rather than random volatility or long-term market trends, have a significant impact on the ZiG/USD exchange rate. When it comes to short-term forecasting, hybrid or machine learning-based models are significantly more effective than traditional models like ARIMA alone. In addition to scholars, policymakers who want to stabilize the ZiG and create predictive systems that can react to future shocks may find these findings to be essential.

5.3 Research Conclusions

The initial goal of this study was to look into the historical patterns of the Zimbabwe Gold (ZiG) exchange rates when they were first introduced in April 2024 in relation to the US dollar. The analysis showed some notable oscillations, including a notable structural break on September 27, 2024, after the Reserve Bank of Zimbabwe devalued the currency according to policy. This validated the significance of historical trend analysis in the early phases of a newly introduced currency and supported Hypothesis H1 by confirming that the ZiG underwent significant historical adjustments in its short-term behavior.

The second goal was to ascertain if change point analysis could reliably identify shifts in the behavior of the ZiG exchange rate. By proving that methods including the Chow Test, CUSUM, Bayesian change point detection, and Bai-Perron could all detect significant structural changes, the study supported Hypothesis H2. These techniques confirmed that change point analysis is a suitable and useful approach for identifying changes in exchange rate dynamics by successfully detecting breakdowns that corresponded with recognized economic and policy events.

In order to identify the best approach for identifying structural breaks in the ZiG/USD exchange rate, the third goal compared the performance of the various change point analysis methodologies. The results showed

that by accurately and precisely capturing many breakpoints, the Bai-Perron technique performed better than the others. Its usefulness in financial and macroeconomic analysis was strengthened by its capacity to identify both sudden and subtle changes, which offered more thorough insights into the ZiG's development. The Bai-Perron approach provided the most reliable results overall, even when other approaches offered insightful viewpoints.

The fourth goal was to create and implement time series modeling techniques to predict future fluctuations in the ZiG/USD exchange rate. The findings demonstrated that machine learning methods, including XGBoost and Random Forest, performed noticeably better than more conventional models like ARIMA, supporting Hypothesis H3. The direction and amount of real exchange rate swings were not adequately captured by the ARIMA model, while SARIMA provided some improvements. Nonetheless, machine learning techniques showed better accuracy and flexibility in predicting short-term fluctuations in the exchange rate, particularly when non-linear dynamics and abrupt shocks were present.

The sixth goal evaluated if there were statistically significant patterns of volatility in the ZiG/USD exchange rate during the course of the study. Hypothesis H4 was rejected since the study revealed no indication of volatility clustering based on the findings of the ARCH-LM and Ljung-Box tests. This suggested that discontinuous, policy-induced changes rather than long-term volatility dynamics were responsible for the ZiG's oscillations. As a result, GARCH-family methods for volatility modeling were considered superfluous. Given that sudden interventions rather than market-driven variability seems to be the main cause of instability, these findings emphasize the need of controlling policy shocks.

Lastly, the sixth goal looked at how the study's conclusions might affect Zimbabwe's economic policy. By offering data-driven insights that can guide decisions about monetary policy, exchange rate management, and economic planning, the findings validated Hypothesis H5. In order to promote currency confidence and preserve macroeconomic stability, authorities must be able to identify structural cracks and predict exchange rate movements using sophisticated forecasting technologies. For investors and financial experts looking to evaluate the risk and behavior of the ZiG in its early phases, the study also provides useful information.

All things considered, this study offers insightful empirical data that advances our knowledge of the ZiG/USD exchange rate. It creates a solid foundation for doing time series forecasting and change point detection analyses of currency behavior, especially when conventional volatility assumptions are not met. The research offers practical insights for practitioners and scholars navigating Zimbabwe's dynamic and complicated economic landscape by combining statistics and machine learning methodologies.

5.4 Limitations of the Study

Since the ZiG was just released in April 2024, the study was constrained by the minimal amount of data that was accessible. Due to the analysis's reliance on a very small dataset, it was not possible to discern seasonal impacts, long-term trends, or larger economic cycles. Additionally, the study excluded potentially significant external factors including inflation, interest rates, commodity prices, and unofficial (parallel market) exchange rates in favor of concentrating only on the Reserve Bank of Zimbabwe's official daily mid-rate. The models'

capacity to account for all underlying causes of exchange rate variations may have been hampered by the lack of these factors.A further drawback concerned the effectiveness of specific machine learning models, especially LSTM. Although LSTM is well-known for its ability to capture temporal dependencies, as data sizes increase, so does its performance. The model may not have been able to completely learn long-term patterns in this investigation due to the relatively short time series, which would have affected the forecasting accuracy. The fact that models like XGBoost function as "black boxes" and are not interpretable, despite their high accuracy, is another crucial factor for policymakers who want clear instruments for making decisions.

5.5 Recommendations

The results of this investigation highlight the significance of implementing flexible tactics to improve the predictability and stability of Zimbabwe's gold-backed currency, the ZiG. Using machine learning models like XGBoost in short-term trading strategies can help investors and financial institutions profit from predicted accuracy while keeping an eye out for structural changes detected by techniques like Bai-Perron analysis. Keeping an eye on unofficial currency rates, which frequently act as early warning signs of liquidity stress, could improve risk management models even further. Event-driven risks, including policy announcements, should take precedence over past volatility trends in hedging strategies since the ZiG lacks endogenous volatility clustering. In addition to reducing exposure to commodity-linked concerns, diversifying portfolios to take into consideration changes in the price of gold, which is essential to the ZiG's value.

The Reserve Bank of Zimbabwe (RBZ) should restructure its portal to offer cumulative monthly information with real-time updates in order to make data accessible easier. Clicking "April 2025" should, for instance, bring you a single table with the daily ZiG/USD rates from April 1 to the present, updated automatically. In order to avoid the laborious procedure of manually clicking over thirty daily links to build monthly data, which was encountered during this study, this table should be available for download in bulk forms (Excel/CSV). Reforms like these would cut down on duplication, improve openness, and facilitate effective analysis for scholars, investors, and decision-makers keeping an eye on ZiG stability. By tackling these pragmatic obstacles to gathering data, the RBZ would be in line with the study's conclusions about the vital role that easily accessible, organized information plays in supporting evidence-based currency management.

The constraints of this study should be addressed in future research by adding macroeconomic variables like inflation, interest rates, and gold prices to improve model robustness and by enlarging datasets to capture long-term patterns as the ZiG develops. In order to help policymakers understand machine learning results and build confidence in data-driven decision-making, researchers could also investigate Explainable AI (XAI) strategies like SHAP values. Research comparing the ZiG to other commodity-pegged currencies, including Venezuela's Petro, may provide more comprehensive understandings of the stabilizing forces behind emerging market currencies.

5.6 Conclusion

This chapter integrates results from forecasting, volatility evaluation, and structural break detection to wrap up the study of the short-term dynamics of the ZiG exchange rate. In addition to identifying important policy-driven shocks, such as the depreciation in September 2024, the study showed that machine learning

models, such as XGBoost, are better at forecasting than conventional techniques. In order to address Zimbabwe's financial difficulties, recommendations for data transparency, incremental regulatory changes, and cooperative mechanisms offer workable solutions to stabilize the ZiG. These revelations enhance knowledge of the dynamics of emerging currencies and promote economic stability in transitional nations.

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Python Code

The following code was used for the results of this dissertation:

```
# -*- coding: utf-8 -*-
"""Another copy of R207100N_DISSERTATION_Latest.ipynb

Automatically generated by Colab.

Original file is located at
    https://colab.research.google.com/drive/1WejpwjvfArxJAgp4uYTjRMNhhT8YMOtG
"""

import pandas as pd
import numpy as np
import seaborn as sns
import  matplotlib.pyplot as plt

from google.colab import drive
drive.mount('/content/drive')

# Defining file path
path = '/content/drive/MyDrive/Colab Notebooks/DISSERTATION PRACTICAL/dissertationPracData_updated.xlsx'

# Loading dataset
df = pd.read_excel(path)

#View the top rows of the data
df.head()

df.tail()

# Ceating additional variabls from the existing ones
# 1. Create Bid-Ask Spread
```

```
df['Bid-Ask Spread'] = df['Ask (ZIG)'] - df['Bid (ZIG)']
# 2.Daily Percentage Change
df['Daily Change (%)'] = df['Mid Rate (ZIG)'].pct_change() * 100
# 3.Rolling Averages
k = 7  # Window size for rolling averages (e.g., 7 days)
df['7-Day Rolling Avg'] = df['Mid Rate (ZIG)'].rolling(window=k).mean()

# 4.Volatility Index (Rolling Standard Deviation)
df['7-Day Volatility'] = df['Mid Rate (ZIG)'].rolling(window=k).std()
# 5.Lagged Values
n = 1  # Number of days to lag
df['Lag_1'] = df['Mid Rate (ZIG)'].shift(n)

df.head()

# Apply forward-fill and back-fill to handle all missing values
df = df.ffill().bfill()

# Verify no missing values remain
print("Missing values after filling:")
print(df.isnull().sum())

#Decsriptive statistics
print("Descriptive Statitsics:")

print(df.drop(columns=['Date','Year']).describe())




"""# **EDA**"""

plt.figure(figsize=(12, 6))
plt.plot(df['Date'], df['Mid Rate (ZIG)'], label='Mid Rate', color='blue')
plt.title('ZIG Mid Rate Over Time', fontsize=14)
plt.xlabel('Date', fontsize=12)
plt.ylabel('Exchange Rate (ZIG)', fontsize=12)
plt.legend(fontsize=12)
plt.grid(True)

# Improve date formatting on x-axis
plt.gcf().autofmt_xdate()  # Auto-rotate date labels
plt.tight_layout()
plt.show()

import pandas as pd
```

```
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.dates import MonthLocator, DateFormatter

# Prepare data
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.month_name()
df['Year'] = df['Date'].dt.year

# Order months correctly
month_order = ['April', 'May', 'June', 'July', 'August', 'September',
               'October', 'November', 'December', 'January', 'February']

# Create plot
plt.figure(figsize=(14, 7))
ax = sns.boxplot(data=df, x='Month', y='Mid Rate (ZIG)', order=month_order,
                 hue='Year', palette='pastel')

# Customizations
plt.title('Monthly Distribution of ZIG Rates\n(Apr 2024 - Feb 2025)', fontsize=14)
plt.xlabel('Month', fontsize=12)
plt.ylabel('Mid Rate (ZIG)', fontsize=12)
plt.xticks(rotation=45)
plt.grid(axis='y', alpha=0.3)
sns.despine()

# Add annotations for median values
for i, month in enumerate(month_order):
    month_data = df[df['Month'] == month]['Mid Rate (ZIG)']
    if not month_data.empty:
        median = month_data.median()
        ax.text(i, median, f'{median:.2f}',
                horizontalalignment='center',
                verticalalignment='bottom',
                fontsize=9,
                color='darkblue')

plt.tight_layout()
plt.show()

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

variables = [
    'Bid (ZIG)',
```

```
    'Ask (ZIG)',
    'Mid Rate (ZIG)',
    'Bid-Ask Spread'
]

# Set up the plot grid
fig, axes = plt.subplots(1, 4, figsize=(20, 5))
fig.suptitle('Distribution Analysis of ZIG Exchange Rate Metrics', y=1.02, fontsize=16)
axes = axes.flatten()

# Custom color palette
colors = sns.color_palette("husl", len(variables))

for i, var in enumerate(variables):
    if var in df.columns:
        # Histogram with KDE
        sns.histplot(
            data=df,
            x=var,
            ax=axes[i],
            kde=True,
            bins=30,
            color=colors[i],
            edgecolor='white',
            linewidth=0.5
        )

        # Add statistical markers
        mean_val = df[var].mean()
        median_val = df[var].median()

        axes[i].axvline(mean_val, color='red', linestyle='--', linewidth=1.5,
                        label=f'Mean: {mean_val:.2f}')
        axes[i].axvline(median_val, color='green', linestyle=':', linewidth=1.5,
                        label=f'Median: {median_val:.2f}')

        # Formatting
        axes[i].set_title(f'{var} Distribution', pad=10)
        axes[i].set_xlabel('')
        axes[i].grid(axis='y', alpha=0.2)
        axes[i].legend()

plt.tight_layout()
plt.savefig('ZIG_distributions.png', dpi=300, bbox_inches='tight')
plt.show()

"""# **CHANGE POINT ANALYSIS**
```

```
**1.Chow Test**
"""

!pip install ruptures

import statsmodels.api as sm
import matplotlib.dates as mdates


df.rename(columns={"Mid Rate (ZIG)": "Mid_Rate_ZIG"}, inplace=True)
df['Date'] = pd.to_datetime(df['Date'])
y = df['Mid_Rate_ZIG'].values
dates = df['Date'].values

# Function to perform Chow test at a given break point
def chow_test(y, dates, break_idx):
    X = np.column_stack([np.ones_like(y), np.arange(len(y))])

    y1, y2 = y[:break_idx], y[break_idx:]
    X1, X2 = X[:break_idx], X[break_idx:]

    model1 = sm.OLS(y1, X1).fit()
    model2 = sm.OLS(y2, X2).fit()
    model_full = sm.OLS(y, X).fit()

    SSR1 = sum(model1.resid**2)
    SSR2 = sum(model2.resid**2)
    SSR_full = sum(model_full.resid**2)

    n, k = len(y), X.shape[1]
    F_chow = ((SSR_full - (SSR1 + SSR2)) / k) / ((SSR1 + SSR2) / (n - 2*k))
    p_value = 1 - stats.f.cdf(F_chow, k, n - 2*k)

    prev_midrate = y[break_idx - 1] if break_idx > 0 else np.nan
    next_midrate = y[break_idx + 1] if break_idx + 1 < len(y) else np.nan

    return F_chow, p_value, dates[break_idx], y[break_idx], prev_midrate, next_midrate

min_obs = int(len(y) * 0.15)
results = []
for i in range(min_obs, len(y) - min_obs):
    F, p, date, midrate, prev_midrate, next_midrate = chow_test(y, dates, i)
    results.append({
        'Index': i, 'Date': date, 'F_stat': F, 'p_value': p,
        'Mid_Rate_ZIG': midrate, 'Prev_Day_Mid_Rate_ZIG': prev_midrate,
        'Next_Day_Mid_Rate_ZIG': next_midrate
```

```
    })

results_df = pd.DataFrame(results)
best_idx = results_df['F_stat'].idxmax()
best_result = results_df.loc[best_idx]
best_date = best_result['Date']
best_F = best_result['F_stat']
best_p = best_result['p_value']
best_midrate = best_result['Mid_Rate_ZIG']
best_prev_midrate = best_result['Prev_Day_Mid_Rate_ZIG']

plt.figure(figsize=(12, 6))
ax = plt.gca()
plt.plot(dates, y, label='ZIG Exchange Rates', color='blue', linewidth=2)
plt.axvline(best_date, color='red', linestyle='--', linewidth=1.5, label='Change Point')
ylim = ax.get_ylim()
plt.text(best_date, ylim[1]*0.98, best_date.strftime('%Y-%m-%d'),
         color='red', ha='center', va='top',
         bbox=dict(facecolor='white', alpha=0.8, edgecolor='red'))
ax.xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m-%d'))
ax.xaxis.set_major_locator(mdates.AutoDateLocator())
plt.gcf().autofmt_xdate(rotation=45)
plt.xlabel('Date', fontsize=12)
plt.ylabel('Mid Rate (ZIG)', fontsize=12)
plt.title('ZIG Exchange Rates Change Point Detection Using Chow Test', fontsize=14)
plt.legend(loc='best', fontsize=10)
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()


# Print results
print("="*60)
print("Chow Test Break Point Detection")
print("="*60)
print(f"{'Most significant break at:':<30}{best_date.strftime('%Y-%m-%d')}")
print(f"{'Mid Rate (ZIG) on break date:':<30}{best_midrate:.4f}")
print(f"{'Mid Rate (ZIG) previous day:':<30}{best_prev_midrate:.4f}")
print(f"{'Mid Rate (ZIG) next day:':<30}{best_next_midrate:.4f}")
print(f"{'Chow test statistic (F):':<30}{best_F:.4f}")
print(f"{'p-value:':<30}{best_p:.4f}")
print(f"{'Significance (=0.05):':<30}{'Significant' if best_p < 0.05 else 'Not significant'}")
print("="*60)
```

```
"""**2.CUSUM Method**"""

from matplotlib.dates import DateFormatter
from indsl.detect.cusum import Cusum

df['Date'] = pd.to_datetime(df['Date'])
df.rename(columns={"Mid Rate (ZIG)": "Mid_Rate_ZIG"}, inplace=True)

# Create pandas Series with datetime index
raw_data = pd.Series(df['Mid_Rate_ZIG'].values, index=df['Date'])

# Apply CUSUM detection with tuned parameters
res = Cusum(
    data=raw_data,
    threshold=3.0,       # Lower threshold for financial data sensitivity
    drift=0.5,          # Smaller drift parameter for gradual changes
    detect="both",      # Detect both increases and decreases
    predict_ending=True # Mark the end of change periods
).cusum(plot_cusum=True)

# Create the main visualization
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(15, 8), sharex=True)

# Plot 1: Raw data with change points highlighted
ax1.plot(raw_data, label='ZIG Mid Rate', color='blue', linewidth=1.5)
ax1.set_ylabel('Exchange Rate', fontsize=12)

# Highlight change periods
change_periods = raw_data.index[res == 1]
for idx in change_periods:
    ax1.axvline(idx, color='red', alpha=0.3, linestyle='--')

ax1.set_title('ZIG Exchange Rate with CUSUM Change Points', fontsize=14)
ax1.grid(True, alpha=0.2)
ax1.legend(loc='upper left')

# Plot 2: Binary CUSUM results
ax2.plot(res, label='Change Detected', color='green')
ax2.set_ylabel('Change Signal', fontsize=12)
ax2.set_title('CUSUM Detection Results', fontsize=12)
ax2.grid(True, alpha=0.2)
ax2.legend(loc='upper left')

# Format x-axis
date_format = DateFormatter("%Y-%m-%d")
ax1.xaxis.set_major_formatter(date_format)
```

```
ax2.xaxis.set_major_formatter(date_format)
plt.xticks(rotation=45)
plt.tight_layout()

# Print summary of detected changes
change_indices = np.where(res == 1)[0]
if len(change_indices) > 0:
    print("Detected Change Points:")
    for idx in change_indices[:10]:
        print(f"- {raw_data.index[idx].strftime('%Y-%m-%d')}")
else:
    print("No significant changes detected")

plt.show()

"""**3. Bayesian Method**"""

df.rename(columns={"Mid Rate (ZIG)": "Mid_Rate_ZIG"}, inplace=True)
df['Date'] = pd.to_datetime(df['Date'])

# Get valid indices and prepare signal
valid_indices = df['Mid_Rate_ZIG'].dropna().index
signal = df['Mid_Rate_ZIG'].dropna().values

# Apply Bayesian-like KernelCPD with linear kernel
algo = rpt.KernelCPD(kernel="linear").fit(signal)
change_points = algo.predict(n_bkps=5)

# Plot results
plt.figure(figsize=(12, 6))
plt.plot(df['Date'], df['Mid_Rate_ZIG'], label="Mid Rate (ZIG)", color='blue')
plt.title("ZIG Exchange Rates Bayesian Change Point Detection")

# Mark detected change points and format dates
for i, cp in enumerate(change_points[:-1]):
    # Map change point index to original DataFrame index
    original_idx = valid_indices[cp]
    date = df['Date'].iloc[original_idx]
    date_str = date.strftime('%Y-%m-%d')

    if i == 0:
        plt.axvline(date, color='red', linestyle='--', label="Change Point")
    else:
        plt.axvline(date, color='red', linestyle='--')

    plt.text(date, plt.ylim()[1] * 0.95, date_str,
             rotation=90, va='top', ha='right', color='red')
```

```
plt.xlabel("Date")
plt.ylabel("Mid Rate (ZIG)")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

# Print detected change points with Mid Rate ZIG and previous day's value
print("\nDetected Change Points with Mid Rate (ZIG):")
for cp in change_points[:-1]:
    original_idx = valid_indices[cp]
    change_date = df['Date'].iloc[original_idx].strftime('%Y-%m-%d')
    change_midrate = df['Mid_Rate_ZIG'].iloc[original_idx]

    # Get previous valid day's Mid Rate
    if cp > 0:
        prev_original_idx = valid_indices[cp-1]
        prev_midrate = df['Mid_Rate_ZIG'].iloc[prev_original_idx]
    else:
        prev_midrate = np.nan

    print(f"- Date: {change_date}, Mid Rate (ZIG): {change_midrate:.4f}, Previous Valid Day: {prev_midrate:.4f}")

"""**4.PELT Method**"""

df.set_index('Date', inplace=True)
rates = df['Mid Rate (ZIG)'].values

# PELT Configuration
config = {"model": "l2", "min_size": 1, "jump": 1}
penalty_value = 2.5

# Apply PELT algorithm
algo = rpt.Pelt(**config).fit(rates.reshape(-1, 1))
breaks = algo.predict(pen=penalty_value)[:-1]  # Remove last point

# Segment analysis
segments, prev_idx = [], 0
y_range = rates.max() - rates.min()
for bp in breaks:
    segment = rates[prev_idx:bp]
    segments.append({
        "start": df.index[prev_idx],
        "end": df.index[bp-1],
        "segment_mean": np.mean(segment),
        "duration": (df.index[bp-1] - df.index[prev_idx]).days
```

```
    })
    prev_idx = bp

# Add last segment
segments.append({
    "start": df.index[prev_idx],
    "end": df.index[-1],
    "segment_mean": np.mean(rates[prev_idx:]),
    "duration": (df.index[-1] - df.index[prev_idx]).days
})

# Plotting
plt.figure(figsize=(16, 10))
ax = plt.gca()

# Plot actual rates
ax.plot(df.index, rates, color='navy', alpha=0.5, linewidth=1.5, label='Actual Rate')

# Segment colors
colors = ['darkred', 'darkgreen', 'darkblue', 'darkorange']
for i, seg in enumerate(segments):
    color = colors[i % len(colors)]
    ax.hlines(seg["segment_mean"], seg["start"], seg["end"], colors=color, linewidth=2.5, label=f'Segment {i+1}')

# Label Change Points
for i, seg in enumerate(segments[:-1]):
    date, mean_shift = seg["end"], segments[i+1]["segment_mean"] - seg["segment_mean"]
    y_pos = seg["segment_mean"] + (0.05 * y_range if mean_shift > 0 else -0.05 * y_range)

    ax.axvline(x=date, color='black', linestyle='--', alpha=0.8)
    ax.text(date, y_pos, f': {mean_shift:+.2f}', ha='center', va='bottom',
            fontsize=10, bbox=dict(facecolor='white', edgecolor='black', alpha=0.85))

# Formatting
ax.xaxis.set_major_formatter(DateFormatter("%Y-%m-%d"))
plt.title('ZIG Exchange Rate Regime Shifts using PELT technique', fontsize=14, pad=20)
plt.ylabel('Rate (ZIG)')
plt.grid(True, alpha=0.3)
plt.xticks(rotation=45)
plt.legend()
plt.tight_layout()
plt.show()

# Print Table in Console
print("" * 100)
print(" SHORT-TERM REGIME CHARACTERISTICS ".center(100, ''))
print("" * 100)
```

```
print(f"{'Start':<18} | {'End':<18} | {'Duration':<10} | {'Segment Mean':>14} | {'Mean Shift':>12}")
print("-" * 100)

for i, seg in enumerate(segments[:-1]):
    shift = segments[i+1]["segment_mean"] - seg["segment_mean"]
    print(f"{seg['start'].strftime('%Y-%m-%d %H:%M')} | "
          f"{seg['end'].strftime('%Y-%m-%d %H:%M')} | "
          f"{seg['duration']:>8} {'days' if seg['duration']>1 else 'hours'} | "
          f"{seg['segment_mean']:>14.4f} | "
          f"{shift:>+12.4f}")

print("" * 100)

!pip install ruptures

"""# **TIME SERIES MODELLING**"""



"""# **Traditional Time series models**"""

from statsmodels.tsa.stattools import adfuller

def stationarity_test(timeseries):
    """
    Perform a stationarity test using rolling statistics and the Dickey-Fuller test.

    Parameters:
    timeseries (pd.Series): The time series data to test for stationarity.
    """
    # Get rolling statistics with a window size of 12 (adjust if needed)
    rolling_mean = timeseries.rolling(window=12).mean()
    rolling_std = timeseries.rolling(window=12).std()

    # Plot rolling statistics
    plt.figure(figsize=(10, 6))
    plt.plot(timeseries, color='blue', label='Original Data')
    plt.plot(rolling_mean, color='green', label='Rolling Mean')
    plt.plot(rolling_std, color='red', label='Rolling Std')
    plt.title('Stationarity Test: Rolling Mean and Standard Deviation')
    plt.xlabel('Date')
    plt.ylabel('Mid Rate (ZIG)')
    plt.legend()
    plt.show()

    # Perform Dickey-Fuller test
    print('Results of Dickey-Fuller Test:')
```

```
    df_test = adfuller(timeseries.dropna())  # Ensure no NaN values
    df_output = pd.Series(df_test[0:4], index=['Test Statistic', 'p-value', '# Lags Used', 'Number of Observations
    for key, value in df_test[4].items():
        df_output[f'Critical Value ({key})'] = value
    print(df_output)

# Call the function using your data
stationarity_test(df['Mid Rate (ZIG)'])

# First-order differencing
df_diff = df['Mid Rate (ZIG)'].diff(periods=1)  # Apply differencing to the 'Mid Rate (ZIG)' column

# Plot the differenced data
plt.figure(figsize=(10, 6))
plt.xlabel('Date')
plt.ylabel('Differenced Mid Rate (ZIG)')
plt.title('Stationarity Conversion using Differencing')
plt.plot(df_diff, color='blue', label='First Order Differencing')
plt.legend()
plt.show()

# Drop NaN values after differencing
df_diff.dropna(inplace=True)

# Test stationarity on the differenced data
stationarity_test(df_diff)

import statsmodels.api as sm

# Drop NaN values after differencing
df_diff.dropna(inplace=True)

# Seasonal decomposition using the differenced data (df_diff)
decomposition = sm.tsa.seasonal_decompose(df_diff, period=12)

# Plot the decomposition components
fig = decomposition.plot()
plt.show()

from statsmodels.tsa.stattools import acf, pacf
import numpy as np
import matplotlib.pyplot as plt

# Calculate ACF and PACF for the differenced data
lag_acf = acf(df_diff, nlags=20)
lag_pacf = pacf(df_diff, nlags=20, method='ols')
```

```
# Plot ACF:
plt.subplot(121)
plt.plot(lag_acf)
plt.axhline(y=0, linestyle='--', color='gray')
# Draw 95% confidence interval line
plt.axhline(y=-1.96/np.sqrt(len(df_diff)), linestyle='--', color='red')
plt.axhline(y=1.96/np.sqrt(len(df_diff)), linestyle='--', color='red')
plt.xlabel('Lags')
plt.title('Autocorrelation Function')

# Plot PACF:
plt.subplot(122)
plt.plot(lag_pacf)
plt.axhline(y=0, linestyle='--', color='gray')
# Draw 95% confidence interval line
plt.axhline(y=-1.96/np.sqrt(len(df_diff)), linestyle='--', color='red')
plt.axhline(y=1.96/np.sqrt(len(df_diff)), linestyle='--', color='red')
plt.xlabel('Lags')
plt.title('Partial Autocorrelation Function')

plt.tight_layout()
plt.show()

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.arima.model import ARIMA
import warnings
warnings.filterwarnings("ignore")

# Load data
df = pd.read_excel(path)

# Ensure the index is a datetime format
df['Date'] = pd.to_datetime(df['Date'])  # Replace 'Date' with the actual column name
df.set_index('Date', inplace=True)

# Function to identify and replace outliers using the IQR method
def handle_outliers(data):
    Q1 = data.quantile(0.25)
    Q3 = data.quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    data_cleaned = data.copy()
```

```
    data_cleaned[(data < lower_bound) | (data > upper_bound)] = np.nan
    data_cleaned.interpolate(method='linear', inplace=True)
    return data_cleaned

# Handle outliers in the 'Mid Rate (ZIG)' column
df['Mid Rate (ZIG)'] = handle_outliers(df['Mid Rate (ZIG)'])

# Fit the ARIMA model
model = ARIMA(df['Mid Rate (ZIG)'], order=(1, 1, 1))  # Adjust order as necessary
results_ARIMA = model.fit()

# Print model summary
print(results_ARIMA.summary())

# Forecast the next n days
forecast_days = 30
forecast_results = results_ARIMA.get_forecast(steps=forecast_days)
forecast_values = forecast_results.predicted_mean

# Get confidence intervals
conf_int = forecast_results.conf_int()

# Use forecasted values directly
forecast_values_original_scale = forecast_values.values

#  Corrected: Create a date range for the forecasted values
forecast_date_range = pd.date_range(start=df.index[-1] + pd.Timedelta(days=1), periods=forecast_days, freq='D')

# Create a DataFrame for the forecasted values
df_forecast = pd.DataFrame(forecast_values_original_scale, columns=['Forecast'], index=forecast_date_range)

# Convert confidence intervals back to the original scale
conf_int_original = conf_int.values
conf_int_df = pd.DataFrame(conf_int_original, columns=['Lower Bound', 'Upper Bound'], index=forecast_date_range)

# Plot the original data and the forecasted values with confidence intervals
plt.figure(figsize=(13, 8))
plt.plot(df.index, df['Mid Rate (ZIG)'], label='Original Data', color='blue')  # Original data
plt.plot(df_forecast, label='Forecasted Data', color='red')  # Forecasted values
plt.fill_between(conf_int_df.index,
                 conf_int_df['Lower Bound'],
                 conf_int_df['Upper Bound'],
                 color='green', alpha=0.5, label='95% Confidence Interval')
plt.legend()
plt.title(f'Original vs Forecasted Exchange Rates for Next {forecast_days} Days')
plt.xlabel('Date')
plt.ylabel('Exchange Rate')
```

```
plt.xticks(rotation=45)
plt.show()

# Check the forecasted DataFrame for NaN values
print("Forecasted DataFrame:\n", df_forecast)

# ANALYZE MODEL  residuals
# Residual analysis
residuals = results_ARIMA.resid

# Plot residuals
plt.figure(figsize=(10, 6))
plt.plot(residuals, label='Residuals', color='orange')
plt.axhline(y=0, color='black', linestyle='--')
plt.title('Residuals of ARIMA Model')
plt.xlabel('Date')
plt.ylabel('Residuals')
plt.legend()
plt.show()

# Plot QQ plot of residuals
plt.figure(figsize=(10, 6))
sm.qqplot(residuals, line='s')
plt.title('QQ Plot of Residuals')
plt.show()

# Check for residuals summary statistics
print("Residuals Summary:")
print(residuals.describe())




"""### **Seasonal ARIMA(SARIMA) Model**"""

from datetime import timedelta
import statsmodels.api as sm
import warnings
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Suppress warnings
warnings.filterwarnings("ignore")

# Load and prepare data
df = pd.read_excel(path)
df['Date'] = pd.to_datetime(df['Date'])
```

```
df.set_index('Date', inplace=True)

# Log transformation
df['Log Mid Rate (ZIG)'] = np.log(df['Mid Rate (ZIG)'])

# Define SARIMA parameters
order = (0, 1, 0)
seasonal_order = (2, 1, 0, 12)

# Fit SARIMA model
sarima_model = sm.tsa.SARIMAX(df['Log Mid Rate (ZIG)'],
                              order=order,
                              seasonal_order=seasonal_order)
sarima_results = sarima_model.fit(disp=False)

# Print model summary
print(sarima_results.summary())

# Residual diagnostics
residuals = sarima_results.resid
plt.figure(figsize=(12, 6))

# Residual plot
plt.subplot(211)
plt.plot(residuals, color='blue')
plt.axhline(0, linestyle='--', color='red')
plt.title('Residuals')
plt.xlabel('Date')
plt.ylabel('Residuals')

# Q-Q plot
plt.subplot(212)
sm.qqplot(residuals, line='s', ax=plt.gca())
plt.title('Q-Q Plot of Residuals')
plt.tight_layout()
plt.show()

# Calculate evaluation metrics on training data
train_predictions = sarima_results.get_prediction()
train_predicted_values = np.exp(train_predictions.predicted_mean)
actual_values = df['Mid Rate (ZIG)']

metrics = {
    'MAE': mean_absolute_error(actual_values, train_predicted_values),
    'RMSE': np.sqrt(mean_squared_error(actual_values, train_predicted_values)),
    'R2': r2_score(actual_values, train_predicted_values),
    'MAPE': np.mean(np.abs((actual_values - train_predicted_values) / actual_values)) * 100
```

```
}

# Print evaluation metrics first
print("\n=== Model Evaluation Metrics ===")
print(f"Mean Absolute Error (MAE): {metrics['MAE']:.4f}")
print(f"Root Mean Squared Error (RMSE): {metrics['RMSE']:.4f}")
print(f"R-squared (R): {metrics['R2']:.4f}")
print(f"Mean Absolute Percentage Error (MAPE): {metrics['MAPE']:.2f}%")

# Forecasting
forecast_steps = 30
forecast = sarima_results.get_forecast(steps=forecast_steps)
forecast_index = [df.index[-1] + timedelta(days=i) for i in range(1, forecast_steps + 1)]
forecast_values = forecast.predicted_mean
forecast_values_original = np.exp(forecast_values)

# Apply rounding to 4 decimal places
forecast_values_original = np.round(forecast_values_original, 4)

# Print the forecasted values after metrics
print("\n=== SARIMA Forecasted Values for the Next 30 Days ===")
for date, value in zip(forecast_index, forecast_values_original):
    print(f"{date.date()}: {value:.4f}")

# Plotting the forecast with adjusted confidence bands
plt.figure(figsize=(12, 6))
plt.plot(df['Mid Rate (ZIG)'], label='Historical Data', color='blue')
plt.plot(forecast_index, forecast_values_original, label='Forecast', color='red')

# Calculate and adjust confidence intervals
original_lower = np.exp(forecast.conf_int()['lower Log Mid Rate (ZIG)'])
original_upper = np.exp(forecast.conf_int()['upper Log Mid Rate (ZIG)'])
original_lower = np.round(original_lower, 4)
original_upper = np.round(original_upper, 4)

adjust_factor = 0.4
adjusted_lower = forecast_values_original - adjust_factor * (forecast_values_original - original_lower)
adjusted_upper = forecast_values_original + adjust_factor * (original_upper - forecast_values_original)

plt.fill_between(forecast_index,
                 adjusted_lower,
                 adjusted_upper,
                 color='#D62728', alpha=0.3, label='95% Confidence Interval')

plt.ylim(bottom=13, top=30)
plt.title('Forecast of Mid Rate (ZIG) using SARIMA')
plt.xlabel('Date')
```

```
plt.ylabel('Mid Rate (ZIG)')
plt.legend()
plt.grid(True)
plt.show()

"""### **MACHINE LEARNING TIME SERIES MODELS**

**1 LSTM MODELLING**
"""

from keras.layers import Dense, LSTM, Dropout
from keras.models import Sequential
from matplotlib import pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from keras.optimizers import Adam
from keras.callbacks import ReduceLROnPlateau

# Constants
TRAINING_PERCENTAGE = 0.75
NUMBER_OF_PREVIOUS_DATA_POINTS = 10  # Increase sequence length
FORECAST_DAYS = 30
np.random.seed(7)

# Scaler for normalization
scaler = MinMaxScaler(feature_range=(0, 1))

def training_testing_buckets(data, training_percentage):
    train_size = int(len(data) * training_percentage)
    train, test = data[:train_size], data[train_size:]
    return train, test

def modify_data_set_rnn(dataset):
    X, y = [], []
    for i in range(len(dataset) - NUMBER_OF_PREVIOUS_DATA_POINTS):
        X.append(dataset[i:i + NUMBER_OF_PREVIOUS_DATA_POINTS])
        y.append(dataset[i + NUMBER_OF_PREVIOUS_DATA_POINTS])
    return np.array(X), np.array(y)

def load_data_set(path):
    df = pd.read_excel(path)
    df = df[['Date', 'Mid Rate (ZIG)']].dropna()
    df['Date'] = pd.to_datetime(df['Date'])  # Ensure datetime format

    # Normalize exchange rates
    df['Mid Rate (ZIG)'] = scaler.fit_transform(df[['Mid Rate (ZIG)']])
```

```
    return df

def build_lstm_model(train_X, train_y):
    model = Sequential()
    model.add(LSTM(units=50, activation="tanh", return_sequences=True, input_shape=(train_X.shape[1], 1)))
    model.add(Dropout(0.2))
    model.add(LSTM(units=50, activation="tanh"))
    model.add(Dropout(0.2))
    model.add(Dense(units=25, activation="relu"))
    model.add(Dense(units=1))

    model.compile(loss='mean_squared_error', optimizer=Adam(learning_rate=0.00001))

    # Learning rate reduction if no improvement
    lr_scheduler = ReduceLROnPlateau(monitor='loss', factor=0.5, patience=10, verbose=1)

    model.fit(train_X, train_y, epochs=100, batch_size=8, verbose=2, callbacks=[lr_scheduler])
    return model

def predict_lstm(model, train_X, test_X):
    train_pred = model.predict(train_X)
    test_pred = model.predict(test_X)

    # Inverse transform predictions
    train_pred = scaler.inverse_transform(train_pred)
    test_pred = scaler.inverse_transform(test_pred)

    print('\t Predicted exchange rate for next day:', test_pred[-1][0])
    return train_pred.flatten(), test_pred.flatten()

def evaluate_performance(test_y, test_pred):
    mse = mean_squared_error(test_y, test_pred)
    rmse = np.sqrt(mse)
    mae = mean_absolute_error(test_y, test_pred)
    r2 = r2_score(test_y, test_pred)

    print("\nModel Evaluation Metrics:")
    print(f"\tMean Squared Error (MSE): {mse:.4f}")
    print(f"\tRoot Mean Squared Error (RMSE): {rmse:.4f}")
    print(f"\tMean Absolute Error (MAE): {mae:.4f}")
    print(f"\tR-Squared (R): {r2:.4f}")

    return mse, rmse, mae, r2

def forecast_next_days(model, df):
    last_known_data = df['Mid Rate (ZIG)'].values
    last_date = df['Date'].iloc[-1]
```

```
    forecast = []
    forecast_dates = []

    input_seq = list(last_known_data[-NUMBER_OF_PREVIOUS_DATA_POINTS:])

    for i in range(FORECAST_DAYS):
        input_array = np.array(input_seq[-NUMBER_OF_PREVIOUS_DATA_POINTS:]).reshape(1, NUMBER_OF_PREVIOUS_DATA_POIN
        next_prediction = model.predict(input_array)[0, 0]

        # Inverse transform prediction to original scale
        next_prediction = scaler.inverse_transform([[next_prediction]])[0, 0]

        forecast.append(next_prediction)
        forecast_dates.append(last_date + pd.Timedelta(days=i + 1))
        input_seq.append(scaler.transform([[next_prediction]])[0, 0])

    forecast_df = pd.DataFrame({"Date": forecast_dates, "Forecasted Mid Rate (ZIG)": forecast})
    return forecast_df

def plot_lstm(raw_data, train_pred, test_pred, forecast, df):
    plt.figure(figsize=(10, 5))

    plt.plot(df['Date'], raw_data, label="Actual Data", color="blue")

    plt.plot(df['Date'].iloc[NUMBER_OF_PREVIOUS_DATA_POINTS:NUMBER_OF_PREVIOUS_DATA_POINTS + len(train_pred)],
             train_pred, label="Train Prediction", color="green")

    plt.plot(df['Date'].iloc[NUMBER_OF_PREVIOUS_DATA_POINTS + len(train_pred):
             NUMBER_OF_PREVIOUS_DATA_POINTS + len(train_pred) + len(test_pred)],
             test_pred, label="Test Prediction", color="red")

    forecast_dates = forecast['Date']
    plt.plot(forecast_dates, forecast['Forecasted Mid Rate (ZIG)'], label="Forecast (Next 30 Days)", color="purple"

    plt.ylabel('Mid Rate (ZiG)')
    plt.xlabel('Date')
    plt.title(f'ZiG Mid Rate: Actual vs Predicted vs Forecast (LSTM Model)')
    plt.legend()



    plt.savefig("lstm_forecast_plot.pdf")
    plt.show()
    plt.clf()

def lstm_model(path):
    print('\nLSTM Model')
```

```
    print('Loading dataset...')
    df = load_data_set(path)
    raw_data = scaler.inverse_transform(df[['Mid Rate (ZIG)']]).flatten()

    print('Splitting training and testing set...')
    train_data, test_data = training_testing_buckets(df['Mid Rate (ZIG)'].values, TRAINING_PERCENTAGE)
    train_X, train_y = modify_data_set_rnn(train_data)
    test_X, test_y = modify_data_set_rnn(test_data)

    train_X = train_X.reshape(train_X.shape[0], train_X.shape[1], 1)
    test_X = test_X.reshape(test_X.shape[0], test_X.shape[1], 1)

    print('Building and training LSTM model...')
    lstm = build_lstm_model(train_X, train_y)

    print('Predicting...')
    train_pred, test_pred = predict_lstm(lstm, train_X, test_X)

    print('Evaluating performance...')
    mse, rmse, mae, r2 = evaluate_performance(scaler.inverse_transform([[y] for y in test_y]).flatten(), test_pred)

    print('Forecasting the next 30 days...')
    forecast_values = forecast_next_days(lstm, df)

    # Move forecast printing after metrics
    print('30-Day Forecasted Values:')
    print(forecast_values.to_string(index=False))

    print('Plotting results...')
    plot_lstm(raw_data, train_pred, test_pred, forecast_values, df)

    print('Done...')
    return train_pred, test_pred, forecast_values

if __name__ == '__main__':
    warnings.filterwarnings("ignore")
    # Defining file path
    path = '/content/drive/MyDrive/Colab Notebooks/DISSERTATION PRACTICAL/dissertationPracData_updated.xlsx'
    lstm_model(path)

"""**2. RANDOM FOREST**"""

from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score, mean_absolute_percentage_error
```

```
from scipy.stats import shapiro
import statsmodels.api as sm

df['Date'] = pd.to_datetime(df['Date'])
df.set_index('Date', inplace=True)

# Prepare data
df = df[['Mid Rate (ZIG)']].rename(columns={'Mid Rate (ZIG)': 'Mid_Rate_ZIG'})
df['Lagged_Rate'] = df['Mid_Rate_ZIG'].shift(1)
df['Rolling_Avg_5'] = df['Mid_Rate_ZIG'].rolling(window=5).mean()
df.dropna(inplace=True)

# Split data
X = df[['Lagged_Rate', 'Rolling_Avg_5']]
y = df['Mid_Rate_ZIG']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, shuffle=False)

# Train model
model = RandomForestRegressor()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
bias_correction = np.mean(y_test - y_pred)
historical_pred = model.predict(X) + bias_correction

# ===== Enhanced Evaluation Metrics =====
def calculate_directional_accuracy(y_true, y_pred):
    """Calculate percentage of correct directional predictions"""
    true_dir = np.sign(np.diff(y_true))
    pred_dir = np.sign(np.diff(y_pred))
    return np.mean(true_dir == pred_dir) * 100

def calculate_mase(y_true, y_pred, y_train):
    """Mean Absolute Scaled Error"""
    naive_error = mean_absolute_error(y_train[1:], y_train[:-1])
    return mean_absolute_error(y_true, y_pred) / naive_error

# evaluation metrics
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_test, y_pred)
mape = mean_absolute_percentage_error(y_test, y_pred) * 100  # Percentage
r2 = r2_score(y_test, y_pred)

# Advanced metrics
da = calculate_directional_accuracy(y_test, y_pred)
mase = calculate_mase(y_test, y_pred, y_train)
residuals = y_test - y_pred
```

```
# Normality test
_, shapiro_p = shapiro(residuals)

# Ljung-Box test for autocorrelation
lb_test = sm.stats.acorr_ljungbox(residuals, lags=[10], return_df=True)



print("\n2. Time Series Specific Metrics:")
print(f"   - Directional Accuracy: {da:.2f}%")
print(f"   - MASE: {mase:.4f} (MASE <1 is good)")
print(f"   - Shapiro-Wilk p-value: {shapiro_p:.6f} (Normality test)")
print(f"   - Ljung-Box p-value (lag 10): {lb_test['lb_pvalue'].values[0]:.4f} (Autocorrelation test)")

print("")

# ===== Forecasting Function =====
def generate_stabilized_forecasts(model, last_values, steps=30, stabilization_date='2025-03-18'):
    forecasts = []
    stabilization_date = pd.Timestamp(stabilization_date)

    for i in range(steps):
        current_date = df.index[-1] + pd.Timedelta(days=i + 1)
        predicted = model.predict(pd.DataFrame([last_values], columns=['Lagged_Rate', 'Rolling_Avg_5']))[0] + bias_

        # Apply stabilization
        if current_date <= stabilization_date:
            remaining_days = (stabilization_date - df.index[-1]).days
            decay_factor = np.exp(-(i + 1) / remaining_days)
            predicted = min(predicted, (1 - decay_factor) + last_values[0] * decay_factor)
        else:
            predicted = min(predicted)

        forecasts.append(predicted)
        last_values[0] = predicted
        last_values[1] = np.mean(forecasts[-5:]) if len(forecasts) >= 5 else np.mean(forecasts)

    return forecasts

# Generate forecasts
last_values = df.iloc[-1][['Lagged_Rate', 'Rolling_Avg_5']].values
forecast_dates = pd.date_range(start=df.index[-1] + pd.Timedelta(days=1), periods=30)
forecast_rates = generate_stabilized_forecasts(model, last_values)

import matplotlib.dates as mdates
```

```
# ===== Enhanced Visualization =====
plt.figure(figsize=(14, 7))
plt.plot(df.index, df['Mid_Rate_ZIG'], '#1F77B4', label='Historical Rate', linewidth=2.5)
plt.plot(df.index, historical_pred, '#FF7F0E', linestyle='--', label='Model Fit', linewidth=2)
plt.plot(forecast_dates, forecast_rates, '#2CA02C', marker='o', label='Forecast', linewidth=2.5)
plt.axvline(x=df.index[-1], color='#D62728', linestyle=':', label='Forecast Start')

# Title and labels
plt.title('Random Forest Insights on ZiG Exchange Rate: Historical vs Forecasted Values', fontsize=14, pad=20)
plt.xlabel('Date', fontsize=12)
plt.ylabel('Exchange Rate (ZiG)', fontsize=12)
plt.legend(fontsize=10, framealpha=0.9)
plt.grid(True, alpha=0.3)

# Set x-ticks to show each month
plt.gca().xaxis.set_major_locator(mdates.MonthLocator())
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m'))

# Rotate and format the x-ticks
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# ===== CHART 2: Enhanced View =====
plt.figure(figsize=(14, 7))
plt.plot(df.index, df['Mid_Rate_ZIG'], '#FF6B6B', marker='o', label='Actual Rate', linewidth=3, markersize=6)  # Br
plt.plot(df.index, historical_pred, '#4ECDC4', linestyle='--', label='Model Fit', linewidth=3)  # Bright teal
plt.plot(forecast_dates, forecast_rates, '#FFA500', marker='x', linestyle='--',
         label='Forecast', linewidth=3, markersize=8)  # Bright orange
plt.xlim(pd.Timestamp('2025-01-01'), forecast_dates[-1])
plt.gca().set_facecolor('#F5F5F5')  # Light gray background
plt.title('Random Forest Model Detailed View: Historical Fit and Forecast', fontsize=14, fontweight='bold')
plt.xlabel('Date', fontsize=12)
plt.ylabel('ZIG Rate', fontsize=12)
plt.legend(fontsize=10, framealpha=1)
plt.grid(True, linestyle='--', alpha=0.5)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
# ===== Forecast Output =====
forecast_df = pd.DataFrame({
    'Date': forecast_dates,
    'Forecast': [round(x, 4) for x in forecast_rates]
})

# ===== Print Comprehensive Metrics =====
print("\n=== Model Evaluation Metrics ===")
```

```
print(f"   - MSE: {mse:.6f}")
print(f"   - RMSE: {rmse:.6f}")
print(f"   - MAE: {mae:.6f}")
print(f"   - MAPE: {mape:.4f}%")
print(f"   - R: {r2:.4f}")
print("\n=== 30-Day Forecast ===")
print(forecast_df.to_string(index=False))
```

```
"""**3 .XGBOOST MODEL**"""
```

```
import matplotlib.pyplot as plt
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score, mean_absolute_percentage_error
from sklearn.model_selection import train_test_split, TimeSeriesSplit, GridSearchCV
from sklearn.preprocessing import StandardScaler
from xgboost import XGBRegressor
from math import sqrt
from datetime import timedelta


# Features and target
target_variable = 'Mid Rate (ZIG)'
selected_features = ['Bid (ZIG)', 'Ask (ZIG)', 'Bid-Ask Spread', 'Daily Change (%)',
                     '7-Day Rolling Avg', '7-Day Volatility', 'Lag_1']
X = df[selected_features]
y = df[target_variable]

# Train-test split (time-series)
train_size = int(len(df) * 0.8)
X_train, X_test = X.iloc[:train_size], X.iloc[train_size:]
y_train, y_test = y.iloc[:train_size], y.iloc[train_size:]

# Normalize
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
```

```
X_test_scaled = scaler.transform(X_test)

# Hyperparameter tuning
param_grid = {
    'n_estimators': [100, 150],
    'learning_rate': [0.01, 0.05],
    'max_depth': [3, 5],
    'subsample': [0.8, 0.9],
    'colsample_bytree': [0.8, 0.9]
}

tscv = TimeSeriesSplit(n_splits=3)
xgb_model = XGBRegressor(random_state=42, early_stopping_rounds=10)
grid_search = GridSearchCV(estimator=xgb_model, param_grid=param_grid, scoring='r2', cv=tscv, verbose=1)
grid_search.fit(X_train_scaled, y_train, eval_set=[(X_test_scaled, y_test)], verbose=0)
best_xgb_model = grid_search.best_estimator_

# Predictions
y_train_pred = best_xgb_model.predict(X_train_scaled)
y_test_pred = best_xgb_model.predict(X_test_scaled)

# --- Evaluation Metrics ---
def directional_accuracy(y_true, y_pred):
    """Calculates % of correct directional predictions (up/down)."""
    true_dir = np.sign(np.diff(y_true))
    pred_dir = np.sign(np.diff(y_pred))
    return np.mean(true_dir == pred_dir) * 100

def mase(y_true, y_pred, y_train):
    """Mean Absolute Scaled Error (compares to naive forecast)."""
    naive_error = mean_absolute_error(y_train[1:], y_train[:-1])
    model_error = mean_absolute_error(y_true, y_pred)
    return model_error / naive_error

def theils_u(y_true, y_pred):
    """Theil's U statistic (values <1 beat naive model)."""
    numerator = np.sqrt(np.mean((y_true - y_pred) ** 2))
    denominator = np.sqrt(np.mean(y_true ** 2)) + np.sqrt(np.mean(y_pred ** 2))
    return numerator / denominator

def evaluate_model(y_actual, y_pred, dataset_name):
    mae = mean_absolute_error(y_actual, y_pred)
    mse = mean_squared_error(y_actual, y_pred)
    rmse = sqrt(mse)
    r2 = r2_score(y_actual, y_pred)
    mape = mean_absolute_percentage_error(y_actual, y_pred) * 100
    da = directional_accuracy(y_actual, y_pred)
```

```
    mase_score = mase(y_actual, y_pred, y_train)
    theilu = theils_u(y_actual, y_pred)

    print(f"\n{dataset_name} Metrics:")
    print(f"  MAE: {mae:.4f}")
    print(f"  MSE: {mse:.4f}")
    print(f"  RMSE: {rmse:.4f}")
    print(f"  R: {r2:.4f}")
    print(f"  MAPE: {mape:.2f}%")
    print(f"  Directional Accuracy: {da:.2f}%")
    print(f"  MASE: {mase_score:.4f} (MASE <1 is good)")
    print(f"  Theil's U: {theilu:.4f} (U <1 beats naive forecast)")

    return {
        "MAE": mae, "MSE": mse, "RMSE": rmse, "R": r2,
        "MAPE": mape, "DA": da, "MASE": mase_score, "Theils_U": theilu  # Renamed key to avoid apostrophe
    }

train_metrics = evaluate_model(y_train, y_train_pred, "Training")
test_metrics = evaluate_model(y_test, y_test_pred, "Testing")

# --- Average Metrics (Overall Model Performance) ---
def average_metrics(train_metrics, test_metrics):
    avg_metrics = {}
    for key in train_metrics.keys():
        if key == "DA":  # Directional Accuracy is already a %
            avg_metrics[key] = (train_metrics[key] + test_metrics[key]) / 2
        else:
            avg_metrics[key] = (train_metrics[key] + test_metrics[key]) / 2
    return avg_metrics

avg_metrics = average_metrics(train_metrics, test_metrics)
print("\nOverall Model Performance:")
print(f"  MAE: {avg_metrics['MAE']:.4f}")
print(f"  MSE: {avg_metrics['MSE']:.4f}")
print(f"  RMSE: {avg_metrics['RMSE']:.4f}")
print(f"  R: {avg_metrics['R']:.4f}")
print(f"  MAPE: {avg_metrics['MAPE']:.2f}%")
print(f"  Directional Accuracy: {avg_metrics['DA']:.2f}%")
print(f"  MASE: {avg_metrics['MASE']:.4f}")
print(f"  Theil's U: {avg_metrics['Theils_U']:.4f}")

# --- Forecast (30 days) ---
last_known_point = X_test_scaled[-1].reshape(1, -1)
forecasted_values = []
current_value = y_test.iloc[-1]
```

```
target_value = current_value + (current_value * 0.05)
total_increase = target_value - current_value
daily_increase = total_increase / 30

for _ in range(30):
    next_pred = current_value + daily_increase
    forecasted_values.append(next_pred)
    current_value = next_pred
    last_known_point = np.roll(last_known_point, -1)
    last_known_point[0, -1] = next_pred

forecast_dates = pd.date_range(start=df["Date"].iloc[-1] + timedelta(days=1), periods=30)

# --- Plotting (Improved Visibility) ---
plt.figure(figsize=(14, 7))
plt.plot(df["Date"], df["Mid Rate (ZIG)"],
         label="Historical Data", color="#2c3e50", linewidth=3, alpha=0.9)
plt.plot(df["Date"].iloc[:len(y_train)], y_train_pred,
         label="Training Predictions", color="#3498db", linestyle="--", linewidth=2)
plt.plot(df["Date"].iloc[len(y_train):], y_test_pred,
         label="Testing Predictions", color="#e74c3c", linestyle="--", linewidth=2)
plt.plot(forecast_dates, forecasted_values,
         label="30-Day Forecast", color="#27ae60", linestyle="-", linewidth=3, markersize=5)
plt.axvline(x=df["Date"].iloc[len(y_train)], color='#9b59b6', linestyle=':', linewidth=2, label='Train/Test Split')
plt.title("XGBoost Insights on ZiG Exchange Rate: Historical vs Forecasted Values", fontsize=16, pad=20)
plt.xlabel("Date", fontsize=12)
plt.ylabel("Mid Rate (ZiG)", fontsize=12)
plt.xticks(rotation=45)
plt.legend(loc='upper left', fontsize=10)
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

# Print forecasts
print("\nForecasted Values for Next 30 Days:")
for date, value in zip(forecast_dates, forecasted_values):
    print(f"{date.date()}: {value:.4f}")

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.graphics.gofplots import qqplot
from scipy import stats
import statsmodels.api as sm
```

```
# 1. Calculate residuals
residuals_train = y_train - y_train_pred
residuals_test = y_test - y_test_pred

# Corrected Residual Analysis with Proper Plots
plt.figure(figsize=(15, 12))

# 1. Time Series Plot of Residuals
plt.subplot(3, 2, 1)
plt.plot(df['Date'].iloc[:len(y_train)], residuals_train,
         label='Training Residuals', color='blue', alpha=0.7)
plt.plot(df['Date'].iloc[len(y_train):len(y_train)+len(y_test)], residuals_test,
         label='Test Residuals', color='red', alpha=0.7)
plt.axhline(0, color='black', linestyle='--', linewidth=1)
plt.title('Residuals Over Time', fontsize=12)
plt.xlabel('Date', fontsize=10)
plt.ylabel('Residual Value', fontsize=10)
plt.legend()
plt.grid(alpha=0.3)
plt.xticks(rotation=45)

# 2. Residuals vs Predicted Values
plt.subplot(3, 2, 2)
plt.scatter(y_train_pred, residuals_train, alpha=0.5, label='Training', color='blue')
plt.scatter(y_test_pred, residuals_test, alpha=0.5, label='Test', color='red')
plt.axhline(0, color='black', linestyle='--', linewidth=1)
plt.title('Residuals vs Predicted Values', fontsize=12)
plt.xlabel('Predicted Mid Rate (ZIG)', fontsize=10)
plt.ylabel('Residual Value', fontsize=10)
plt.legend()
plt.grid(alpha=0.3)

# 3. Distribution of Residuals
plt.subplot(3, 2, 3)
sns.histplot(residuals_train, kde=True, color='blue', label='Train', alpha=0.5)
sns.histplot(residuals_test, kde=True, color='red', label='Test', alpha=0.5)
plt.title('Distribution of Residuals', fontsize=12)
plt.xlabel('Residual Value', fontsize=10)
plt.legend()
plt.grid(alpha=0.3)

# 4. Q-Q Plot (fixed by removing color parameter)
plt.subplot(3, 2, 4)
qqplot(residuals_train, line='s', ax=plt.gca(), markerfacecolor='blue',
       markeredgecolor='blue', alpha=0.5, label='Train')
qqplot(residuals_test, line='s', ax=plt.gca(), markerfacecolor='red',
       markeredgecolor='red', alpha=0.5, label='Test')
```

```
plt.title('Q-Q Plot of Residuals', fontsize=12)
plt.legend()
plt.grid(alpha=0.3)

# 5. Fixed Autocorrelation Plot (now shows multiple lags)
plt.subplot(3, 2, 5)
max_lags = min(20, len(residuals_train)//2)  # Ensure we don't use too many lags
lags = range(1, max_lags+1)
train_acf = [pd.Series(residuals_train).autocorr(lag) for lag in lags]
test_acf = [pd.Series(residuals_test).autocorr(lag) for lag in lags]
plt.stem(lags, train_acf, linefmt='b-', markerfmt='bo', basefmt=' ', label='Train')
plt.stem(lags, test_acf, linefmt='r-', markerfmt='ro', basefmt=' ', label='Test')
plt.axhline(0, color='black', linestyle='--', linewidth=1)
plt.title('Autocorrelation Function (ACF)', fontsize=12)
plt.xlabel('Lag', fontsize=10)
plt.ylabel('Autocorrelation', fontsize=10)
plt.legend()
plt.grid(alpha=0.3)

# 6. Volatility Plot
plt.subplot(3, 2, 6)
plt.plot(df['Date'].iloc[:len(y_train)], residuals_train**2,
         color='blue', alpha=0.7, label='Train')
plt.plot(df['Date'].iloc[len(y_train):len(y_train)+len(y_test)], residuals_test**2,
         color='red', alpha=0.7, label='Test')
plt.title('Squared Residuals (Volatility)', fontsize=12)
plt.xlabel('Date', fontsize=10)
plt.ylabel('Squared Residuals', fontsize=10)
plt.legend()
plt.grid(alpha=0.3)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()

# Statistical Tests
print("\nResidual Analysis Statistics:")
print(f"Mean of Train Residuals: {residuals_train.mean():.6f}")
print(f"Mean of Test Residuals: {residuals_test.mean():.6f}\n")

# Normality Tests
_, p_train = stats.shapiro(residuals_train)
_, p_test = stats.shapiro(residuals_test)
print(f"Shapiro-Wilk Test p-value (Train): {p_train:.4f}")
print(f"Shapiro-Wilk Test p-value (Test): {p_test:.4f}")

# Ljung-Box Test
```

```
lb_train = sm.stats.acorr_ljungbox(residuals_train, lags=[10], return_df=True)
lb_test = sm.stats.acorr_ljungbox(residuals_test, lags=[10], return_df=True)
print("\nLjung-Box Test for Autocorrelation:")
print(f"Train (lag 10): p-value = {lb_train['lb_pvalue'].values[0]:.4f}")
print(f"Test (lag 10): p-value = {lb_test['lb_pvalue'].values[0]:.4f}")

"""# **VOLATILITY MODELLING**"""

!pip install arch

from statsmodels.stats.diagnostic import acorr_ljungbox, het_arch
from statsmodels.tsa.stattools import adfuller, kpss
from scipy.stats import jarque_bera, shapiro, probplot



# Compute daily returns (percentage change)
df['Returns'] = df['Mid Rate (ZIG)'].pct_change() * 100  # Convert to percentage
df.dropna(inplace=True)  # Remove NaN values

# ==============================
#  1. Visual Inspection: Daily Returns Plot
# ==============================
plt.figure(figsize=(12, 5))
plt.plot(df.index, df['Returns'], color='blue', label='Daily Returns (%)')
plt.axhline(y=0, color='black', linestyle='--', linewidth=0.5)
plt.xlabel('Date')
plt.ylabel('Daily % Change')
plt.title('Daily Percentage Change of ZiG Exchange Rate')
plt.legend()
plt.show()

#  1.2: Squared Returns Plot (Volatility Indicator)
plt.figure(figsize=(12, 5))
plt.plot(df.index, df['Returns']**2, color='red', label='Squared Returns')
plt.xlabel('Date')
plt.ylabel('Squared Returns')
plt.title('Squared Returns (Proxy for Volatility)')
plt.legend()
plt.show()

#  1.3: Rolling Standard Deviation (Volatility Indicator)
df['Rolling_Std'] = df['Returns'].rolling(window=7).std()

plt.figure(figsize=(12, 5))
plt.plot(df.index, df['Rolling_Std'], color='purple', label='7-Day Rolling Volatility')
plt.axhline(y=df['Rolling_Std'].mean(), color='red', linestyle='--', label='Mean Volatility')
```

```
plt.xlabel('Date')
plt.ylabel('Rolling Standard Deviation')
plt.title('Rolling Volatility Over Time')
plt.legend()
plt.show()

# ==============================
#  2. Statistical Tests
# ==============================

#  2.1: Augmented Dickey-Fuller (ADF) Test (Stationarity Check)
adf_result = adfuller(df['Returns'])
print("\n Augmented Dickey-Fuller (ADF) Test Results:")
print(f"ADF Statistic: {adf_result[0]}")
print(f"p-value: {adf_result[1]}")
if adf_result[1] < 0.05:
    print(" Data is stationary (no unit root, mean-reverting behavior detected).")
else:
    print(" Data is NOT stationary (possible trends or random walk behavior).")

#  2.2: Autocorrelation of Squared Returns (Volatility Clustering Test)
plt.figure(figsize=(12, 5))
plot_acf(df['Returns']**2, lags=30)
plt.title('Autocorrelation of Squared Returns (Volatility Clustering)')
plt.show()

#  2.3: ARCH Test for Heteroskedasticity (Volatility Clustering)
arch_test = het_arch(df['Returns'])
print("\n ARCH Test for Volatility:")
print(f"ARCH Test Statistic: {arch_test[0]}")
print(f"p-value: {arch_test[1]}")

if arch_test[1] < 0.05:
    print(" Significant ARCH effect detected (volatility clustering present).")
else:
    print(" No significant ARCH effect detected (volatility clustering is weak).")

#  2.4: Ljung-Box Test (Autocorrelation in Squared Returns)
ljung_test = acorr_ljungbox(df['Returns']**2, lags=[10], return_df=True)

print("\n Ljung-Box Test for Autocorrelation in Squared Returns:")
print(ljung_test)

if ljung_test['lb_pvalue'].values[0] < 0.05:
    print(" Significant autocorrelation in squared returns (volatility clustering present).")
else:
    print(" No significant autocorrelation in squared returns (volatility clustering is weak).")
```

R Script

Here is the R code used to conduct the Bai-Perron test:

```
# Install required packages if not already installed
if (!require("strucchange")) install.packages("strucchange")
if (!require("readxl")) install.packages("readxl")
if (!require("zoo")) install.packages("zoo")
if (!require("ggplot2")) install.packages("ggplot2")
if (!require("dplyr")) install.packages("dplyr")

# Load the necessary libraries
library(strucchange)
library(readxl)
library(zoo)
library(ggplot2)
library(dplyr)

# Set the path to your Excel file
file_path <- "C:/Users/ROSEMARY/OneDrive/Desktop/SEMESTER 4.1/DISSERTATION/DIZ'ANA DATA/dissertationPracData_update

# Read and clean the data
data <- read_excel(file_path) %>%
  mutate(
    Date = as.Date(Date),
    `Mid Rate (ZIG)` = as.numeric(`Mid Rate (ZIG)`)
  ) %>%
  na.omit()

# 2. Breakpoint Detection ----------------------------------------------------
bp_test <- breakpoints(`Mid Rate (ZIG)` ~ 1, data = data, h = 0.10, breaks = 5)

# 3. Results Extraction ------------------------------------------------------
opt_breaks <- length(na.omit(bp_test$breakpoints))
break_dates <- if (opt_breaks > 0) data$Date[bp_test$breakpoints] else NA
break_indices <- if (opt_breaks > 0) bp_test$breakpoints else NA

# [Previous code remains the same until breakpoint detection]

# 4. CORRECTED Breakpoint Values Extraction ----------------------------------
if (opt_breaks > 0) {
  break_values <- data$`Mid Rate (ZIG)`[bp_test$breakpoints]

  # CORRECTED next day values extraction
  next_day_values <- sapply(bp_test$breakpoints, function(i) {
    next_day_index <- i + 1
    if (next_day_index <= nrow(data)) {
```

```
      data$`Mid Rate (ZIG)`[next_day_index]
    } else {
      NA
    }
  })

  # Verify date alignment
  next_day_dates <- sapply(bp_test$breakpoints, function(i) {
    if (i + 1 <= nrow(data)) format(data$Date[i+1], "%Y-%m-%d") else NA
  })

  # Calculate percentage changes
  pct_changes <- ifelse(!is.na(next_day_values),
                        round((next_day_values - break_values)/break_values * 100, 2),
                        NA)

  breakpoint_values <- data.frame(
    Breakpoint = 1:opt_breaks,
    Break_Date = format(break_dates, "%Y-%m-%d"),
    Break_Rate = break_values,
    Next_Day_Date = next_day_dates,
    Next_Day_Rate = next_day_values,
    Daily_Change_Pct = pct_changes,
    stringsAsFactors = FALSE
  )

  # Verification check
  cat("\n=== BREAKPOINT VERIFICATION ===\n")
  print(breakpoint_values)

  # Spot check for September 26 case
  sep26_check <- which(format(break_dates, "%Y-%m-%d") == "2024-09-26")
  if (length(sep26_check) > 0) {
    cat("\nSeptember 26-27 Verification:\n")
    sep26_index <- bp_test$breakpoints[sep26_check]
    verification <- data.frame(
      Date = format(data$Date[sep26_index:(sep26_index+1)], "%Y-%m-%d"),
      Rate = data$`Mid Rate (ZIG)`[sep26_index:(sep26_index+1)]
    )
    print(verification)
  }
} else {
  breakpoint_values <- data.frame(Message = "No breakpoints detected")
}

# [Rest of the code remains the same]
```

```
# 5. Robust Confidence Interval Calculation ------------------------------
safe_confint <- function(bp_obj, break_num, dates) {
  result <- tryCatch({
    ci <- confint(bp_obj, breaks = break_num)
    list(
      lower = dates[ci$confint[break_num, 1]],
      upper = dates[ci$confint[break_num, 3]],
      breakpoint = dates[bp_obj$breakpoints[break_num]]
    )
  }, error = function(e) {
    list(lower = NA, upper = NA, breakpoint = dates[bp_obj$breakpoints[break_num]])
  })
  return(result)
}

ci_results <- if (opt_breaks > 0) {
  lapply(1:opt_breaks, function(i) safe_confint(bp_test, i, data$Date))
} else {
  list()
}

# 6. Enhanced Results Table ----------------------------------------------
generate_results_table <- function(data, break_indices, break_dates, ci_results, break_values) {
  if (opt_breaks == 0) return(data.frame(Message = "No breakpoints detected"))

  break_points <- c(1, break_indices, nrow(data))
  results <- lapply(1:(length(break_points)-1), function(i) {
    seg_data <- data$`Mid Rate (ZIG)`[break_points[i]:break_points[i+1]]
    ci_lower <- if (i <= length(ci_results)) ci_results[[i]]$lower else NA
    ci_upper <- if (i <= length(ci_results)) ci_results[[i]]$upper else NA

    next_day_val <- if (break_points[i] < nrow(data)) data$`Mid Rate (ZIG)`[break_points[i]+1] else NA
    pct_change <- if (!is.na(next_day_val)) round((next_day_val - break_values[i])/break_values[i]*100, 2) else NA

    data.frame(
      Segment = i,
      Start_Date = format(data$Date[break_points[i]], "%Y-%m-%d"),
      End_Date = format(data$Date[break_points[i+1]], "%Y-%m-%d"),
      Break_Rate = break_values[i],
      Next_Day_Rate = next_day_val,
      Daily_Change_Pct = pct_change,
      Duration_Days = as.numeric(data$Date[break_points[i+1]] - data$Date[break_points[i]]) + 1,
      Observations = length(seg_data),
      Segment_Mean = round(mean(seg_data), 4),
      Segment_SD = round(sd(seg_data), 4),
      CI_Lower = if (!is.na(ci_lower)) format(ci_lower, "%Y-%m-%d") else "NA",
      CI_Upper = if (!is.na(ci_upper)) format(ci_upper, "%Y-%m-%d") else "NA",
```

```
      stringsAsFactors = FALSE
    )
  })

  return(do.call(rbind, results))
}

results_table <- generate_results_table(data, break_indices, break_dates, ci_results, break_values)

# 7. Enhanced Visualization ---------------------------------------------
create_breakpoint_plot <- function(data, break_dates, ci_results, break_values, next_day_values) {
  p <- ggplot(data, aes(x = Date, y = `Mid Rate (ZIG)`)) +
    geom_line(color = "blue", linewidth = 0.8) +
    scale_x_date(date_labels = "%Y-%m-%d") +
    labs(title = "ZIG Mid Rate with Structural Breaks Identification uisng Bai Perron Test",
         x = "Date", y = "Mid Rate (ZIG)") +
    theme_minimal(base_size = 12) +
    theme(plot.title = element_text(face = "bold"))

  if (opt_breaks > 0 && !all(is.na(break_dates))) {
    # Add break lines
    p <- p + geom_vline(xintercept = break_dates,
                        color = "red", linetype = "dashed", linewidth = 0.7)

    # Add confidence intervals
    for (i in seq_along(ci_results)) {
      if (!is.na(ci_results[[i]]$lower) && !is.na(ci_results[[i]]$upper)) {
        p <- p + annotate("rect",
                          xmin = ci_results[[i]]$lower,
                          xmax = ci_results[[i]]$upper,
                          ymin = -Inf, ymax = Inf,
                          alpha = 0.2, fill = "orange")
      }
    }

    # Add breakpoint labels with transition arrows
    for (i in seq_along(break_dates)) {
      if (break_indices[i] < nrow(data)) {
        p <- p + annotate("segment",
                          x = break_dates[i],
                          xend = break_dates[i] + 1,
                          y = break_values[i],
                          yend = next_day_values[i],
                          arrow = arrow(length = unit(0.2, "cm")),
                          color = "darkgreen")
      }
    }
```

```
    # Add value labels
    label_data <- data.frame(
      x = break_dates,
      y = rep(max(data$`Mid Rate (ZIG)`) * 0.9, length(break_dates)),
      label = paste0(format(break_dates, "%Y-%m-%d"),
                     "\nRate: ", round(break_values, 4),
                     "\nNext day: ", round(next_day_values, 4))
    )
    p <- p + geom_label(data = label_data,
                        aes(x = x, y = y, label = label),
                        color = "red", size = 3)
  }

  return(p)
}

break_plot <- create_breakpoint_plot(data, break_dates, ci_results, break_values, next_day_values)

# 8. Save Results ------------------------------------------------------
write.csv(breakpoint_values, "breakpoint_transition_values.csv", row.names = FALSE)
write.csv(results_table, "breakpoint_analysis_results.csv", row.names = FALSE)
ggsave("Bai Perron breakpoint_plot.png", break_plot, width = 12, height = 7, dpi = 300)

# 9. Print Results -----------------------------------------------------
cat("\n=== ENHANCED BAI-PERRON STRUCTURAL BREAK ANALYSIS ===\n")
cat("Optimal breakpoints detected:", opt_breaks, "\n\n")

if (opt_breaks > 0) {
  cat("Breakpoint Transition Values:\n")
  print(breakpoint_values, row.names = FALSE)

  cat("\nDetailed Segment Statistics:\n")
  print(results_table, row.names = FALSE)

  cat("\nBreakpoint visualization saved as 'Bai Perron breakpoint_plot.png'\n")
} else {
  cat("No structural breaks detected in the time series.\n")
}

cat("\nResults saved to:\n",
    "- breakpoint_transition_values.csv\n",
    "- breakpoint_analysis_results.csv\n",
    "- breakpoint_plot.png\n")
```

Data set

	A	B	C	D	E	F
1	Date	Month	Bid (ZIG)	Ask (ZIG)	Mid Rate (ZIG)	
2	8-Apr-2024	Apr	13.2226	13.9006	13.5616	
3	9-Apr-2024	Apr	13.0545	14.0151	13.5348	
4	10-Apr-2024	Apr	13.1103	13.892	13.5012	
5	11-Apr-2024	Apr	13.0156	13.8928	13.4542	
6	12-Apr-2024	Apr	13.0824	13.7532	13.4178	
7	15-Apr-2024	Apr	13.0676	13.7378	13.4027	
8	16-Apr-2024	Apr	13.028	13.696	13.362	
9	17-Apr-2024	Apr	13.0102	13.6772	13.3437	
10	19-Apr-2024	Apr	12.9801	13.6457	13.3129	
11	22-Apr-2024	Apr	12.9567	13.6211	13.2889	
12	23-Apr-2024	Apr	12.9223	13.5849	13.2536	
13	24-Apr-2024	Apr	12.9204	13.583	13.2517	
14	25-Apr-2024	Apr	13.0466	13.7156	13.3811	
15	26-Apr-2024	Apr	13.0863	13.7573	13.4218	
16	29-Apr-2024	Apr	13.1047	13.7767	13.4407	
17	30-Apr-2024	Apr	13.0943	13.7658	13.4301	
18	2-May-2024	May	13.0835	13.7545	13.419	
19	3-May-2024	May	13.2151	13.8928	13.554	
20	6-May-2024	May	13.3338	14.0176	13.6757	
21	7-May-2024	May	13.3272	14.0106	13.6689	
22	8-May-2024	May	13.3237	14.0069	13.6653	
23	9-May-2024	May	13.1943	13.8709	13.5326	
24	10-May-2024	May	13.1807	13.8567	13.5187	
25	13-May-2024	May	13.1453	13.8193	13.4823	
26	14-May-2024	May	13.0772	13.7479	13.4126	
27	15-May-2024	May	13.0745	13.7449	13.4097	

Figure 5.1: First Few Columns of the Dataset

⋮

	A	B	C	D	E	F
271	22-Jan-2025	Jan	25.6148	26.9284	26.2716	
272	23-Jan-2025	Jan	25.633	26.9476	26.2903	
273	24-Jan-2025	Jan	25.6429	26.9579	26.3004	
274	25-Jan-2025	Jan	25.6429	26.9579	26.3004	
275	26-Jan-2025	Jan	25.6429	26.9579	26.3004	
276	27-Jan-2025	Jan	25.6631	26.9792	26.3212	
277	28-Jan-2025	Jan	25.6801	26.9971	26.3386	
278	29-Jan-2025	Jan	25.6874	27.0048	26.3461	
279	30-Jan-2025	Jan	25.6914	27.009	26.3502	
280	31-Jan-2025	Jan	25.7065	27.0247	26.3656	
281	1-Feb-2025	Feb	25.7065	27.0247	26.3656	
282	2-Feb-2025	Feb	25.7065	27.0247	26.3656	
283	3-Feb-2025	Feb	25.7218	27.0408	26.3813	
284	4-Feb-2025	Feb	25.7298	27.0492	26.3895	
285	5-Feb-2025	Feb	25.731	27.0506	26.3908	
286	6-Feb-2025	Feb	25.7417	27.0617	26.4017	
287	7-Feb-2025	Feb	25.747	27.0674	26.4072	
288	8-Feb-2025	Feb	25.747	27.0674	26.4072	
289	9-Feb-2025	Feb	25.747	27.0674	26.4072	
290	10-Feb-2025	Feb	25.7523	27.0729	26.4126	
291	11-Feb-2025	Feb	25.7582	27.0792	26.4187	
292	12-Feb-2025	Feb	25.7605	27.0815	26.421	
293	13-Feb-2025	Feb	25.7711	27.0927	26.4319	
294	14-Feb-2025	Feb	25.7756	27.0974	26.4365	
295	15-Feb-2025	Feb	25.7756	27.0974	26.4365	
296	16-Feb-2025	Feb	25.7756	27.0974	26.4365	
297						

Sheet1

Figure 5.2: Last Few Columns of the Dataset